

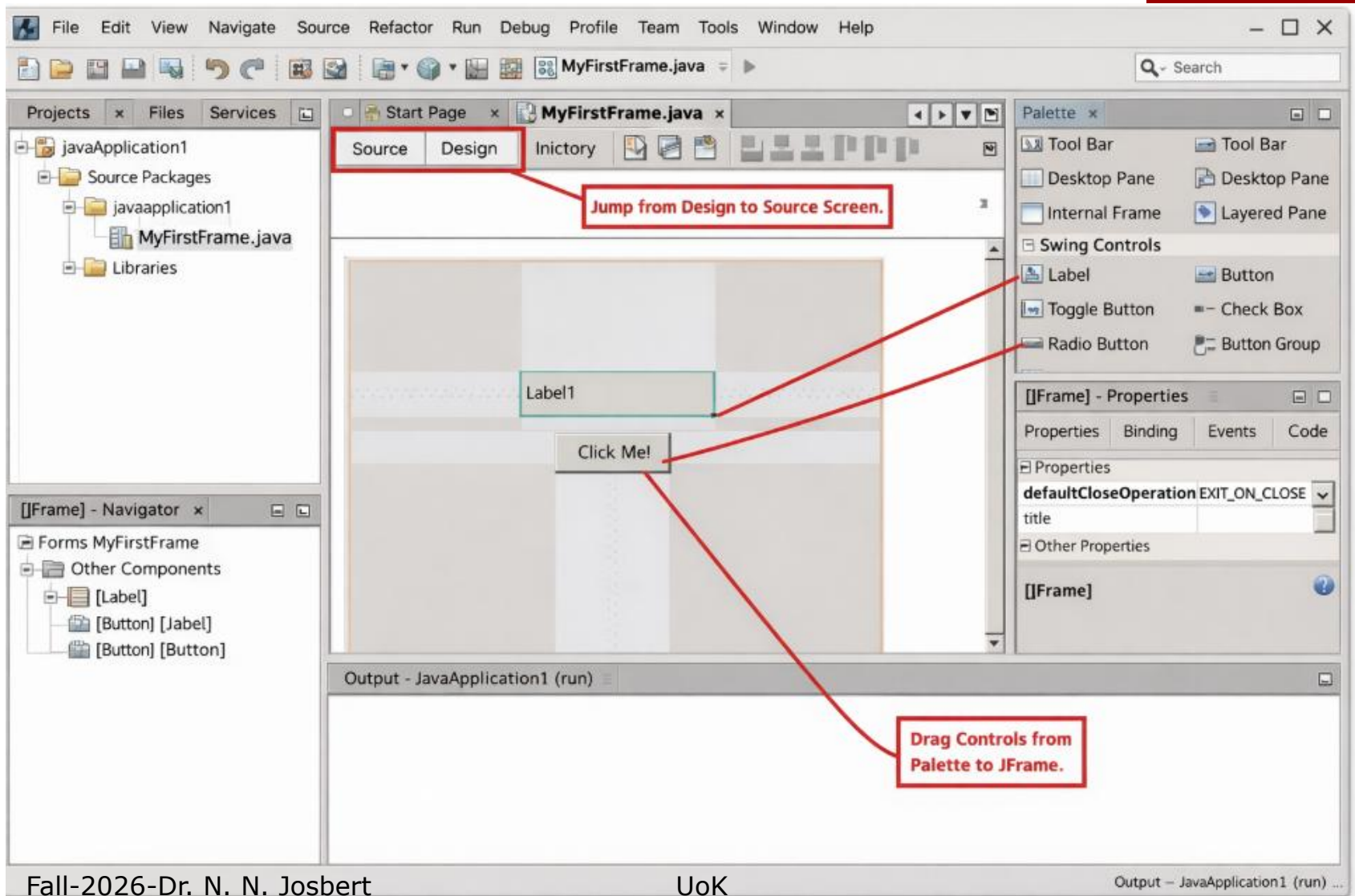
Advanced Java Programming

❑ **UNIT 3: NetBeans IDE GUI builder**

❑ **University of Kigali**

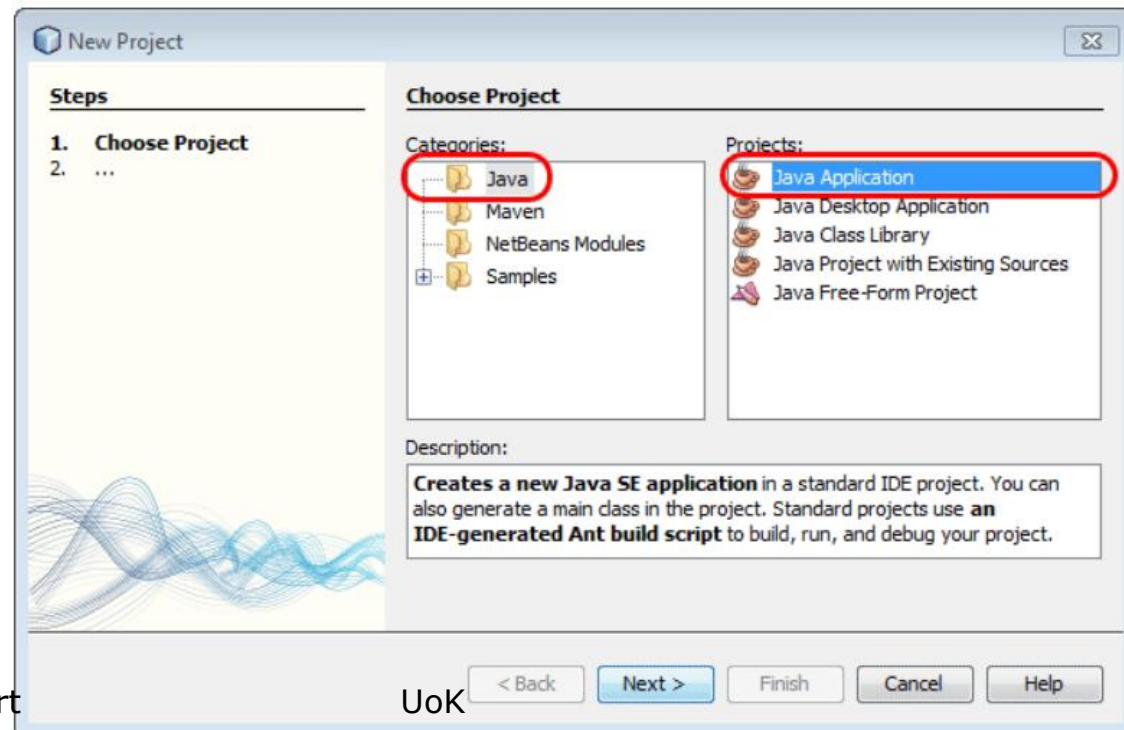
❑ **Trainer name: Dr. NTEZIRIZA NKERABAHIZI Josbert**

NetBeans IDE GUI builder



NetBeans IDE GUI builder

- Open the program named NetBeans.
- If you have not done so already create a folder in your I: drive named Programming 2
- From the File menu choose New Project. Select Java from the Categories list and Java Application from the Projects list. Press the Next button.



NetBeans IDE GUI builder

- Enter a project name. Browse to your I: drive Programming 2 folder. (Note: the computer will create a folder using the project name within the Programming 2 folder) Deselect the "Create Main Class" checkbox and finally press the Finish button.

New Java Application

Steps

1. Choose Project
2. **Name and Location**

Name and Location

Project Name:

Project Location:

Project Folder:

☐ Use Dedicated Folder for Storing Libraries

Libraries Folder:

Different users and projects can share the same compilation libraries (see Help for details).

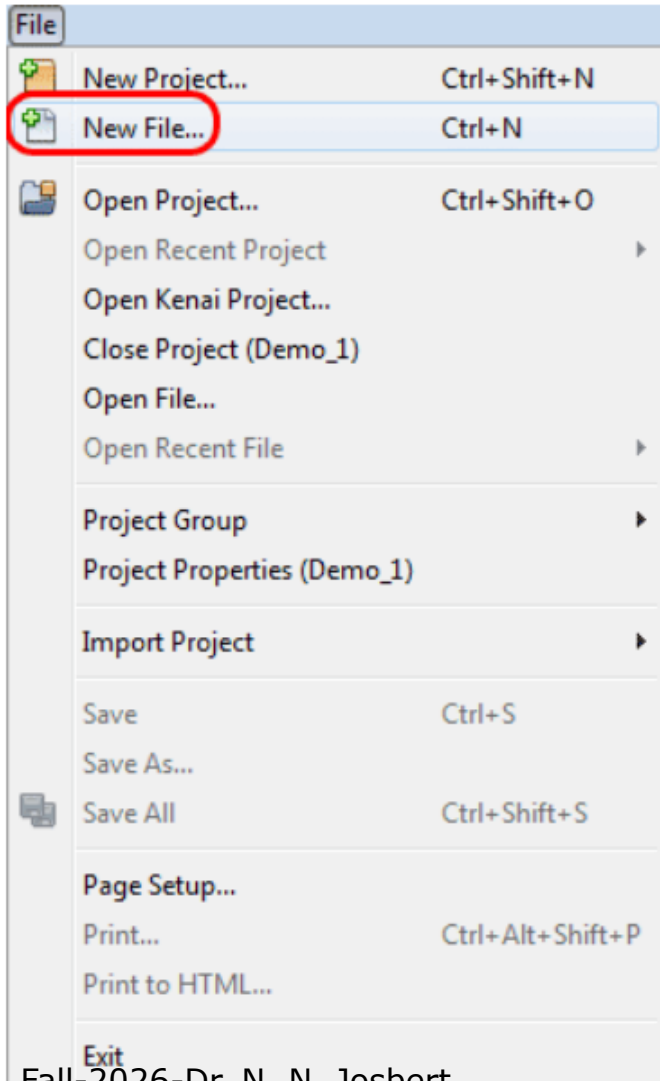
☐ Create Main Class

☒ Set as Main Project

< Back Next > Finish Cancel Help

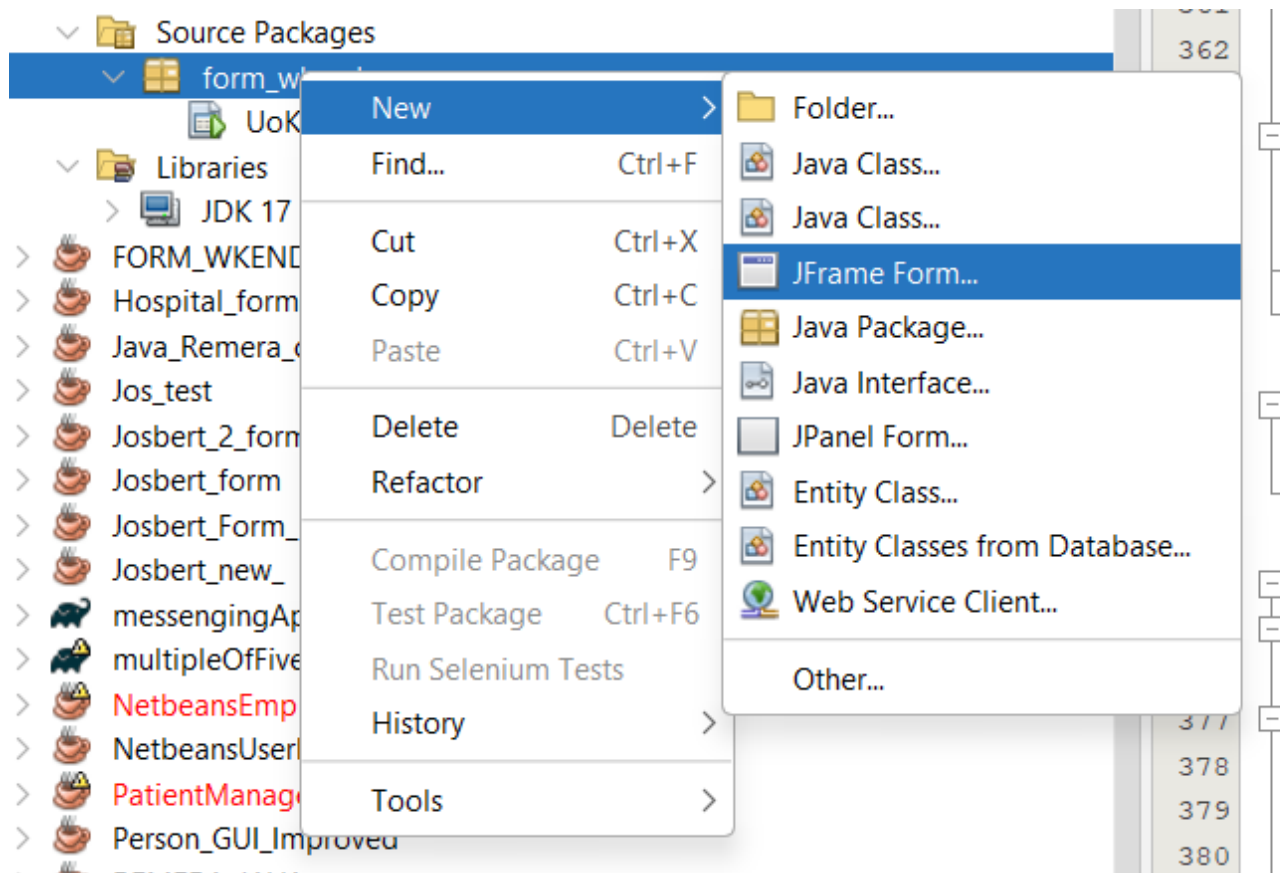
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- In the NetBeans program choose New File from the File menu.



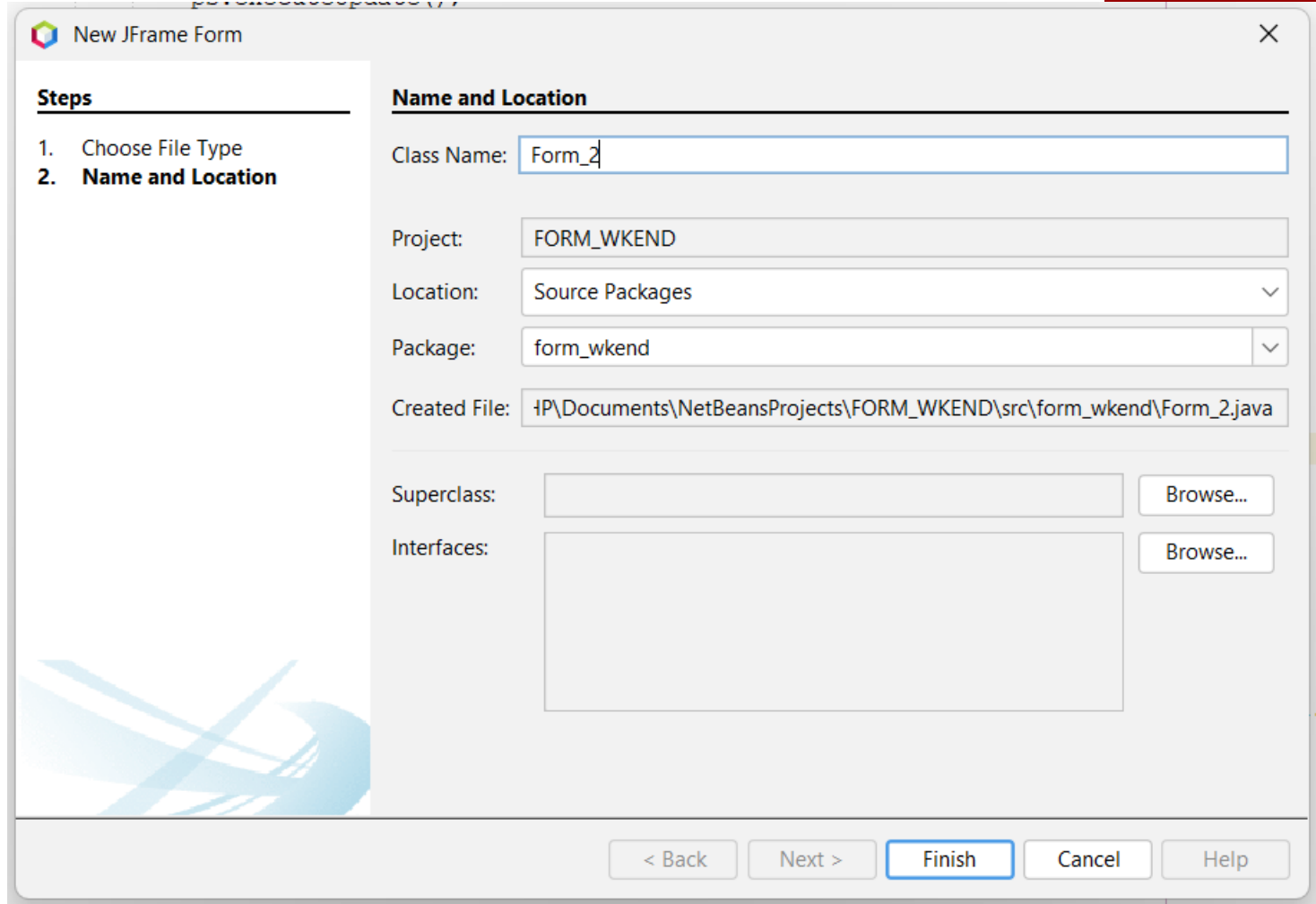
NetBeans IDE GUI builder

- In the New File window select Swing GUI Forms for the Category and JFrame Form for the File Type. Press the Next button. Press the Finish button.



NetBeans IDE GUI builder

- Write the Class Name
- Finish



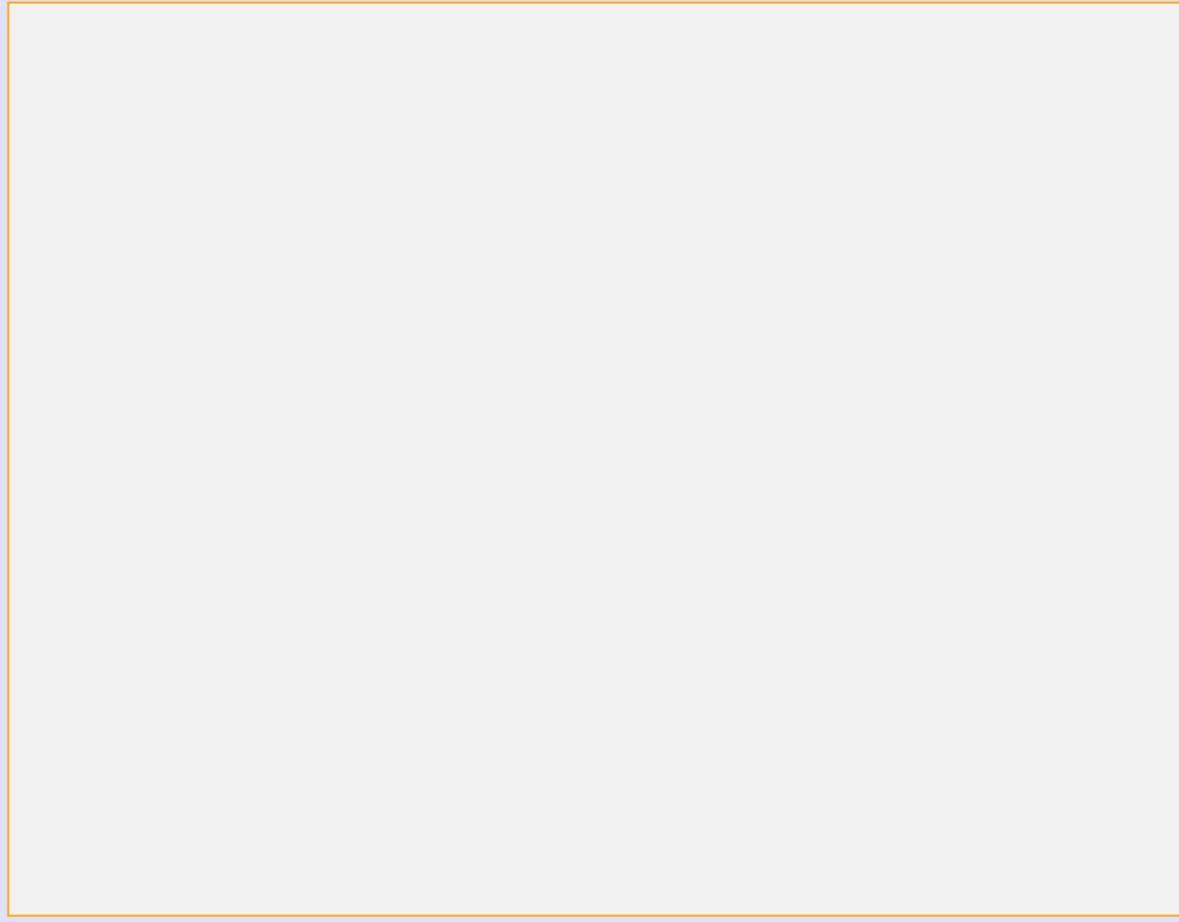
The screenshot shows the 'New JFrame Form' dialog box in NetBeans IDE. The dialog is titled 'New JFrame Form' and has a close button (X) in the top right corner. On the left, there is a 'Steps' panel with two steps: '1. Choose File Type' and '2. Name and Location'. The 'Name and Location' section on the right contains the following fields and controls:

- Class Name:** A text field containing 'Form_2'.
- Project:** A text field containing 'FORM_WKEND'.
- Location:** A dropdown menu showing 'Source Packages'.
- Package:** A dropdown menu showing 'form_wkend'.
- Created File:** A text field showing the full path: 'I:\Documents\NetBeansProjects\FORM_WKEND\src\form_wkend\Form_2.java'.
- Superclass:** A text field with a 'Browse...' button next to it.
- Interfaces:** A large text area with a 'Browse...' button next to it.

At the bottom of the dialog, there are five buttons: '< Back', 'Next >', 'Finish' (which is highlighted with a blue border), 'Cancel', and 'Help'.

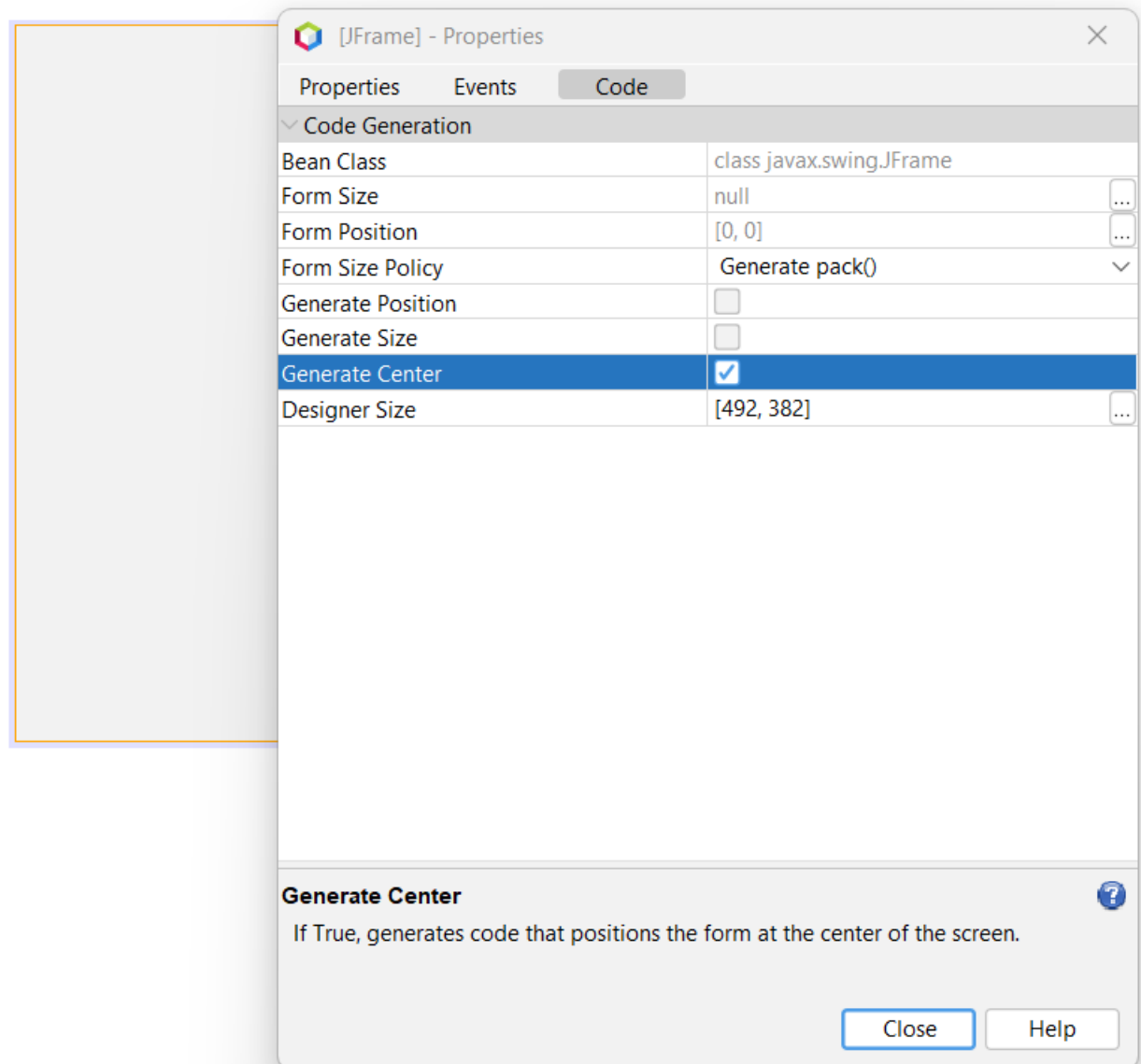
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■ JForm



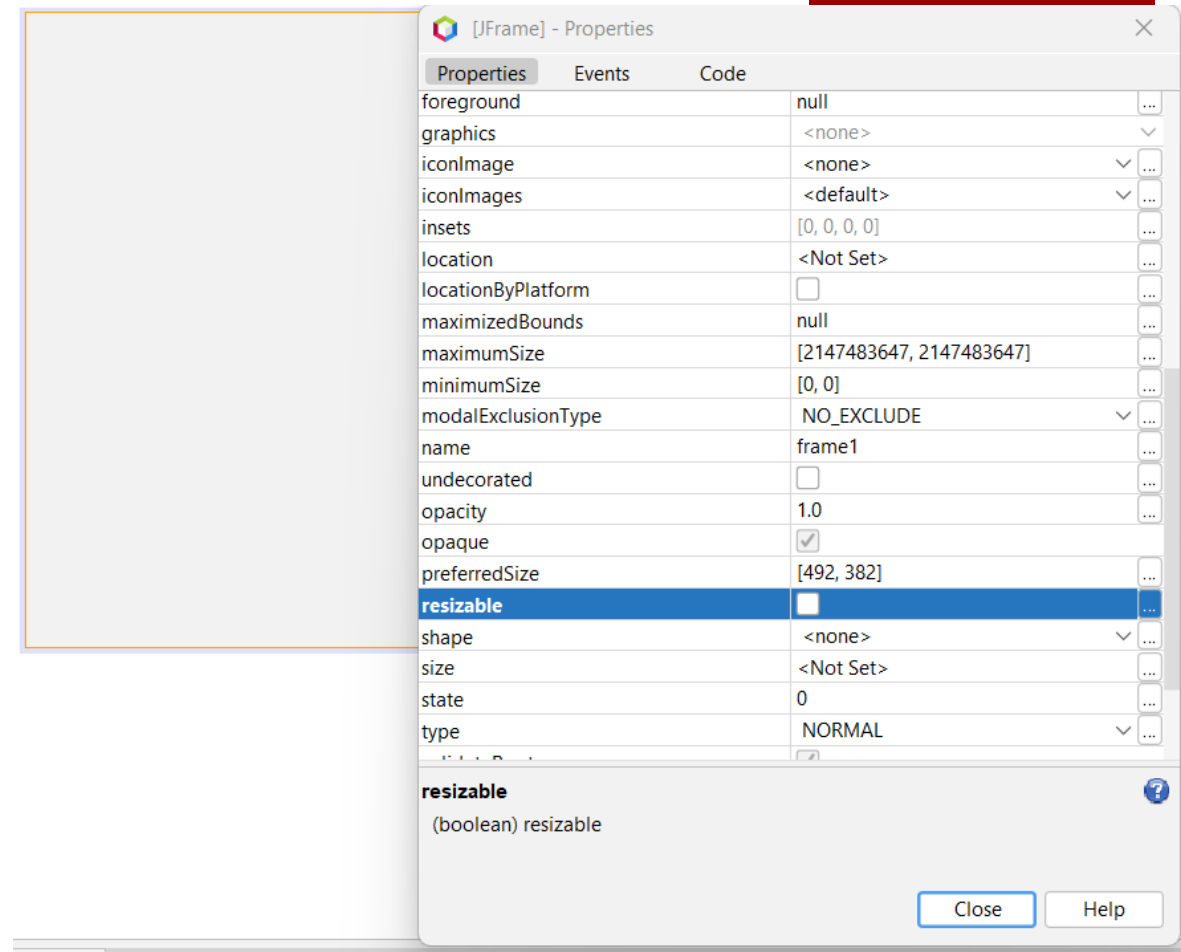
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- Right click
- Properties
- Code
- Generate Center
- Activate
- Close



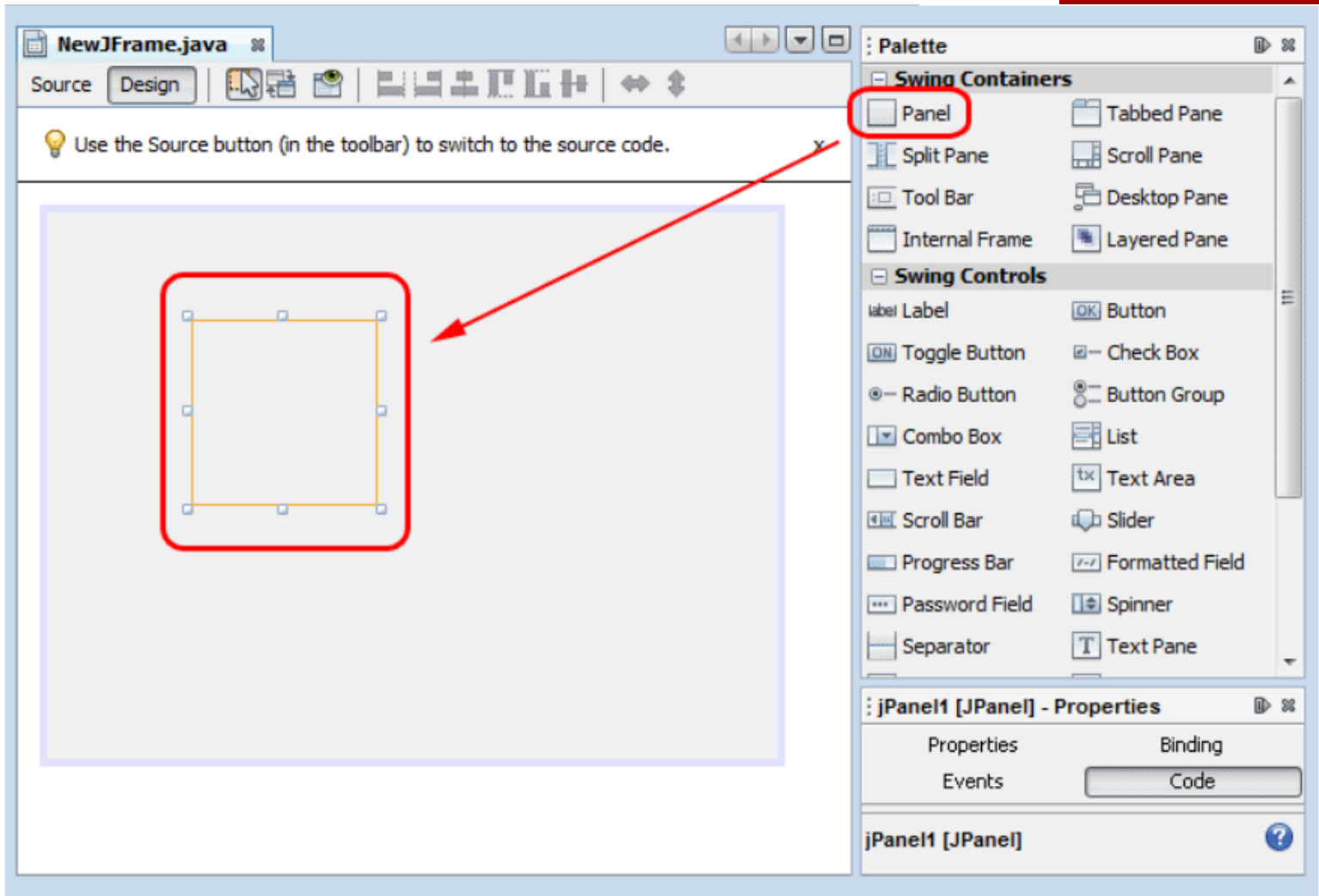
NetBeans IDE GUI builder

- Right click
- Properties
- Properties
- Disactivate resizable



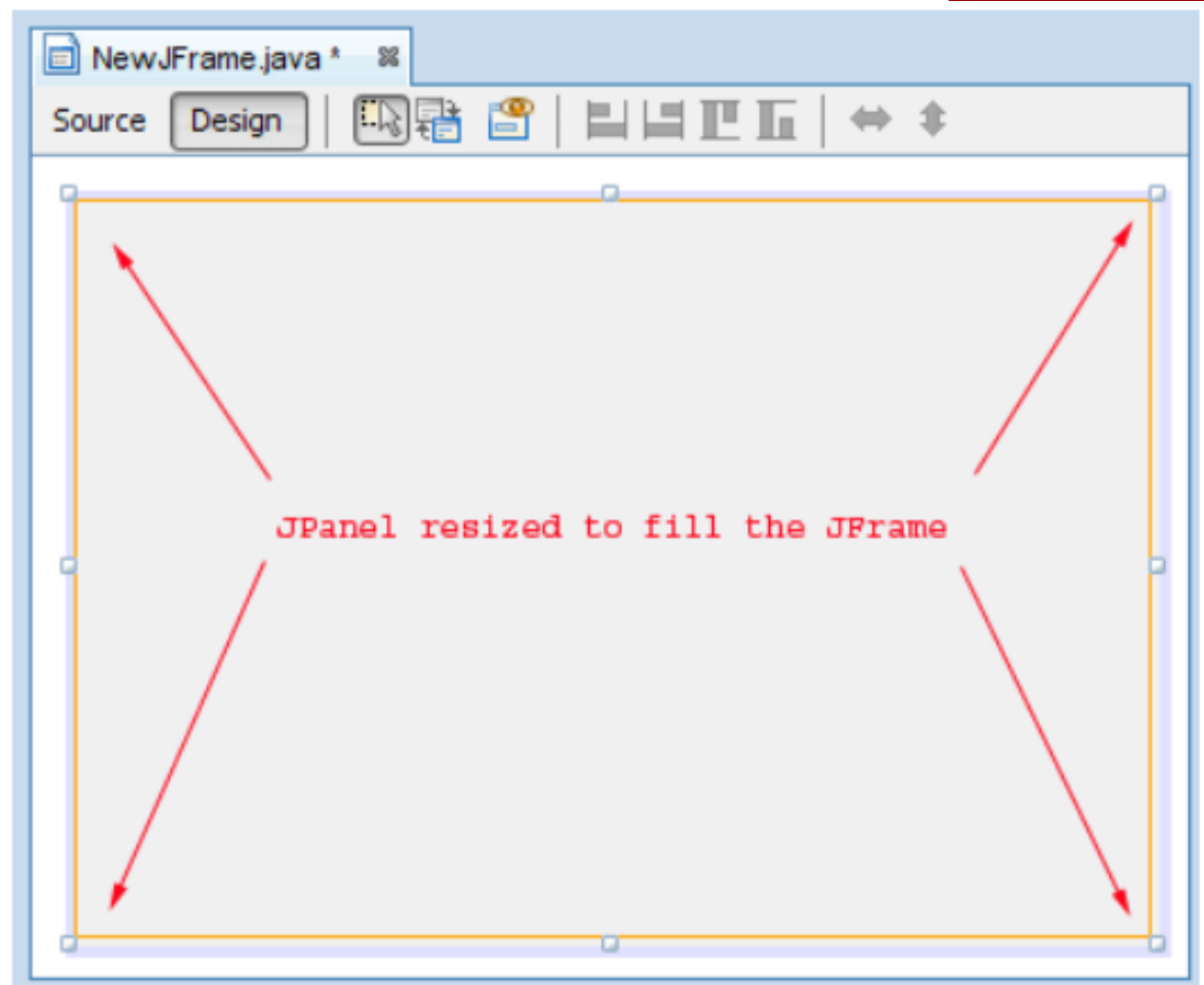
NetBeans IDE GUI builder

- From the Swing Containers palette drag a Panel to the JFrame of the project.



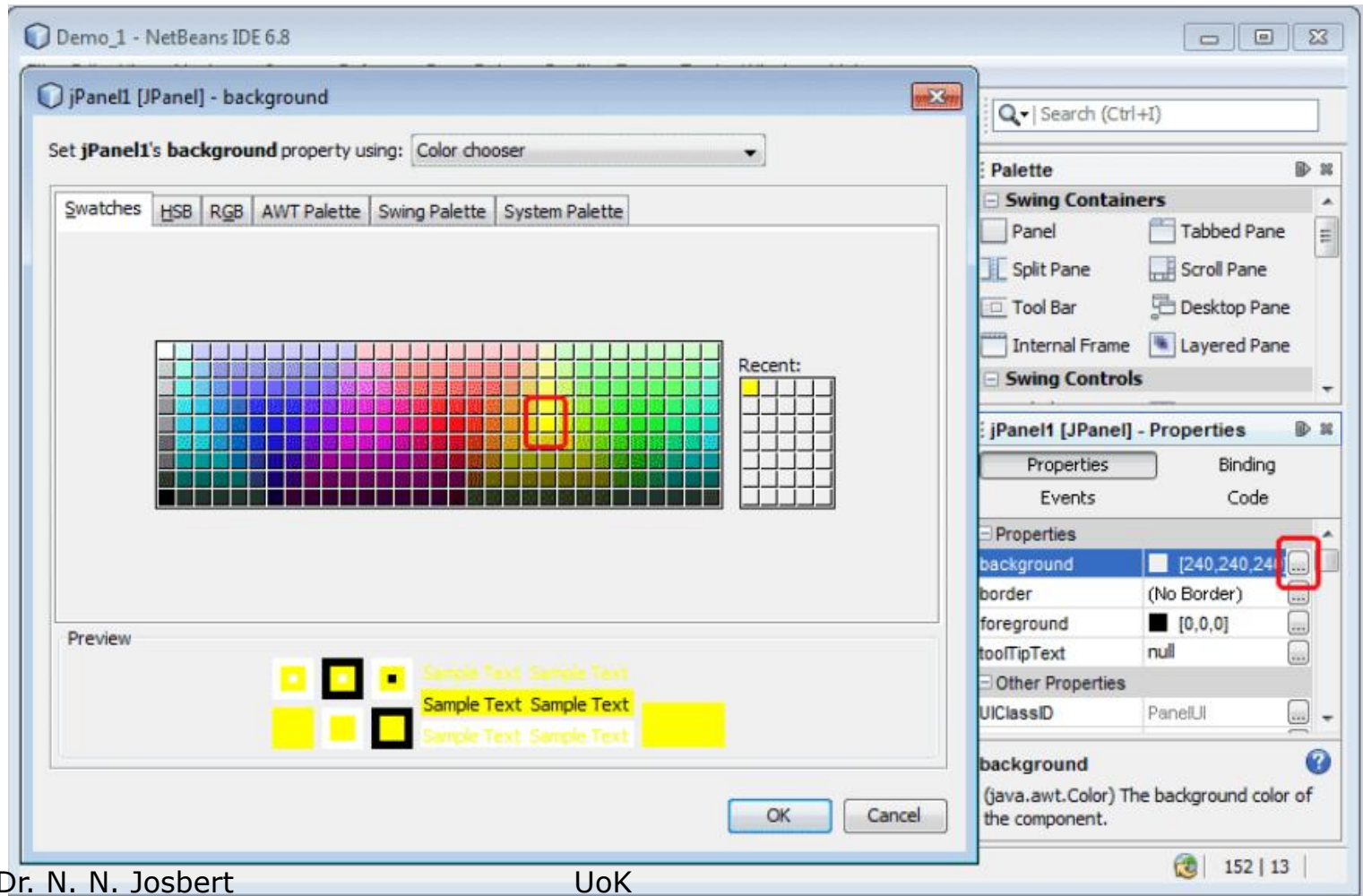
NetBeans IDE GUI builder

- Grab the corner handles and resize the JPanel to fill the entire JFrame.



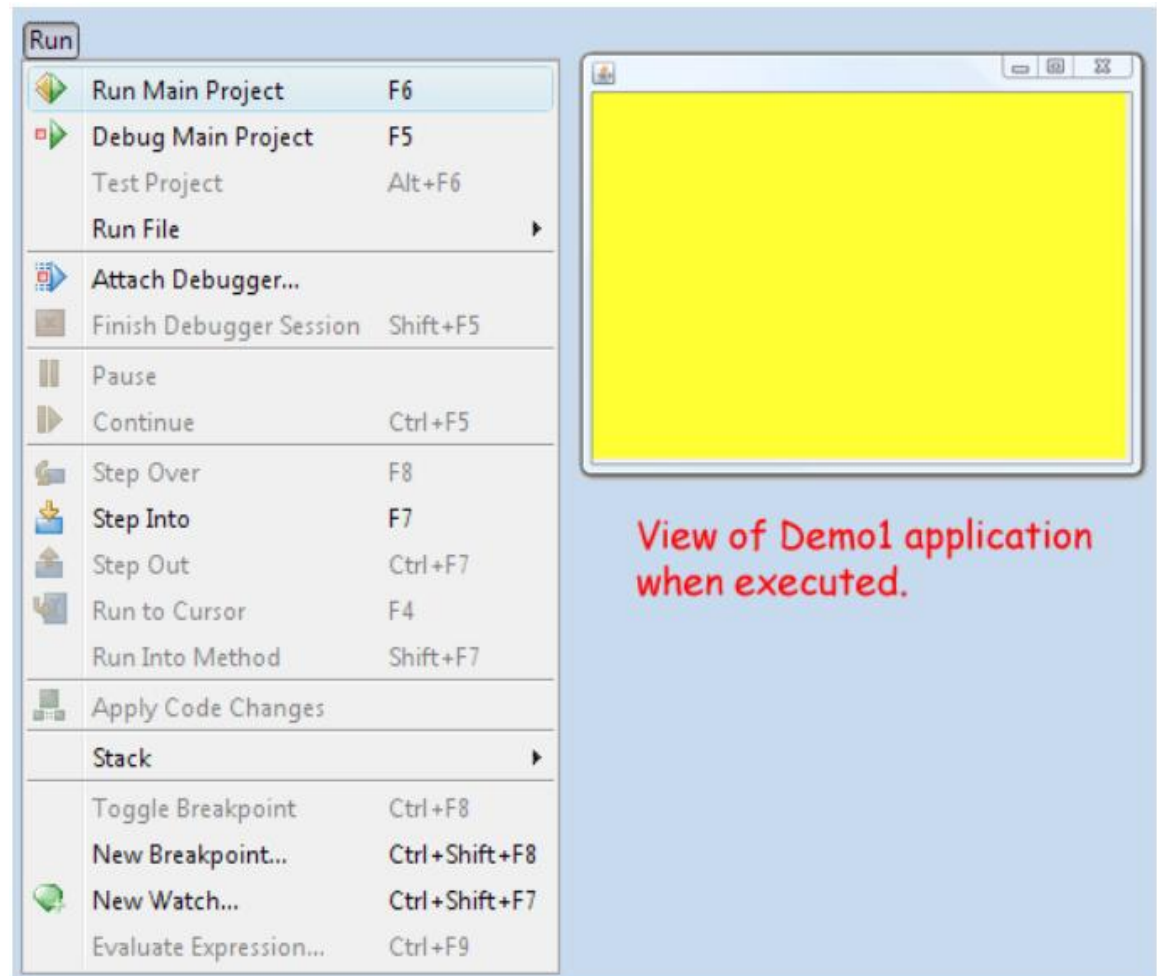
NetBeans IDE GUI builder

- Select the JPanel. In the jPanel1 [JPanel] - Properties palette, select the **background property** and select a yellow color. Press the OK button.



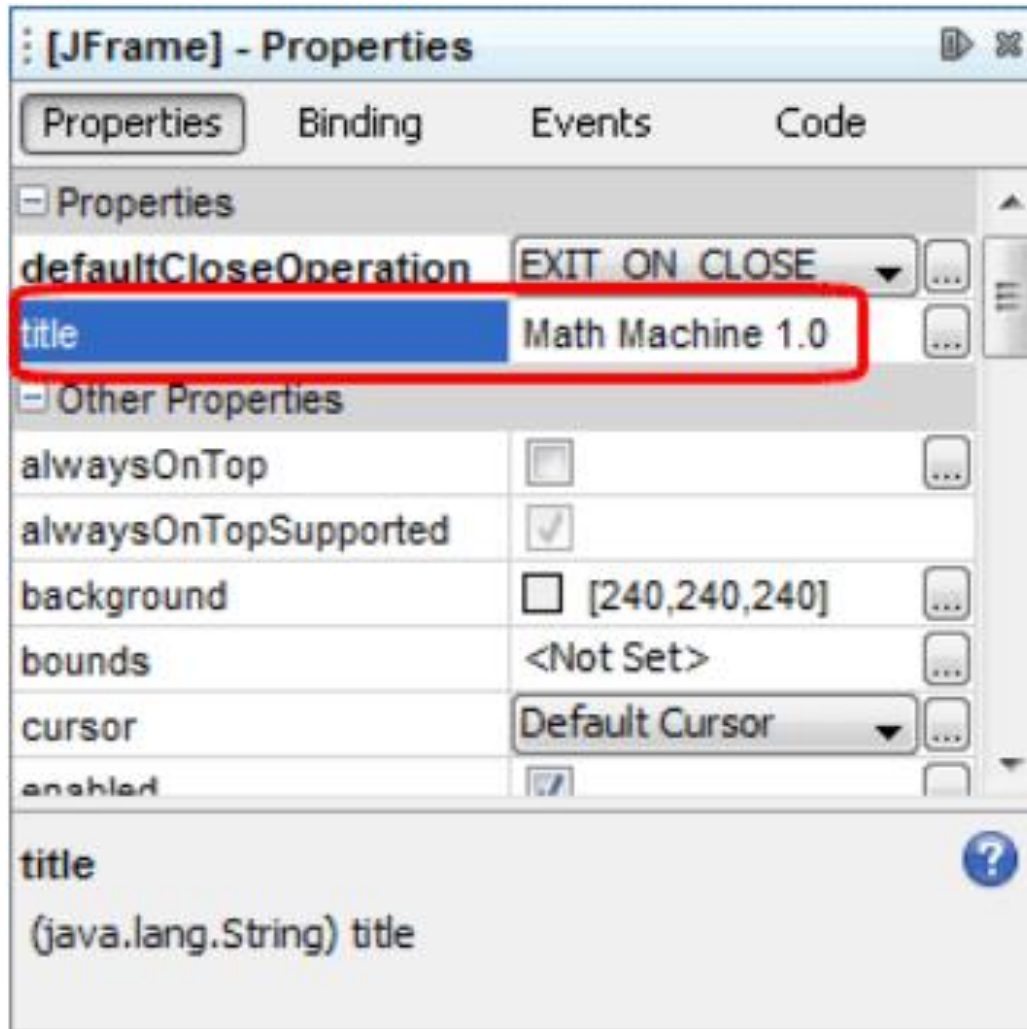
NetBeans IDE GUI builder

- From the Run menu select Run Main Project. Press the OK button when asked to select the main class. Close the application after you are done viewing it.

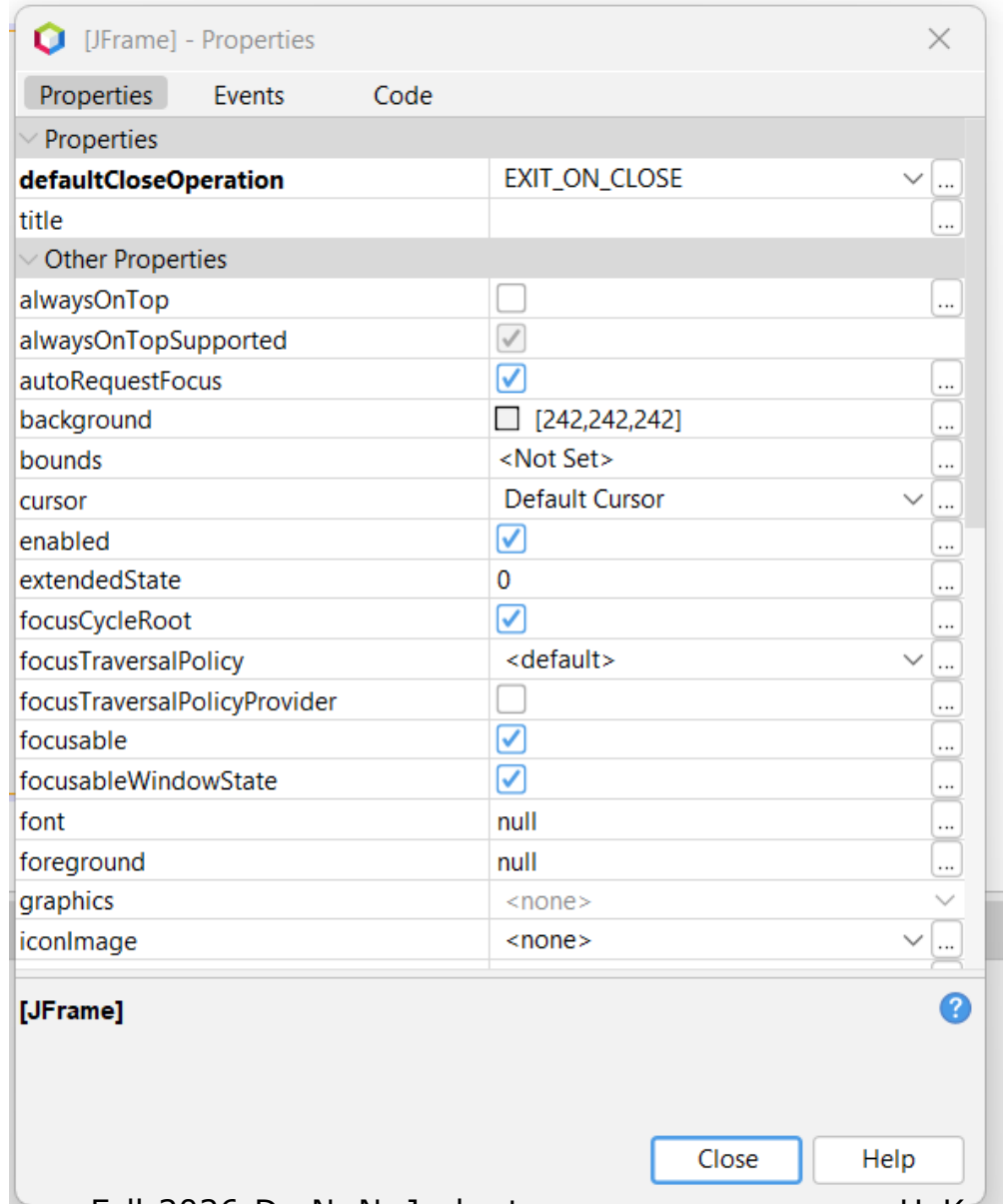


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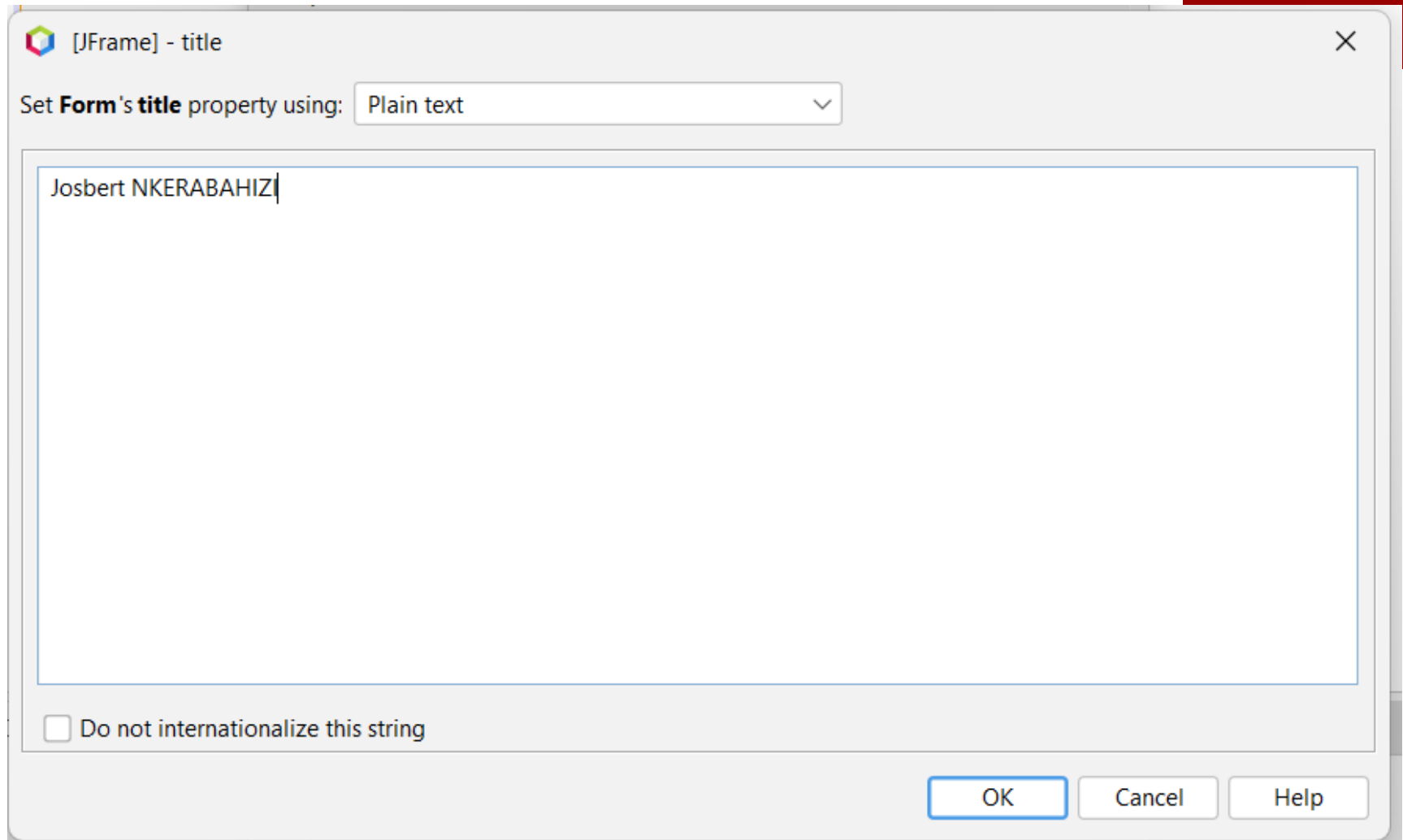
- In the Inspector select the **JFrame**. In the Properties **JFrame** enter a title.



NetBeans IDE GUI builder



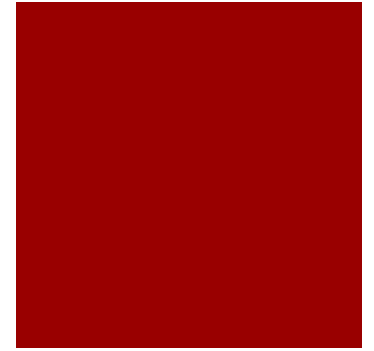
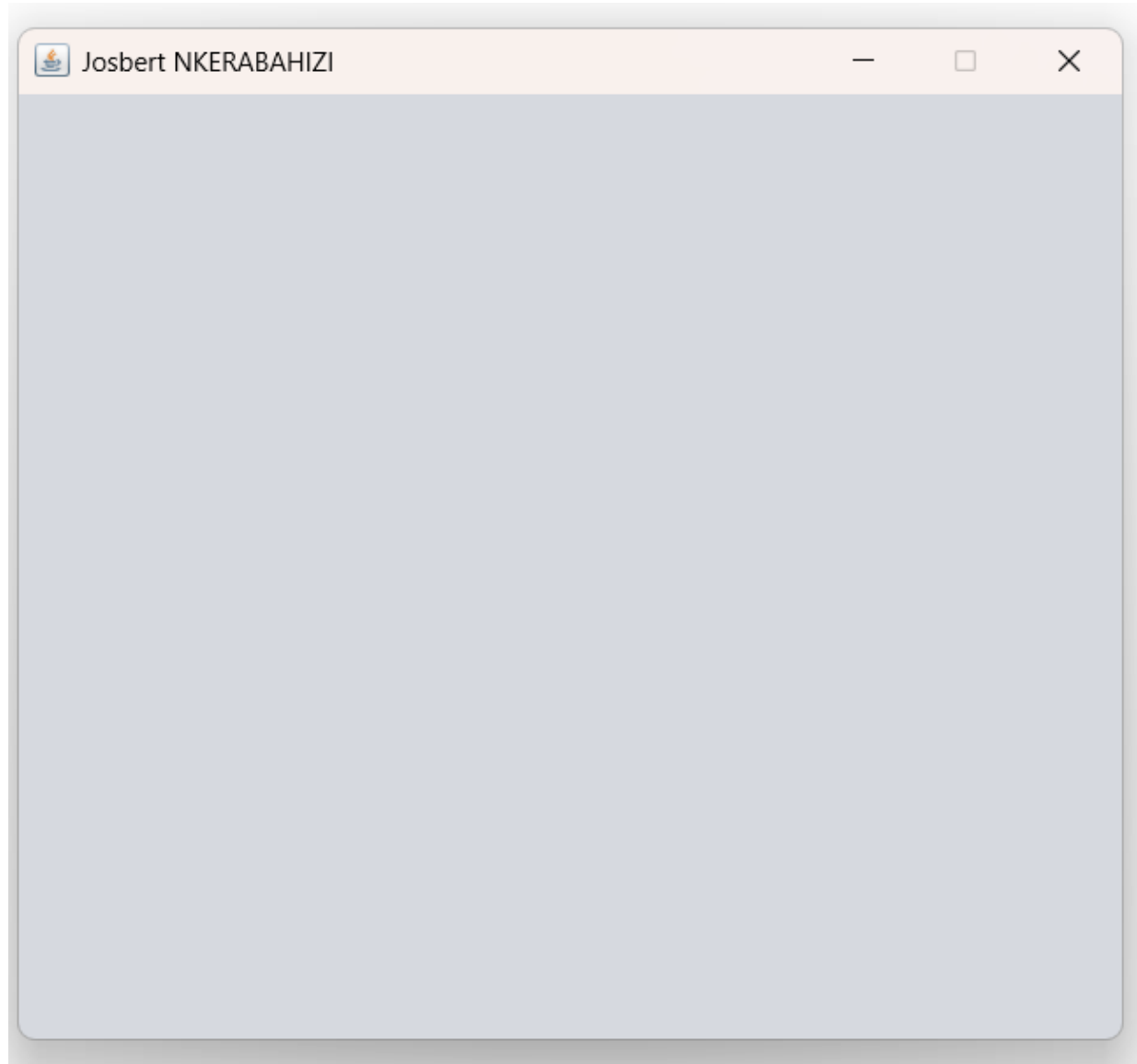
NetBeans IDE GUI builder



NetBeans IDE GUI builder

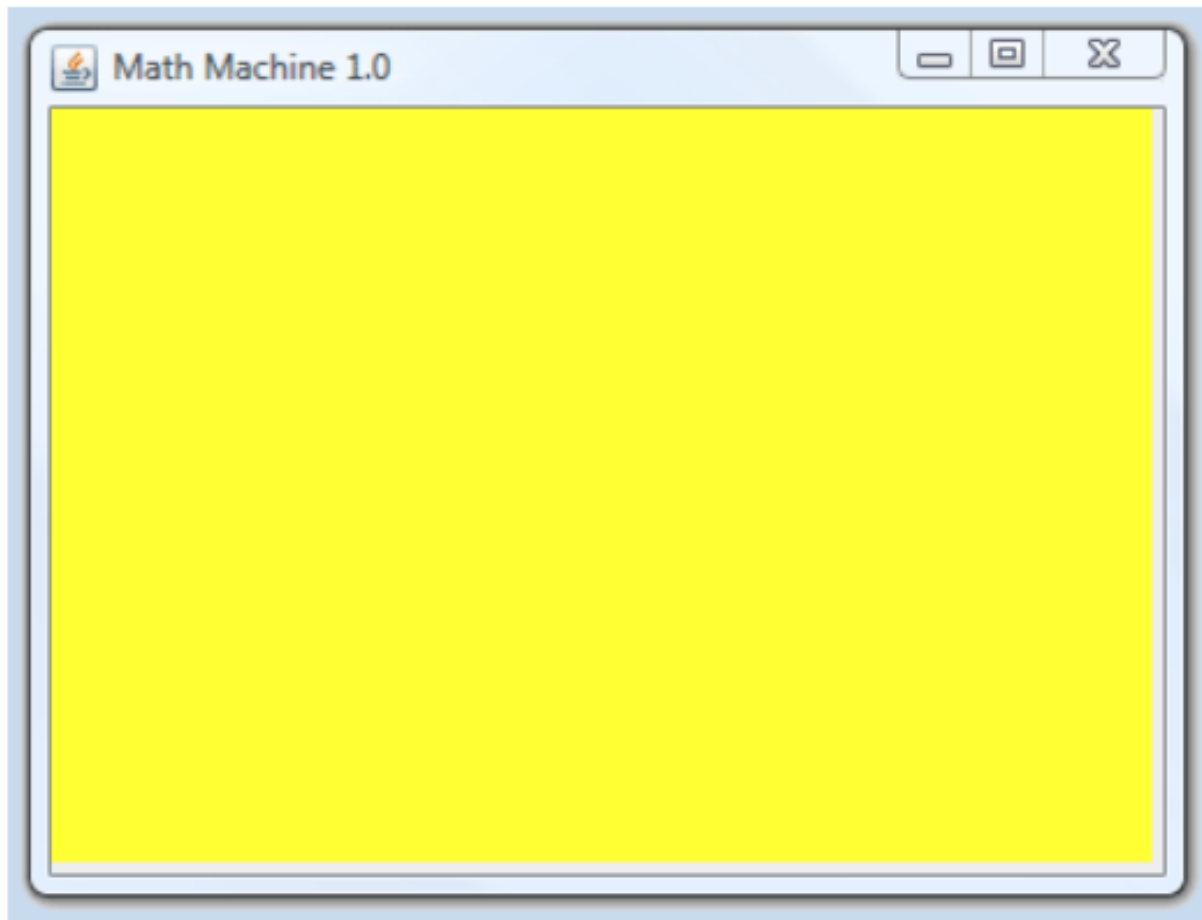


NetBeans IDE GUI builder



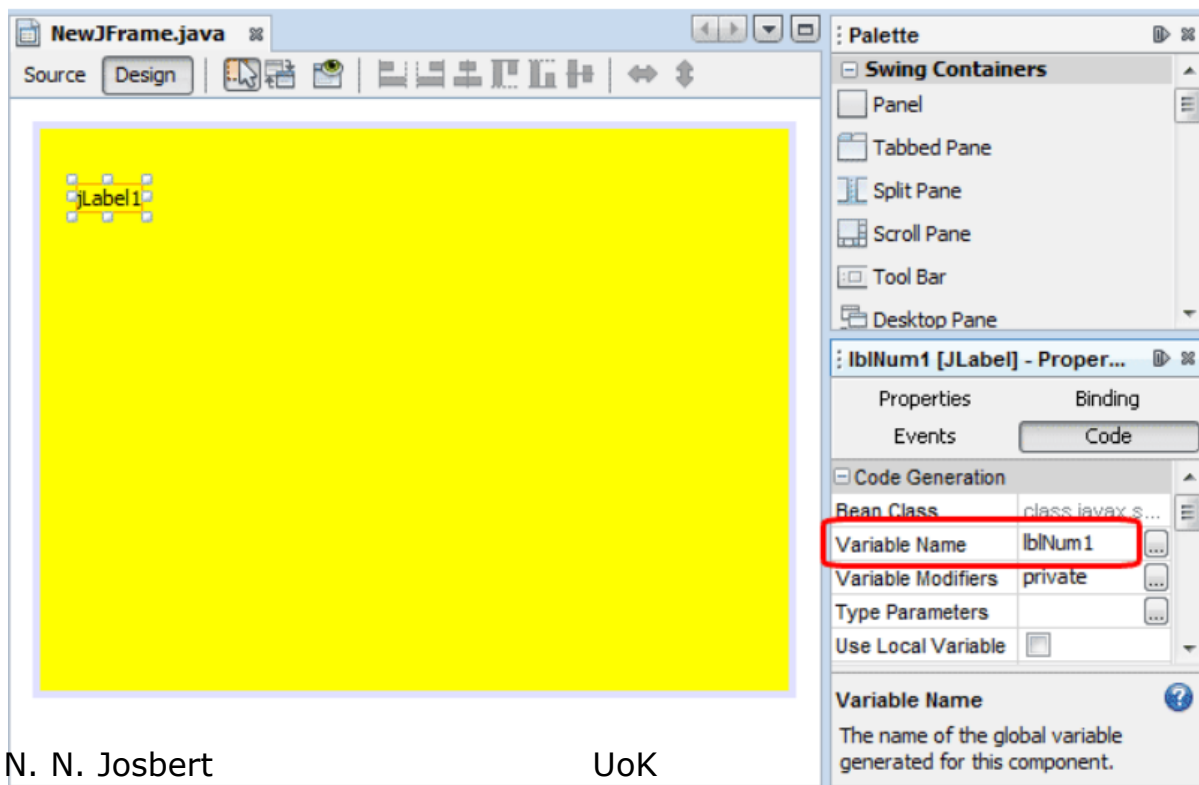
NetBeans IDE GUI builder

- Run the project again and notice that the window has a title. Close the application.



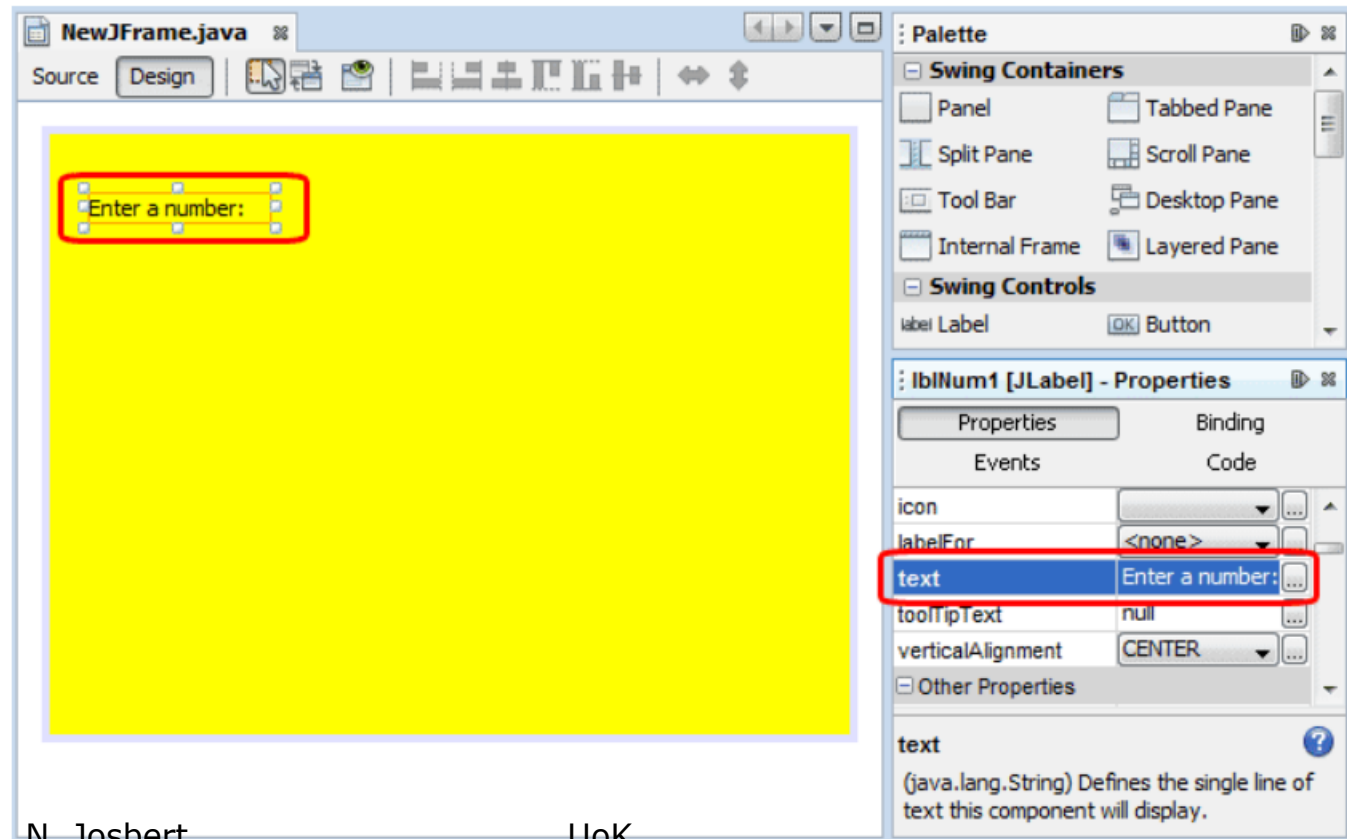
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- Next we will add a label. Labels are used to display text. Drag a Label from the Swing Controls palette onto the Panel of the project. In the Properties palette select the Code button. Change the Variable Name from the default value of jLabel1 to lblNum1. The variable name is the instance name used when coding in Java



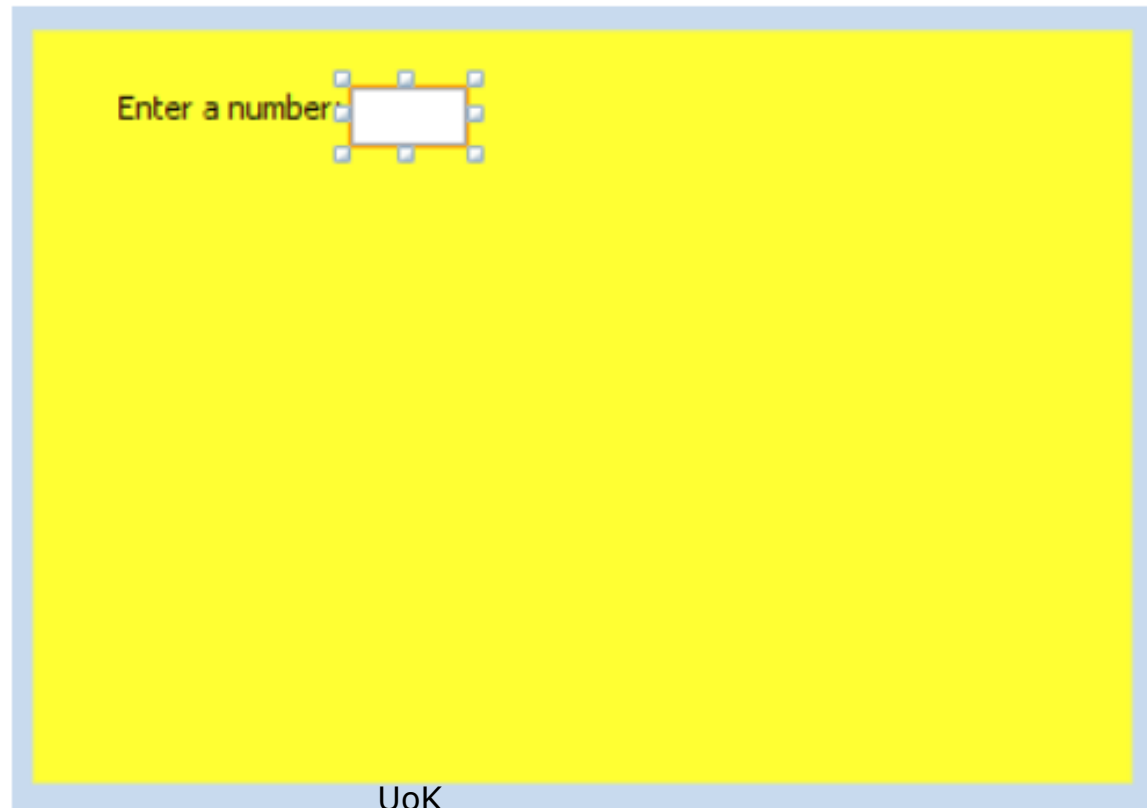
NetBeans IDE GUI builder

- The label currently displays the default text, JLabel1. To change the text select the text property in the Properties palette and enter the text Enter a number: On the JPanel drag the resize handles so that all of the text is visible.



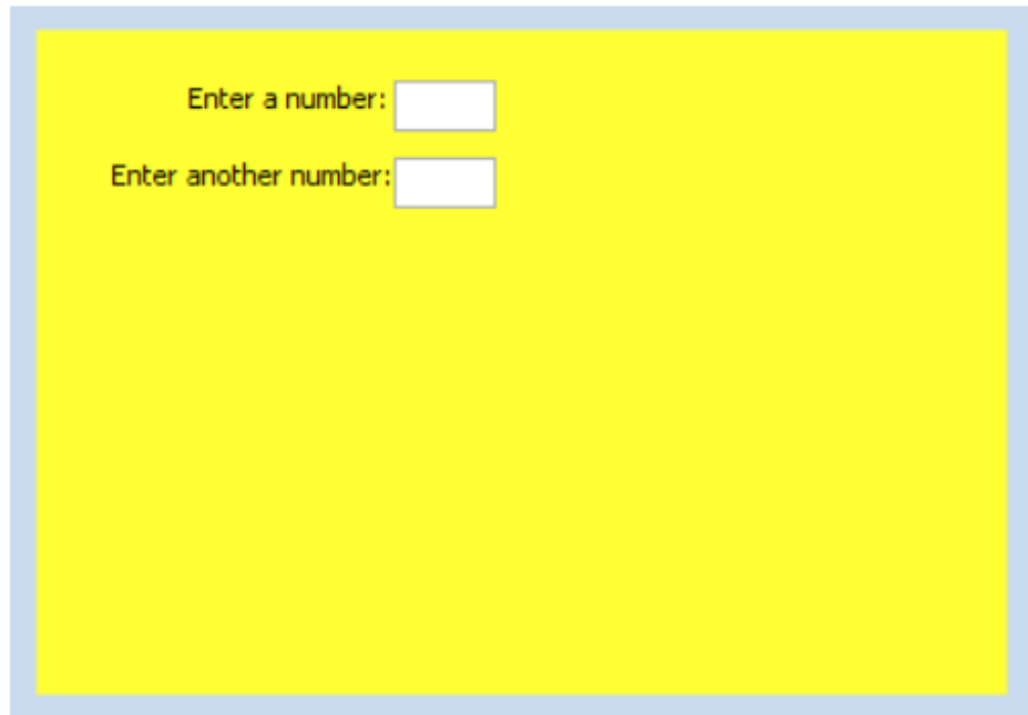
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- From the Swing Controls palette drag a Text Field object onto the project. In the Properties palette clear the default text.
- In the Code area of the Properties palette change the Variable Name to txtNum1.



NetBeans IDE GUI builder

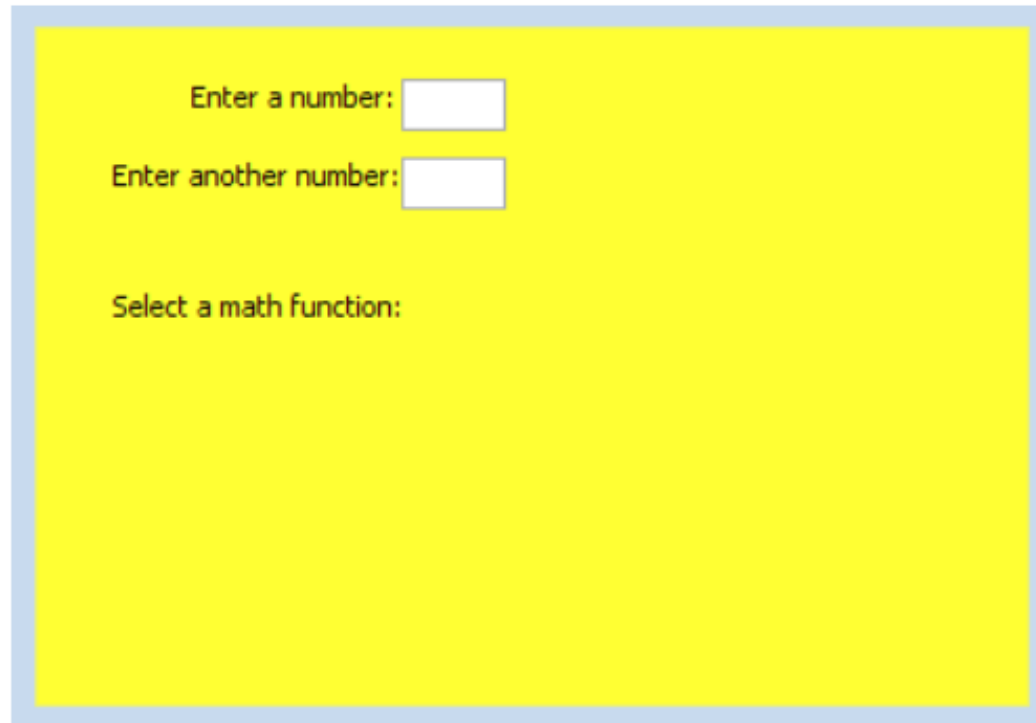
- Add another Label. Give it a Variable Name of lblNum2 and a text value of Enter another number:
- Add another Text Field to the form. Give it a Variable Name of txtNum2 and clear the text so that it is empty.



The image shows a screenshot of a NetBeans IDE GUI builder window. The window has a light blue border. Inside, there is a yellow rectangular area representing the form. On the yellow area, there are two labels and two text fields. The first label is "Enter a number:" followed by a white text field. The second label is "Enter another number:" followed by another white text field. The text fields are empty.

NetBeans IDE GUI builder

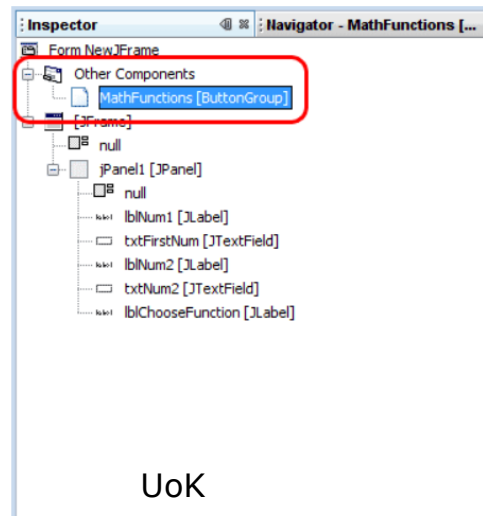
- Add another Label. Give it a Variable Name of lblChooseFunction and a text value of Select a math function:



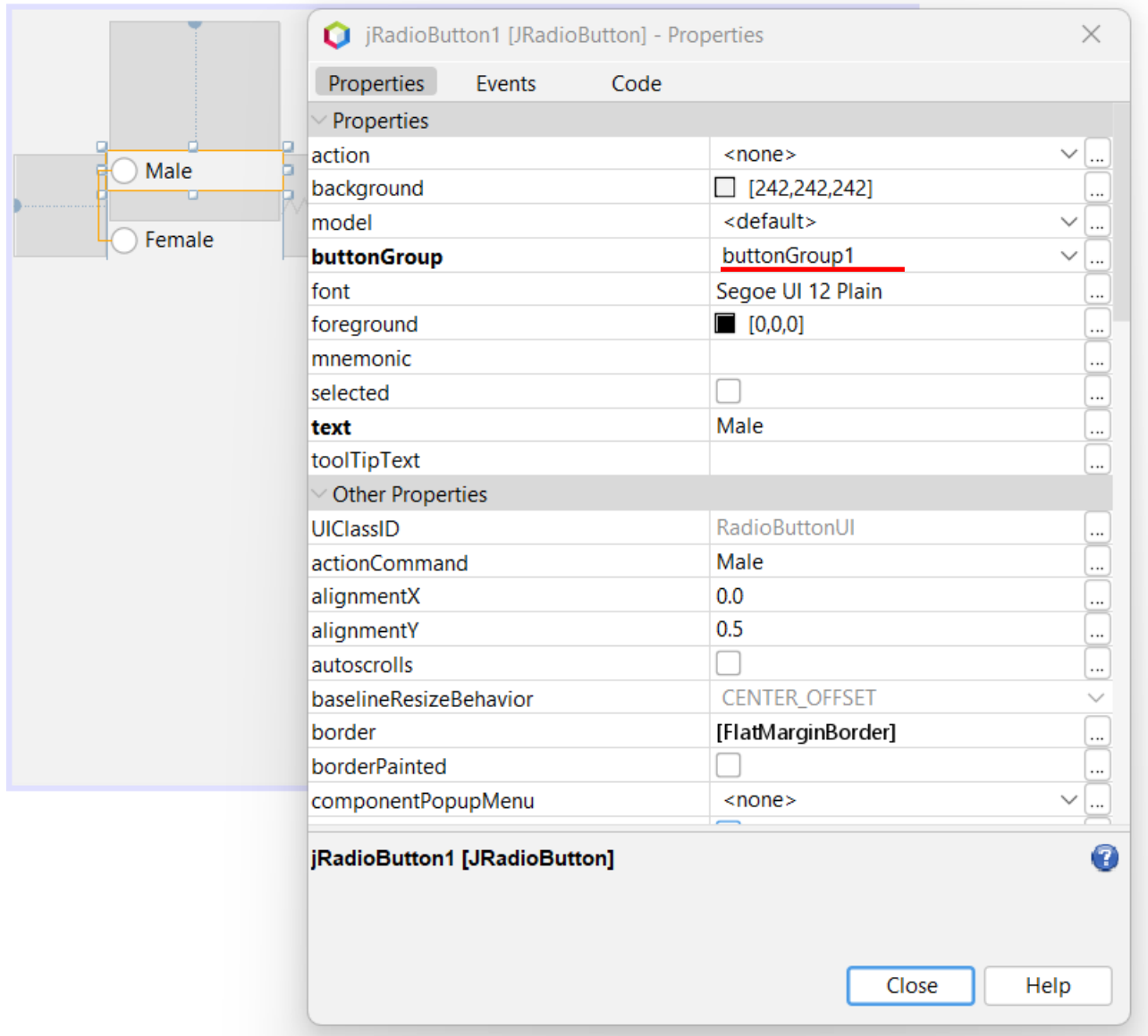
The screenshot shows a NetBeans IDE GUI builder window with a yellow canvas. On the canvas, there are three labels and two text input fields. The first label is "Enter a number:" followed by a text input field. The second label is "Enter another number:" followed by a text input field. The third label is "Select a math function:".

NetBeans IDE GUI builder

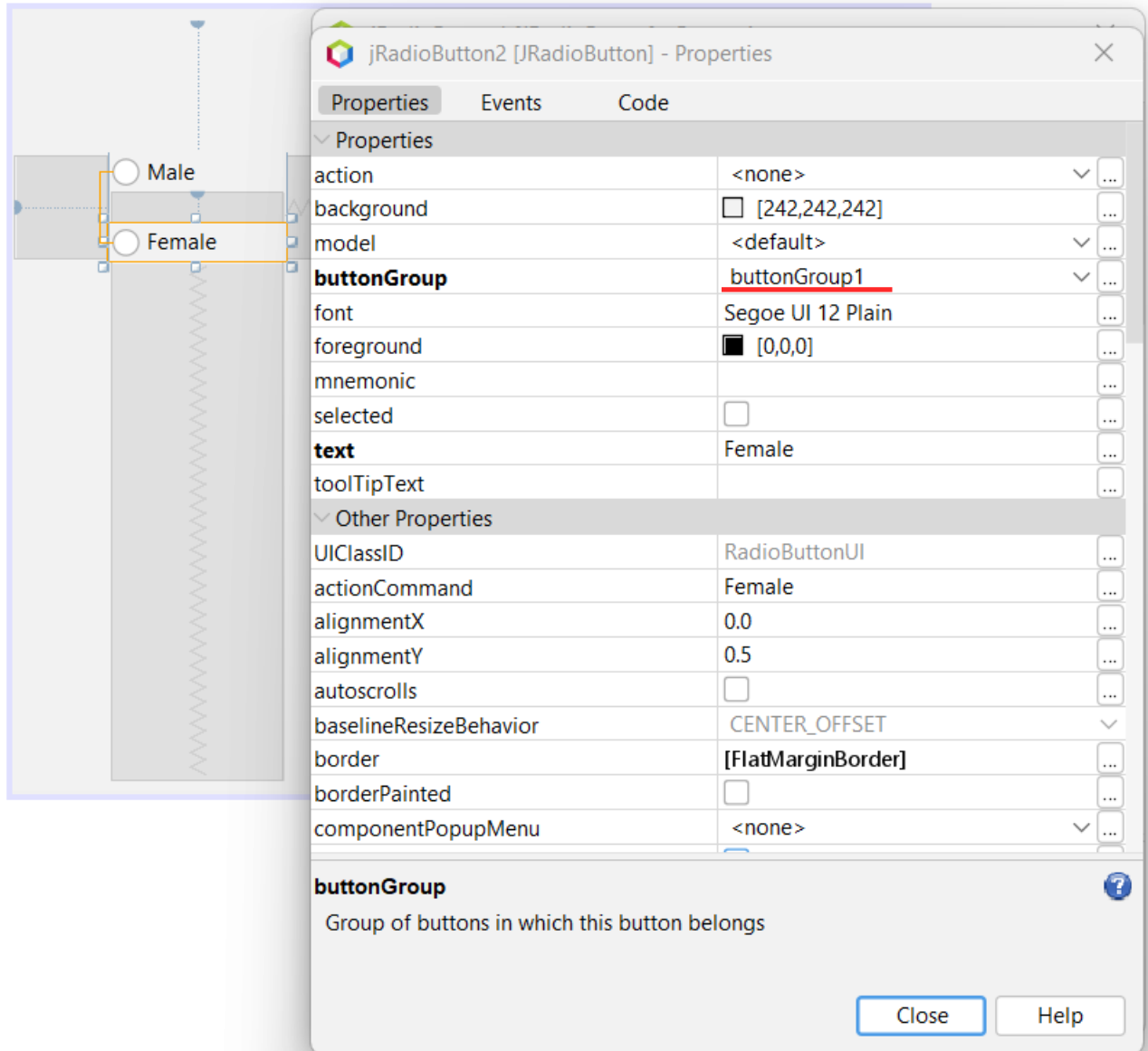
- Drag a Button Group object from the Swing Controls palette onto the form.
- Although nothing visibly changed on the form you can see in the Inspector palette that a Button Group object named `buttonGroup1` was added.
- In the Inspector click on the Button Group object so that it is selected.
- In the Properties palette, code area, change the Variable Name to `MathFunctions`.



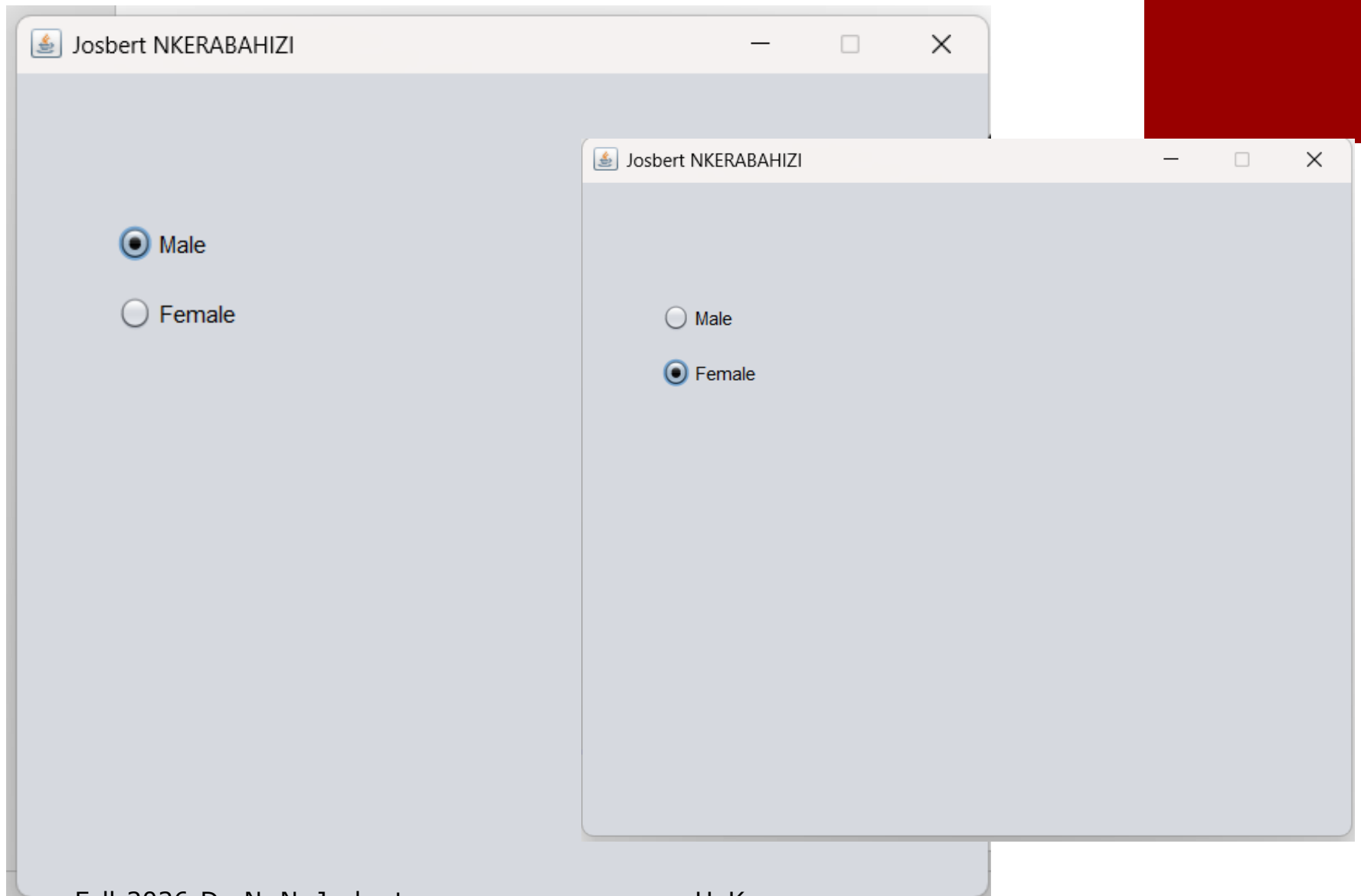
NetBeans IDE GUI builder



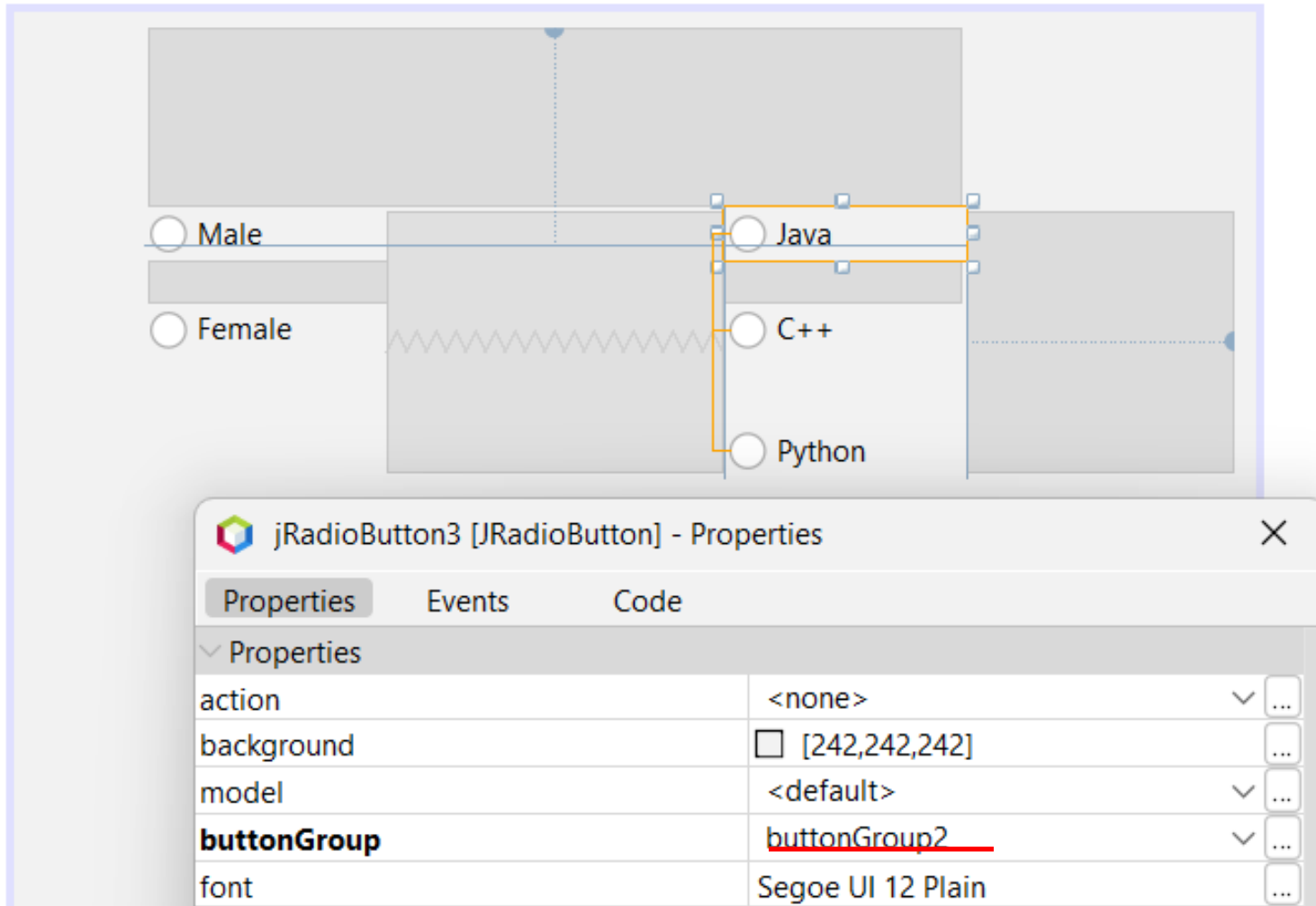
NetBeans IDE GUI builder



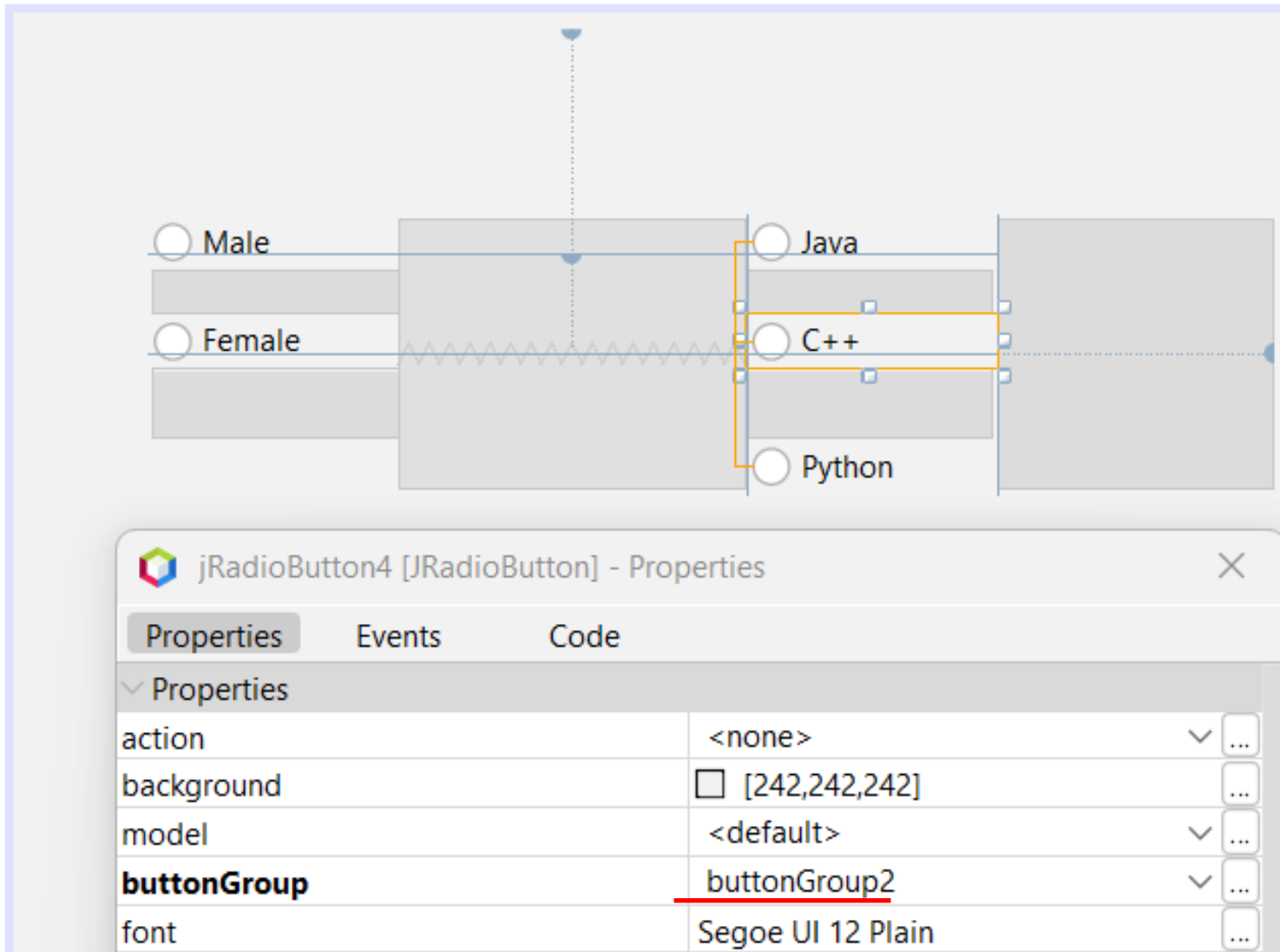
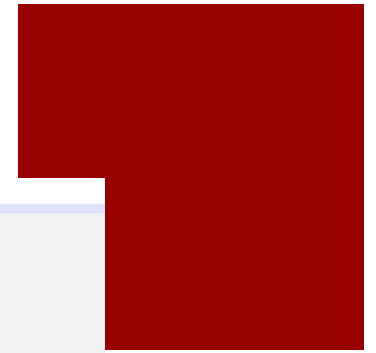
NetBeans IDE GUI builder



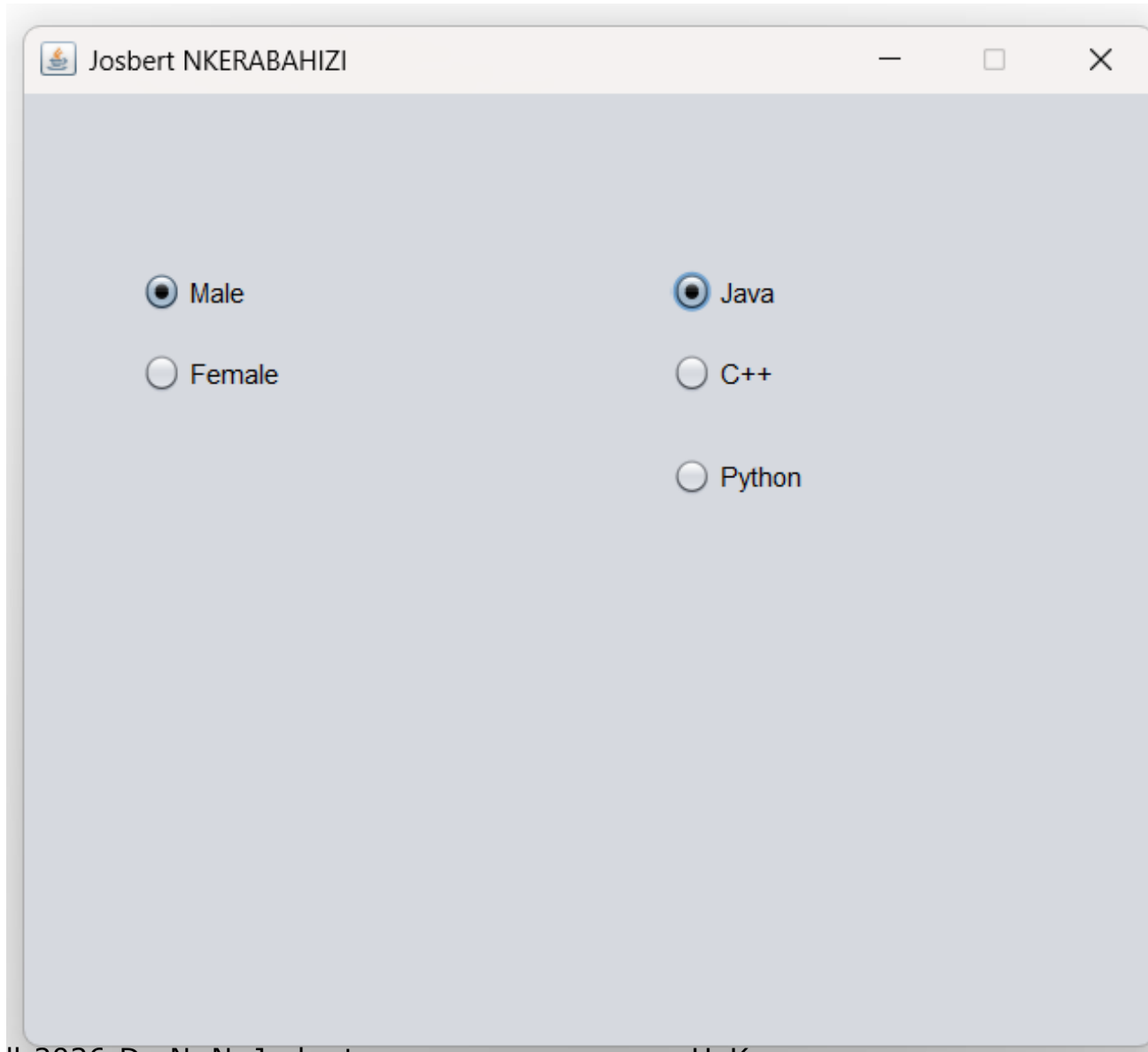
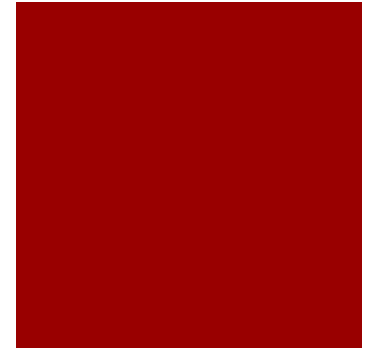
NetBeans IDE GUI builder



NetBeans IDE GUI builder



NetBeans IDE GUI builder



NetBeans IDE GUI builder



Enter a number:

Enter another number:

Select a math function:

Add

Subtract

Multiply

Divide

jRadioButton2 [JRadioButton] - Properties

PropertiesEventsCode

Properties

action	<none>	⌵...
background	☐ [242,242,242]	...
model	<default>	⌵...
buttonGroup	buttonGroup1	⌵...
font	Segoe UI 14 Bold	...
foreground	☒ [0,0,0]	...
mnemonic		...
selected	☐	...
text	Add	...
toolTipText		...

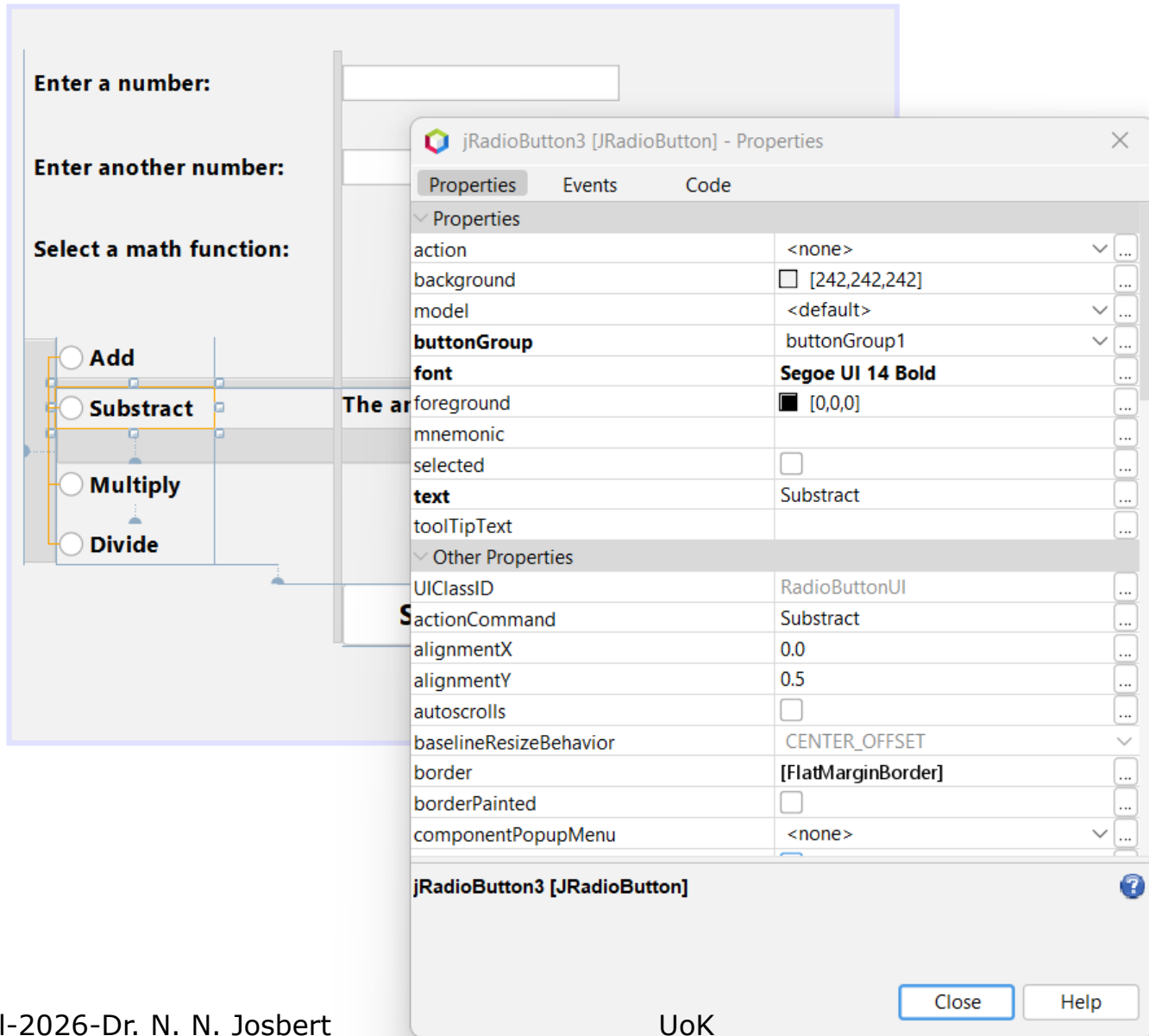
Other Properties

UIClassID	RadioButtonUI	...
actionCommand	Add	...
alignmentX	0.0	...
alignmentY	0.5	...
autoscrolls	☐	...
baselineResizeBehavior	CENTER_OFFSET	⌵
border	[FlatMarginBorder]	...
borderPainted	☐	...
componentPopupMenu	<none>	⌵...

jRadioButton2 [JRadioButton]

CloseHelp

NetBeans IDE GUI builder



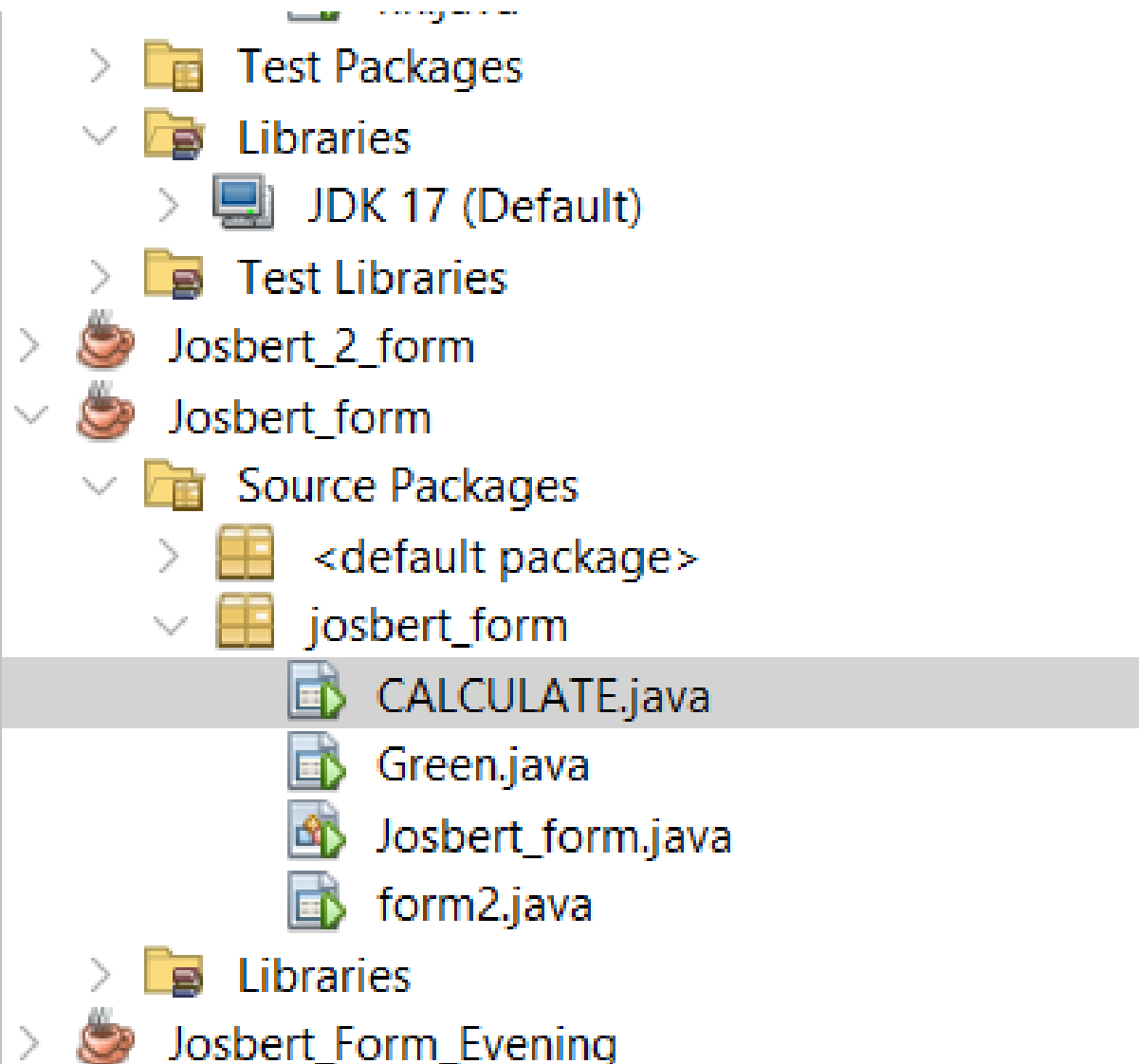
NetBeans IDE GUI builder



The image shows a screenshot of the NetBeans IDE GUI builder window. The window has a light gray background and a thin blue border. It contains the following elements:

- Enter a number:** A text label followed by a white rectangular text input field.
- Enter another number:** A text label followed by a white rectangular text input field.
- Select a math function:** A text label followed by four radio button options:
 - ☐ Add
 - ☐ Subtract
 - ☐ Multiply
 - ☐ Divide
- The answer is:** A text label positioned to the right of the radio button options.
- Solve** and **Exit**: Two white rectangular buttons with black text, located at the bottom center of the window.

NetBeans IDE GUI builder



```
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {  
    // TODO add your handling code here:  
    // =====  
    try {  
        double num1 = Double.parseDouble(jTextField1.getText());  
        double num2 = Double.parseDouble(jTextField2.getText());  
        double result = 0;  
  
        if (jRadioButton2.isSelected()) { // Add  
            result = num1 + num2;  
        }  
        else if (jRadioButton3.isSelected()) { // Subtract  
            result = num1 - num2;  
        }  
        else if (jRadioButton4.isSelected()) { // Multiply  
            result = num1 * num2;  
        }  
        else if (jRadioButton1.isSelected()) { // Divide  
            if (num2 == 0) {  
                jLabel4.setText("Cannot divide by zero!");  
                return;  
            }  
            result = num1 / num2;  
        }  
        else {  
            jLabel4.setText("Please select an operation!");  
            return;  
        }  
  
        jLabel4.setText("The answer is: " + result);  
  
    } catch (NumberFormatException e) {  
        jLabel4.setText("Invalid input!");  
    }  
    // =====
```



Enter a number:

12

Enter another number:

14

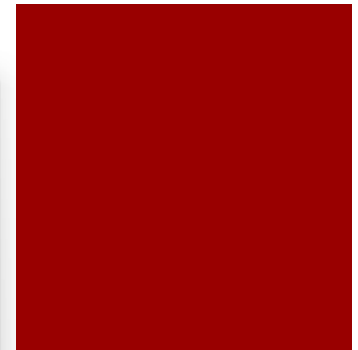
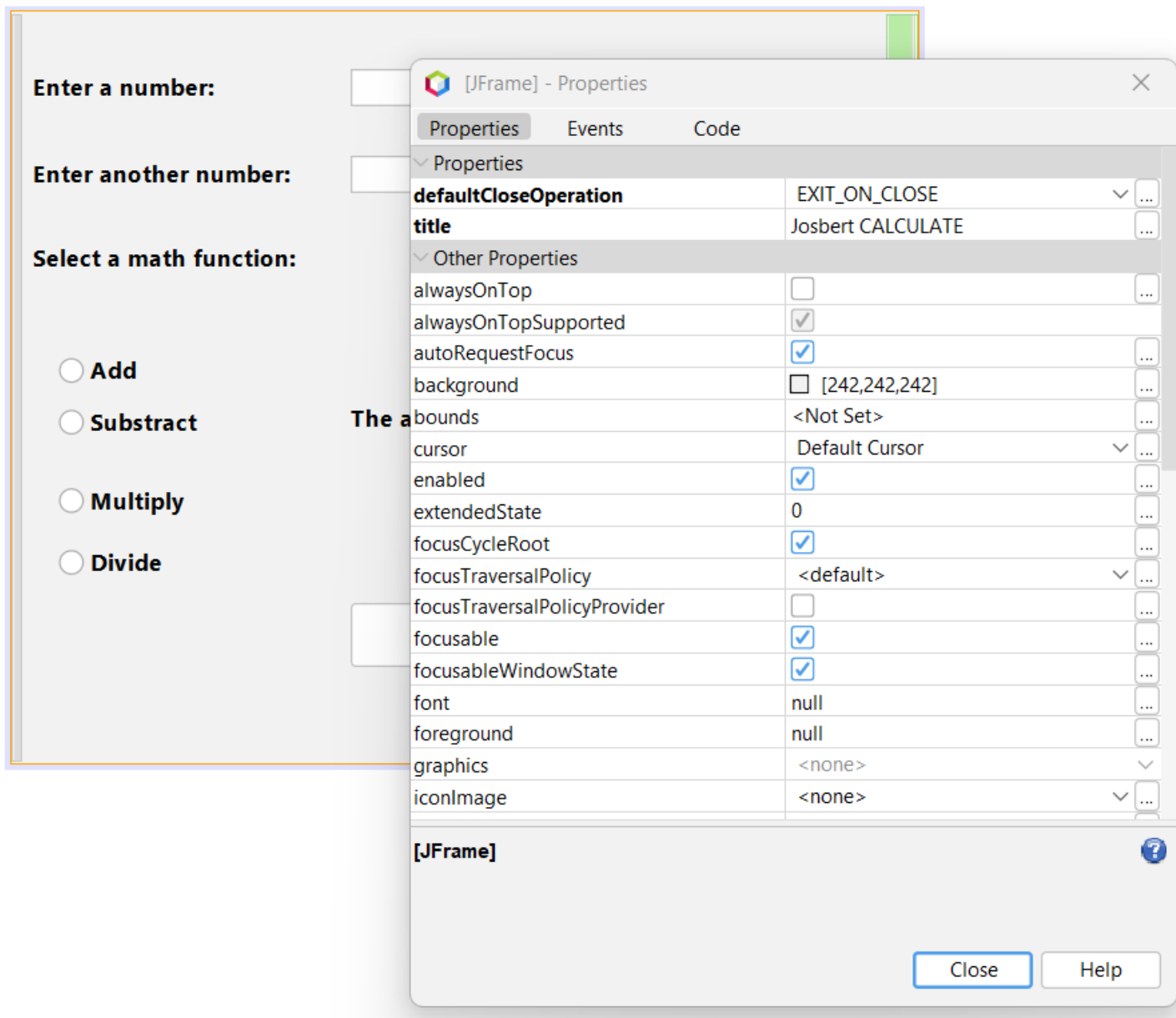
Select a math function:

- ☐ **Add**
- ☐ **Subtract**
- ☒ **Multiply**
- ☐ **Divide**

The answer is: 168.0

Solve

Exit



Josbert CALCULATE

Enter a number: 24

Enter another number: 12

Select a math function:

☒ Add

☐ Subtract

☐ Multiply

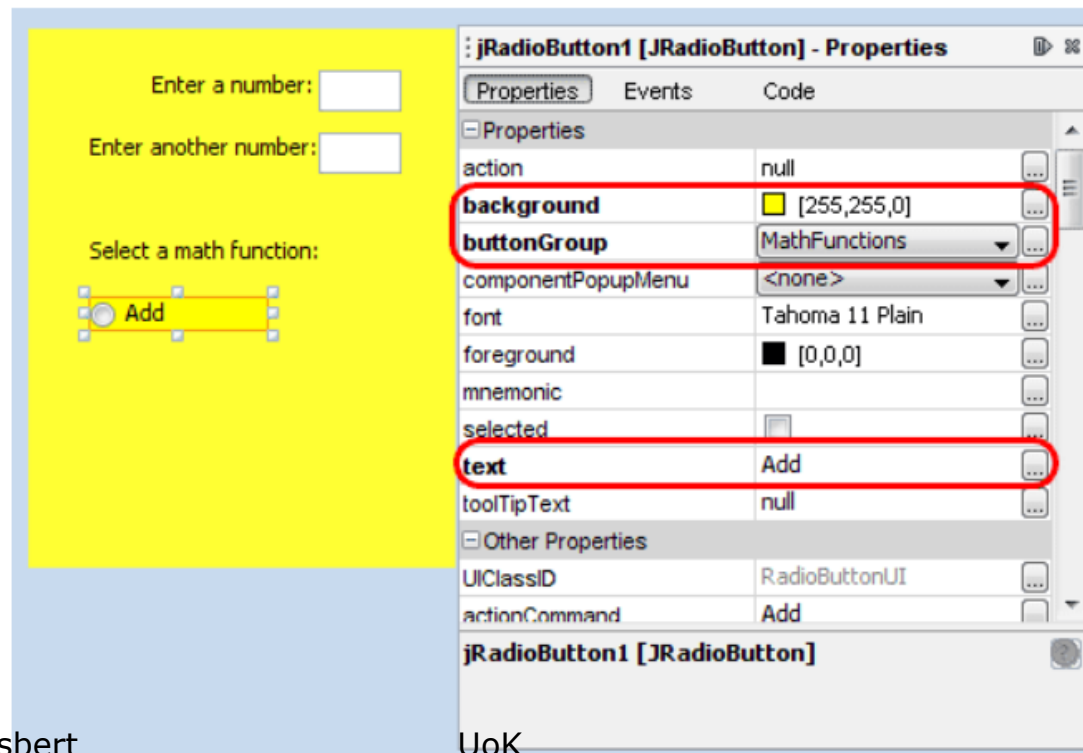
☐ Divide

The answer is: 36.0

Solve Exit

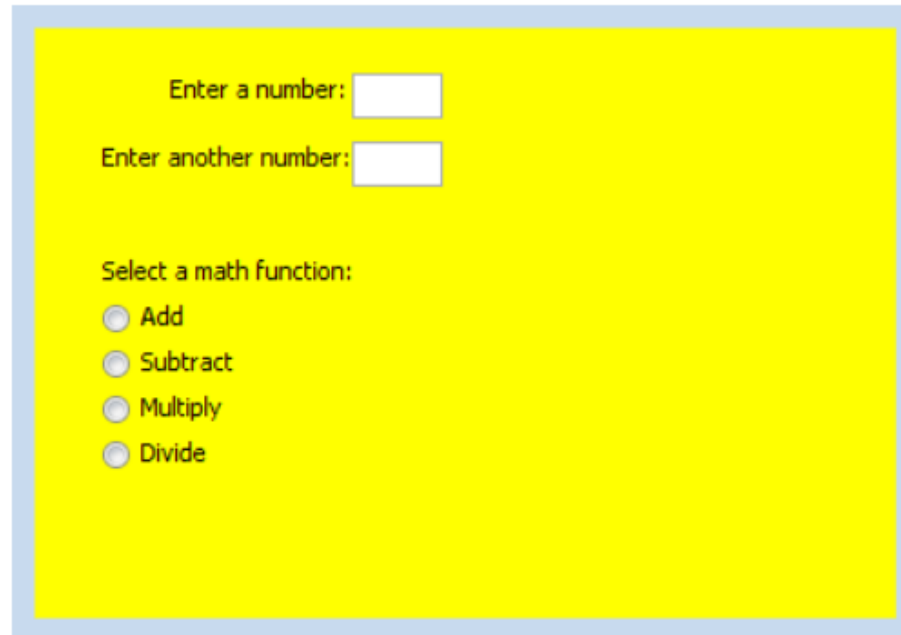
NetBeans IDE GUI builder

- Drag a Radio Button object from the Swing Controls palette onto the form.
- Change the text property to Add. Select the buttonGroup MathFunctions.
- Change the background property to the same yellow color as the JPanel.
- Change the Variable Name to optAdd.



NetBeans IDE GUI builder

- Add three more Radio Buttons to the form. Change their background to yellow.
- Add all of them to the MathFunctions buttonGroup.
- Change the text to Subtract, Multiply and Divide for the three buttons.
- Use the Variable Names optSubtract, optMultiply and optDivide.



Enter a number:

Enter another number:

Select a math function:

☐ Add

☐ Subtract

☐ Multiply

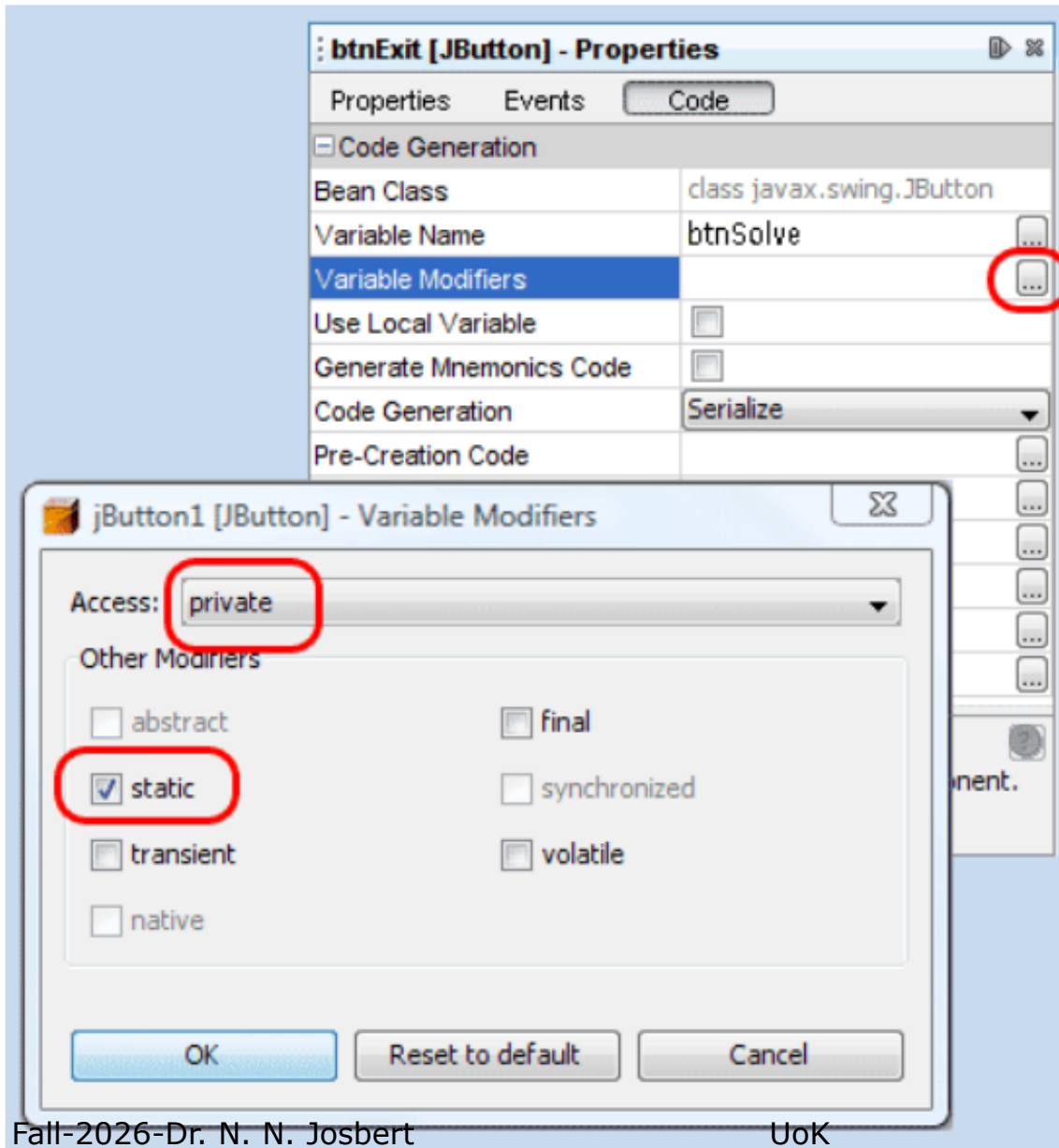
☐ Divide

NetBeans IDE GUI builder

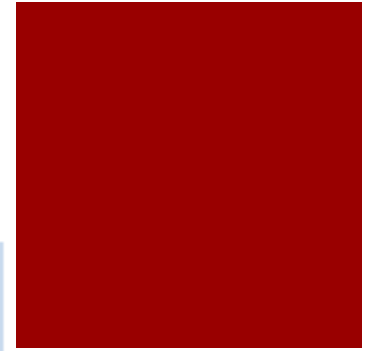
Drag a Button from the Swing Controls palette onto the form.

- In the Code area of the Properties palette change the Variable Name to `btnSolve`.
- Select `private static` for the Variable Modifiers property (if you don't do this the color will not appear when the code is executed).
- In the Properties palette change the text to `Solve`.
- Select the `ContentAreaFilled` property.
- Deselect the `focusPainted` property.
- Select the `Opaque` property.
- Change the background property to give the button some color.
- Change the foreground property to change the color of the text of the button.
- Change the border property to `Bevel Border`.

NetBeans IDE GUI builder



NetBeans IDE GUI builder

A screenshot of a NetBeans IDE GUI builder window. The window has a light blue border and a yellow background. It contains the following elements:

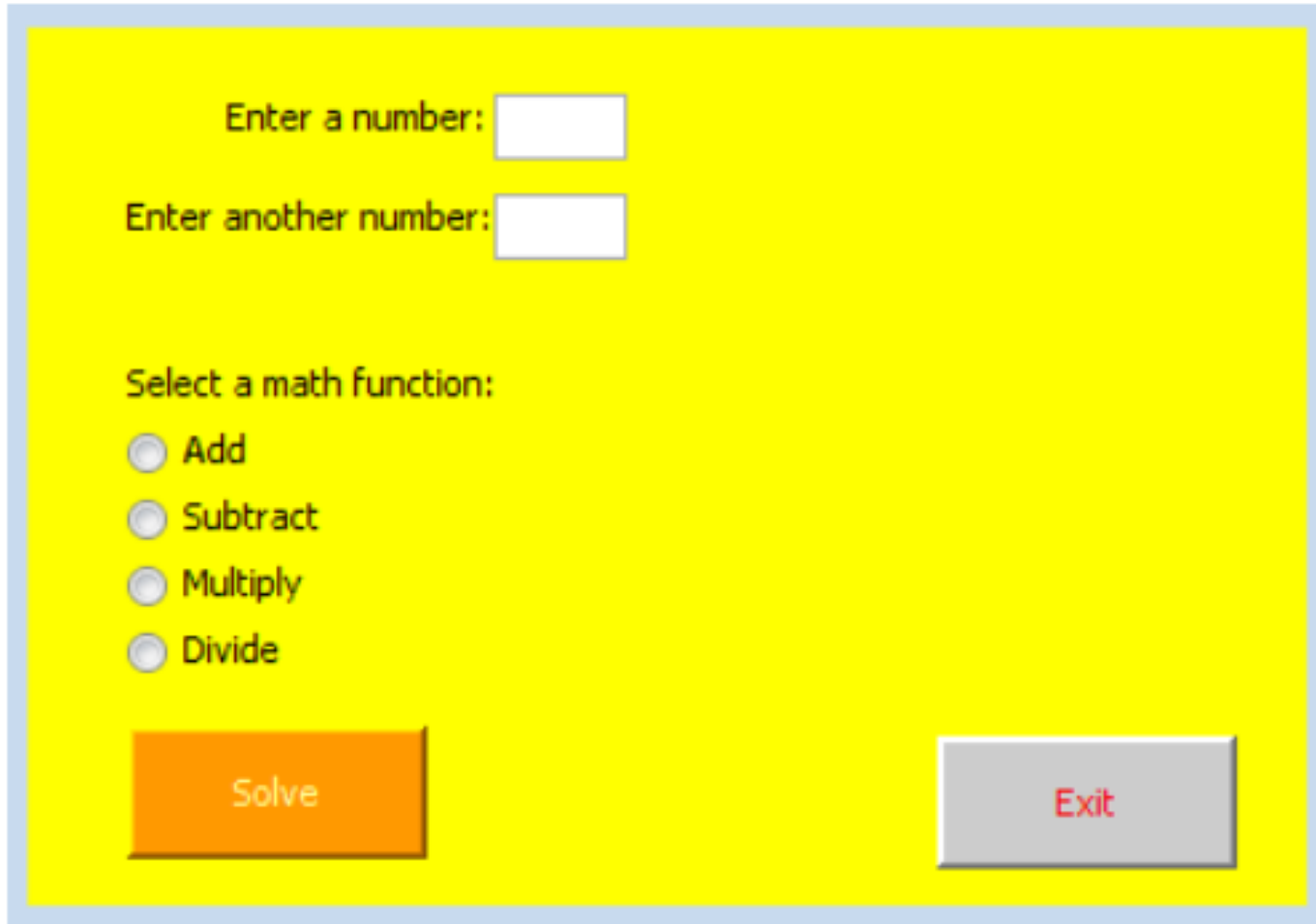
- Text label "Enter a number:" followed by a white text input field.
- Text label "Enter another number:" followed by a white text input field.
- Text label "Select a math function:" followed by four radio button options: "Add", "Subtract", "Multiply", and "Divide".
- An orange button with the text "Solve" in white.

NetBeans IDE GUI builder

Add another Button to the form. In the Code area of the Properties palette change the Variable Name to btnExit.

- Select private static for the Variable Modifiers property
- In the Properties palette change the text to Exit.
- Select the ContentAreaFilled property.
- Deselect the focusPainted property.
- Select the Opaque property.
- Change the background property to give the button some color.
- Change the foreground property to change the color of the text of the button.
- Change the border property to Bevel Border.

NetBeans IDE GUI builder



Enter a number:

Enter another number:

Select a math function:

☐ Add

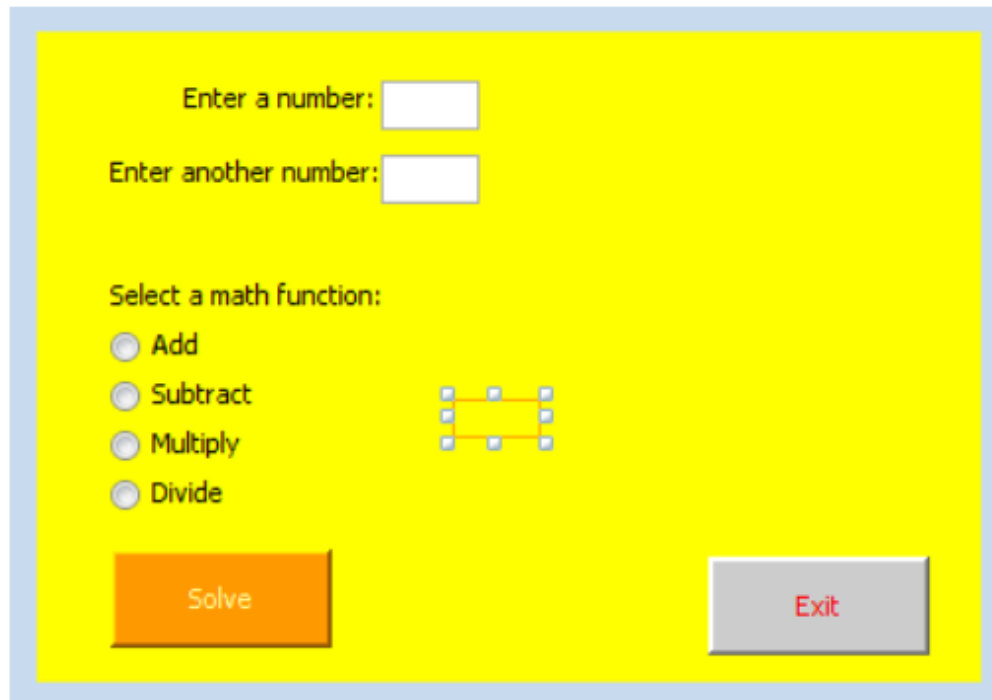
☐ Subtract

☐ Multiply

☐ Divide

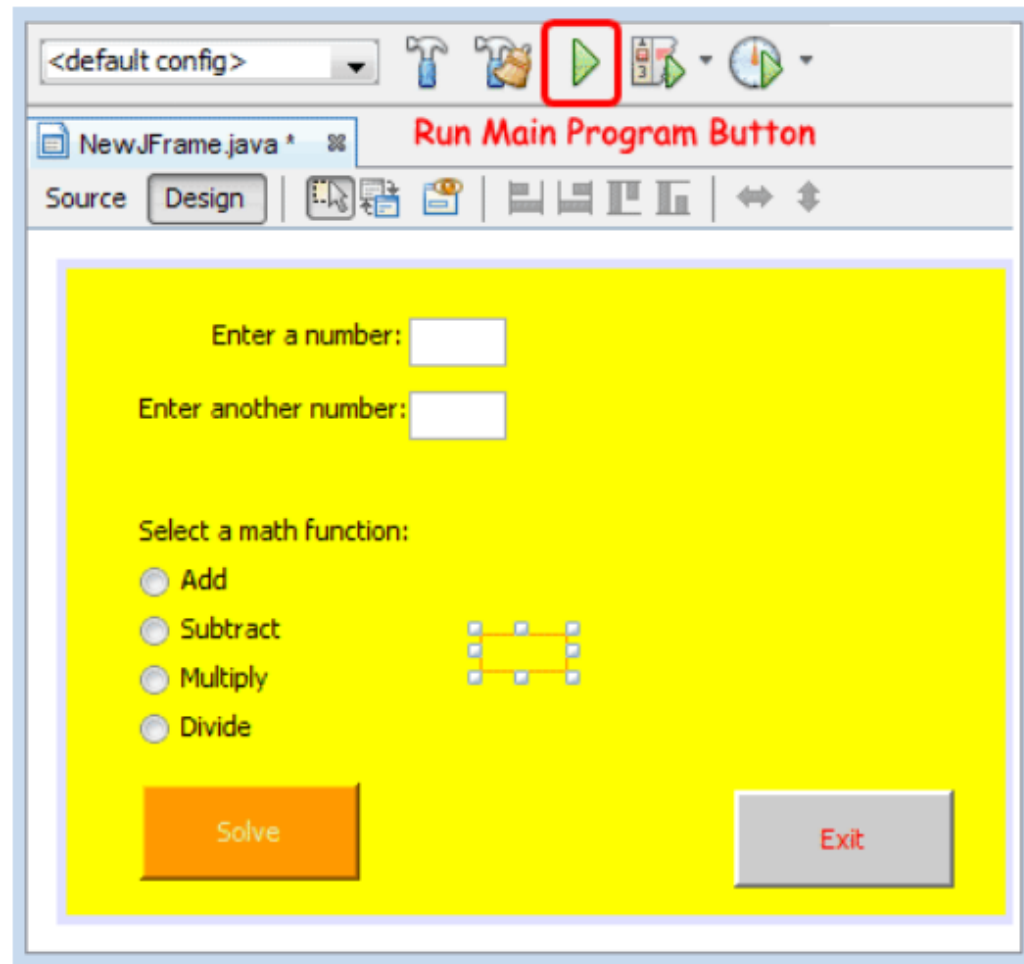
NetBeans IDE GUI builder

- Add a Label to the form. In the Properties palette delete the default text from the text property.
- In the Code area of the Properties palette change the Variable Name to lblResults.



NetBeans IDE GUI builder

- Now that the form is complete we will execute the program to make sure that the appearance is OK. Press the Run Main Project button.



NetBeans IDE GUI builder

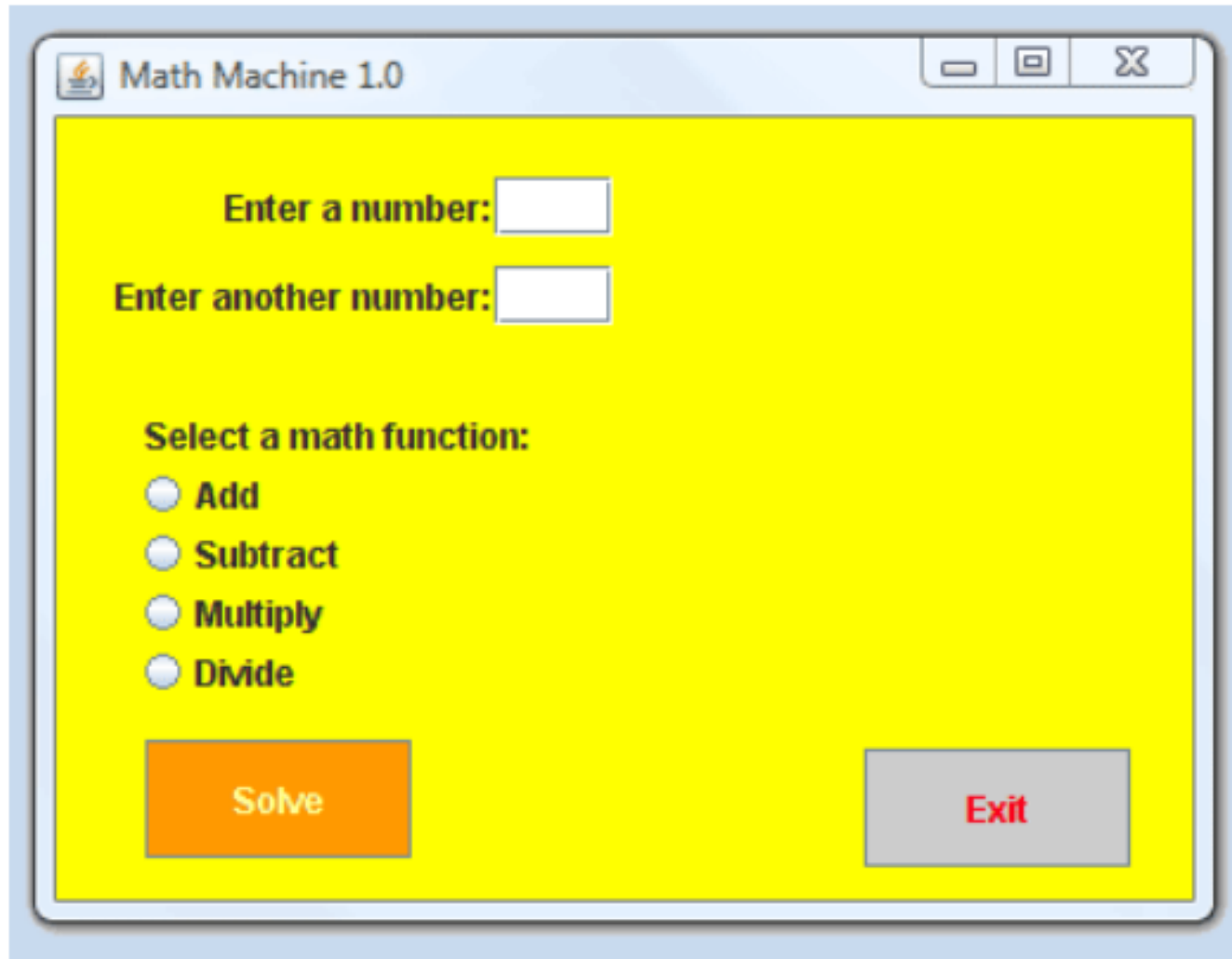


- Notice that the labels all appear as **bold** text.
- This causes the text to increase in size which causes the text to no longer fit in the space assigned. (**Notice: before continuing with this step you can avoid this entire issue by following the directions on how to resolve the 'Look and Feel' issue found at the bottom of this page.**)

If you have not corrected the Look and Feel issue as described at the end of this tutorial then on the form, increase the size of each element so that when executed the text is displayed correctly.

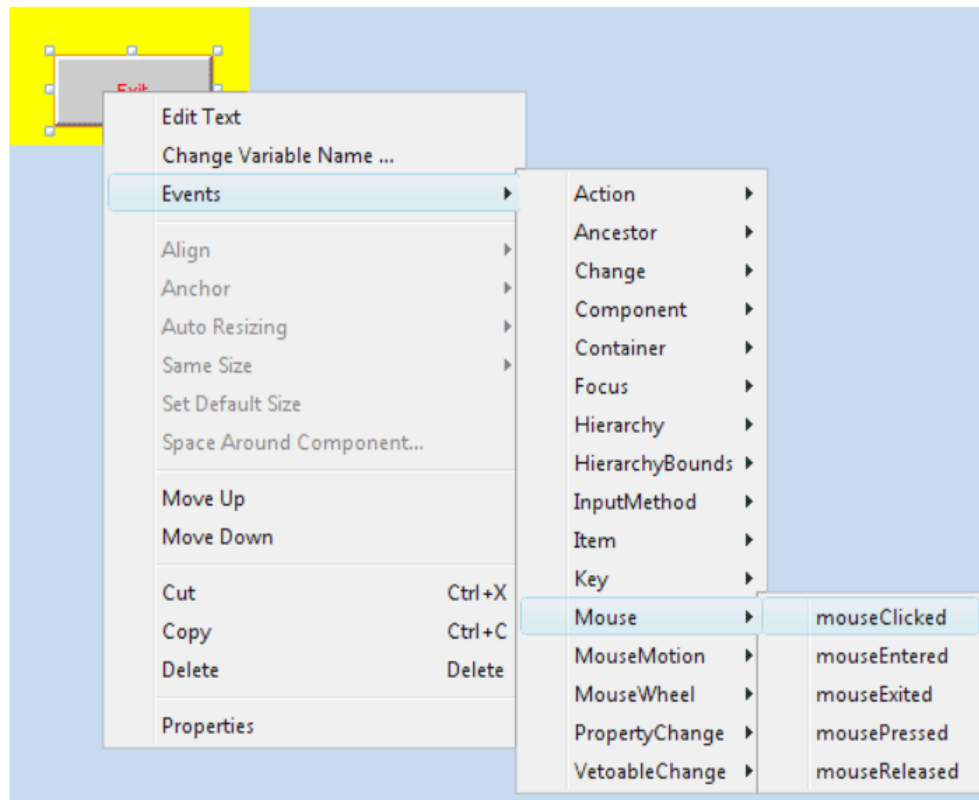
- You may have to make several attempts to get the correct appearance. (**Note:** The lblResults JLabel is transparent. It too needs to be adjusted.
- To select lblResults select it in the Inspector palette so that it appears selected on the form.)

NetBeans IDE GUI builder

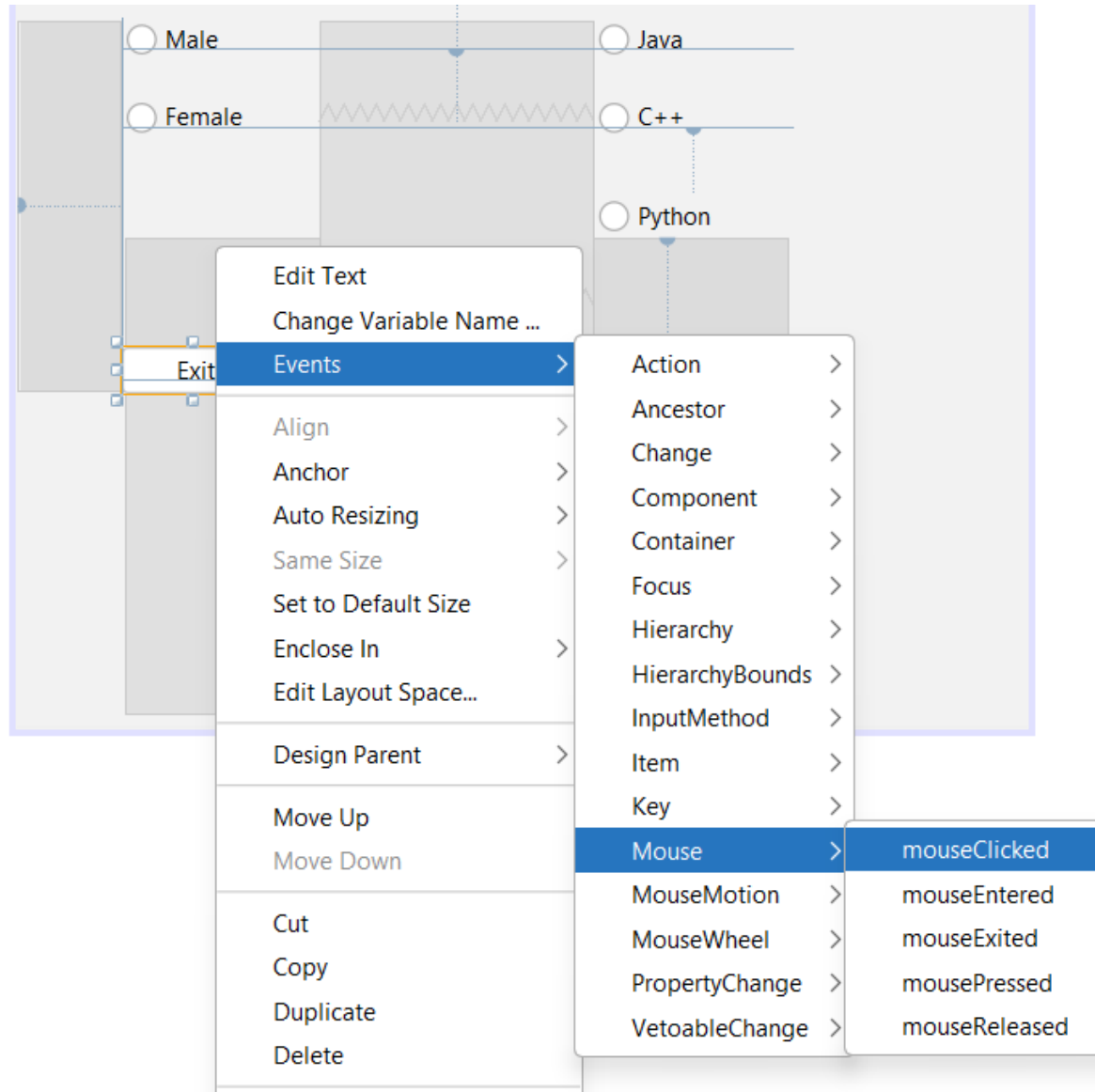


NetBeans IDE GUI builder

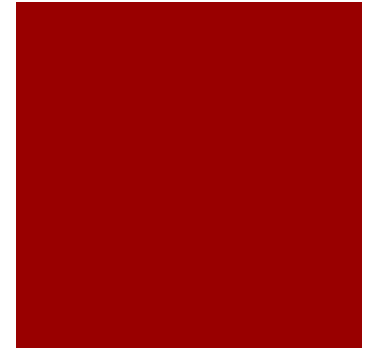
- Now it is time to add the Java code so that program actually does something useful.
- The first thing we will do is add code to the btnExit element. Right-click on the cmdExit button. Select Events, Mouse, mouseClicked.



NetBeans IDE GUI builder



NetBeans IDE GUI builder



```
private void jButton2MouseClicked(java.awt.event.MouseEvent evt) {  
    // TODO add your handling code here:  
    System.exit(0);  
}
```

```
private void jButton2MouseClicked(java.awt.event.MouseEvent evt) {  
    // TODO add your handling code here:  
    System.exit(0);  
}
```

NetBeans IDE GUI builder

- The computer has now left the Design view and entered the Source view.
- If you scroll through the code you will find that all of the necessary code has been written to display your project.
- The computer is now waiting for you to write the code for the btnExit button. The computer has already entered:

```
private void btnExitMouseClicked(java.awt.event.MouseEvent evt) {  
    // TODO add your handling code here:  
}
```

In order to make the program terminate change the code to:

```
private void btnExitMouseClicked(java.awt.event.MouseEvent evt) {  
    System.exit(0);  
}
```

NetBeans IDE GUI builder

- Run the program to confirm that the btnExit button works.
- Next we will add code to the btnSolve button. Right-click on the btnSolve button. Select Events, Mouse, mouseClicked. Add the following code:

```
String enteredNum1,enteredNum2;
int num1=0,num2=0,result=0,remainder=0;
enteredNum1=txtNum1.getText();
enteredNum2=txtNum2.getText();
num1=Integer.parseInt(enteredNum1);
num2=Integer.parseInt(enteredNum2);
if(optAdd.isSelected()==true)
{
    result=num1+num2;
    lblResults.setText("The answer is "+result);
}
else if (optSubtract.isSelected()==true)
{
    result=num1-num2;
    lblResults.setText("The answer is "+result);
}
```

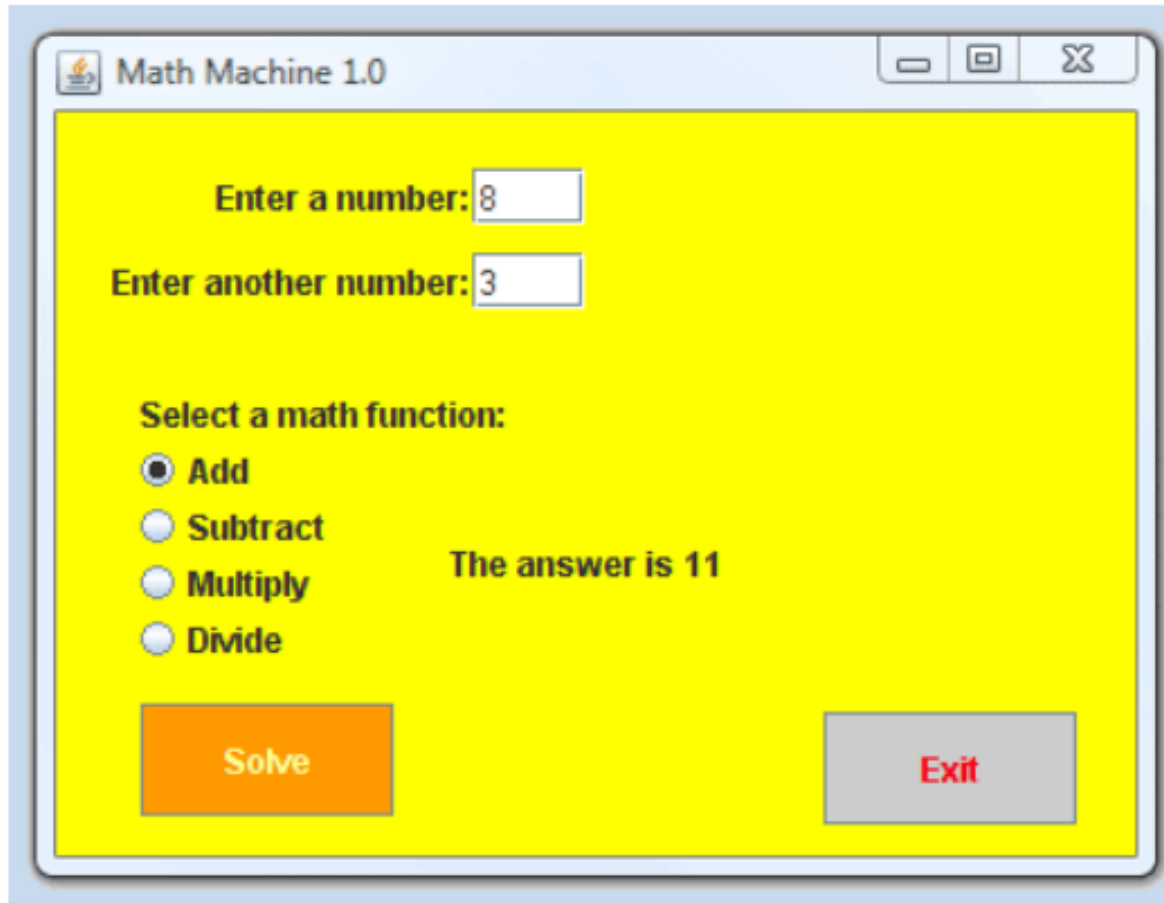
NetBeans IDE GUI builder



```
else if (optMultiply.isSelected()==true)
{
    result=num1*num2;
    lblResults.setText("The answer is "+result);
}
else if (optDivide.isSelected()==true)
{
    result=num1/num2;
    remainder=num1%num2;
    lblResults.setText("The answer is "+result+" with a remainder of "+remainder);
}
else
{
    lblResults.setText("You have not chosen a math function.");
}
```

NetBeans IDE GUI builder

- Run the completed program to make sure that it functions correctly.



NetBeans IDE GUI builder

- Run the completed program to make sure that it functions correctly.
- Directions to fix the Look and Feel issue:
- To stop the font size error you will need to add the code shown below to the main method.
- Press the Source button to view the code for your project. Find the main method. Copy the code below and then paste just after the line `public static void main(String args[]) {`

```
// this code assumes the form is named JFrame which is the  
default name generally
```

NetBeans IDE GUI builder



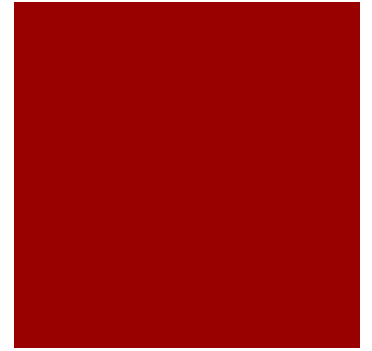
```
try {  
    UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());  
} catch (Exception ex) {  
    Logger.getLogger(NewJFrame.class.getName()).log(Level.SEVERE, "Error setting LaF", ex);}
```

After adding the above code you need to add the following two lines of code at the very top of the source code window:

```
import javax.swing.*;  
import java.util.logging.*;
```

Run your program to verify that the output view is the same as the design view in NetBeans.

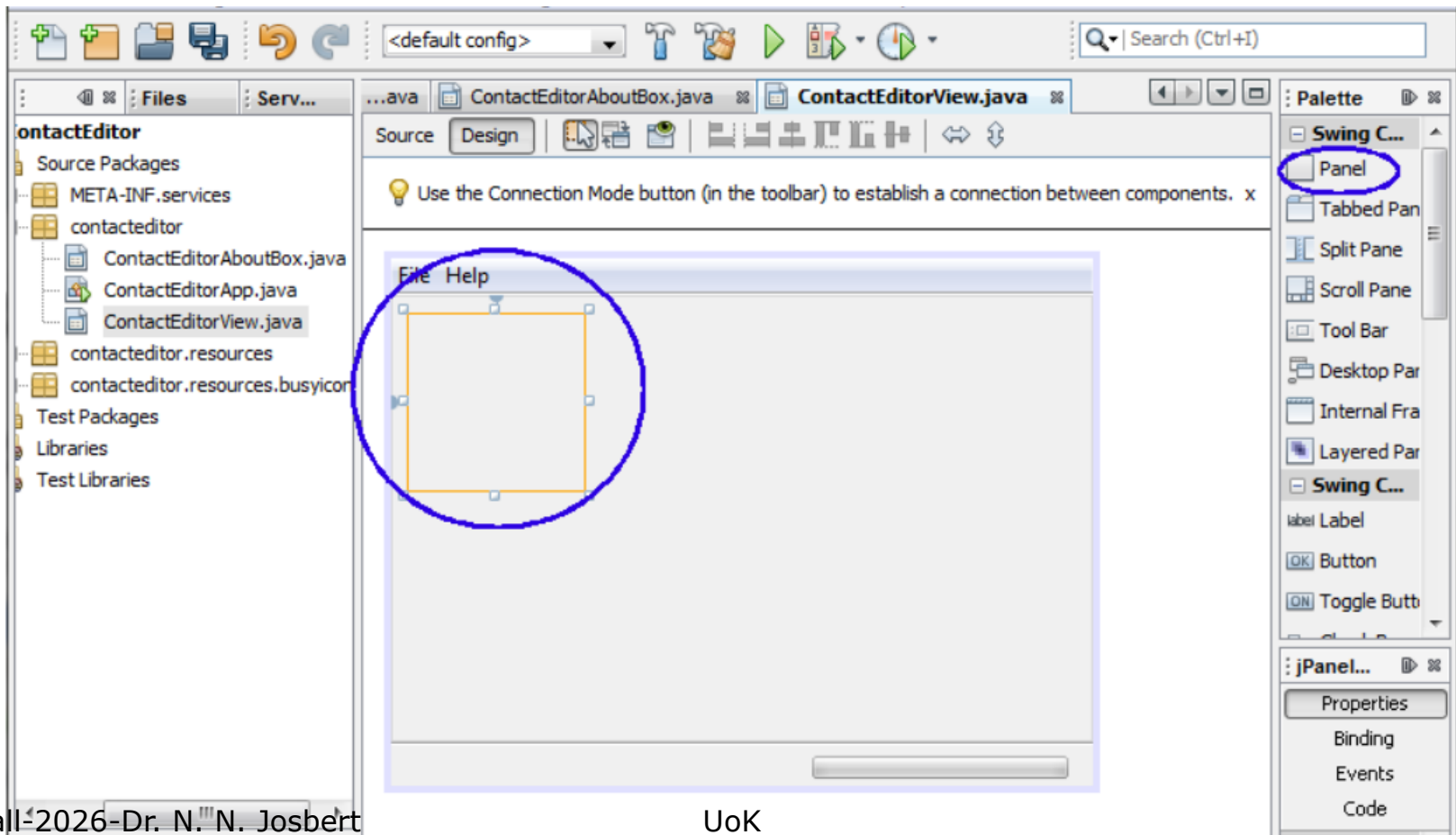
NetBeans IDE GUI builder



A CAS STUDY 2

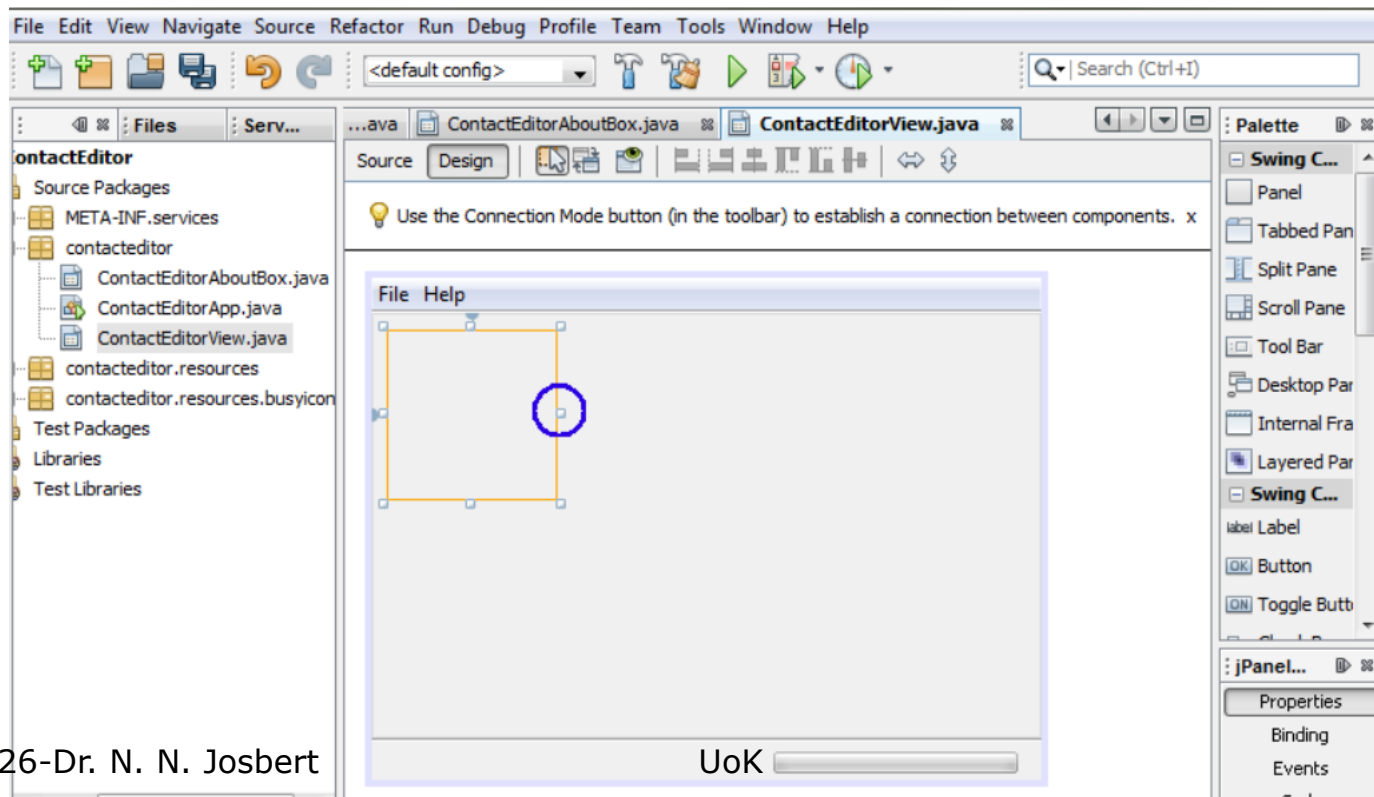
NetBeans IDE GUI builder

- 1. Add Panel to the project
- 1. Add a Panel component to the Design window. (In this document, the terms "Panel" and "JPanel" are used interchangeably.)



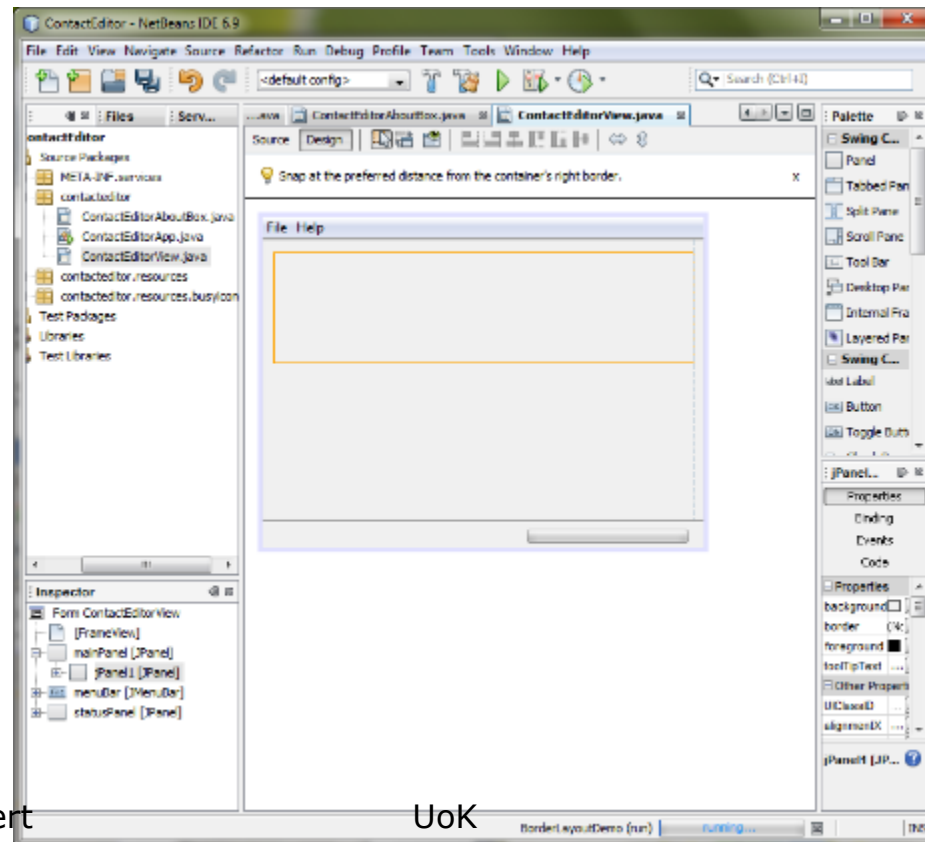
NetBeans IDE GUI builder

- **2. Resize the Panel component.**
- Select the JPanel you just added. The small square resize handles reappear around the component's perimeter.
- Click and hold the resize handle on the right edge of the JPanel and drag until the dotted alignment guideline appears near the form's edge.



NetBeans IDE GUI builder

- Release the resize handle to resize the component.
- Observe that the JPanel component is extended to span between the container's left and right margins in accordance with the recommended offset, as shown in the following illustration.

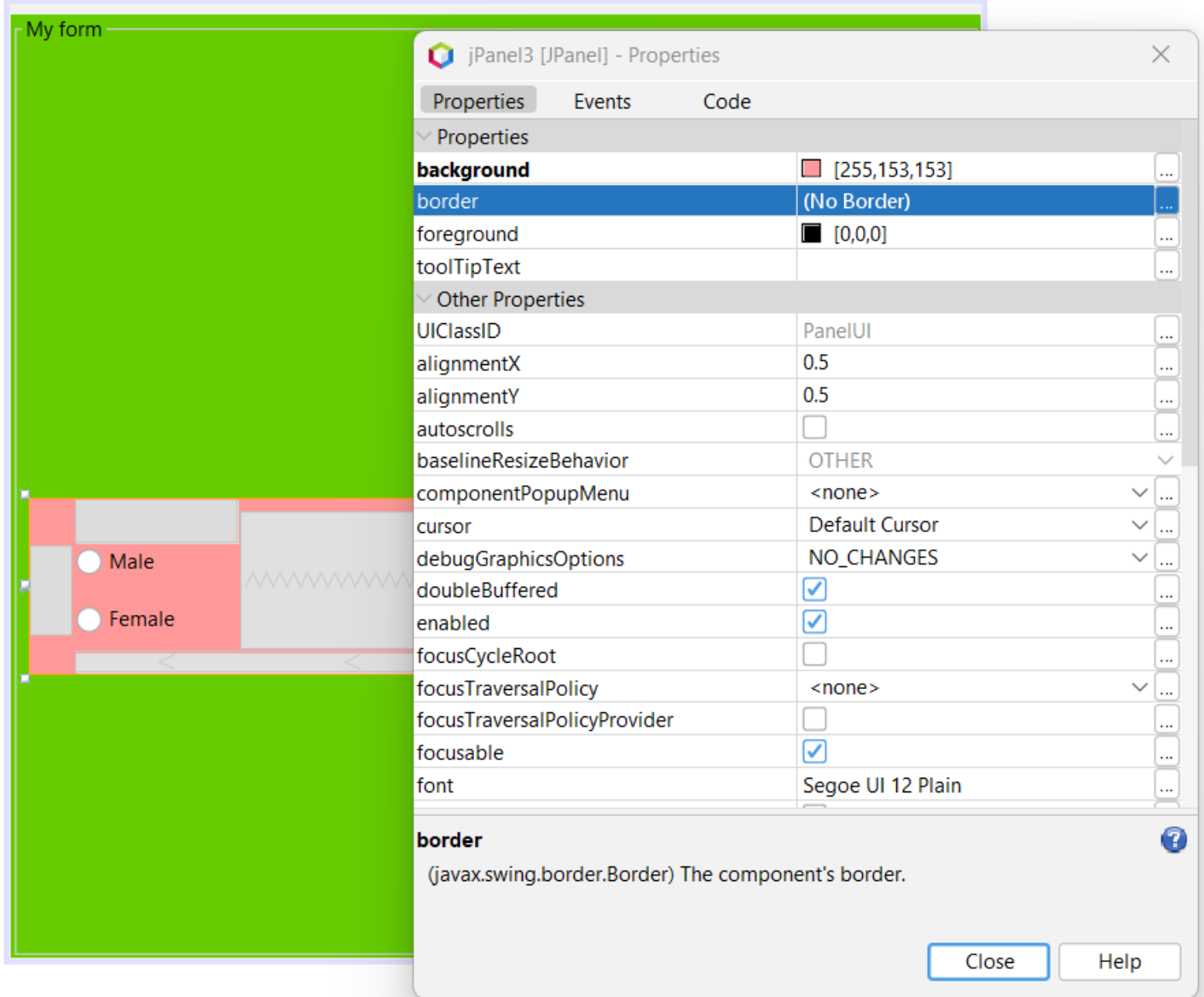


NetBeans IDE GUI builder

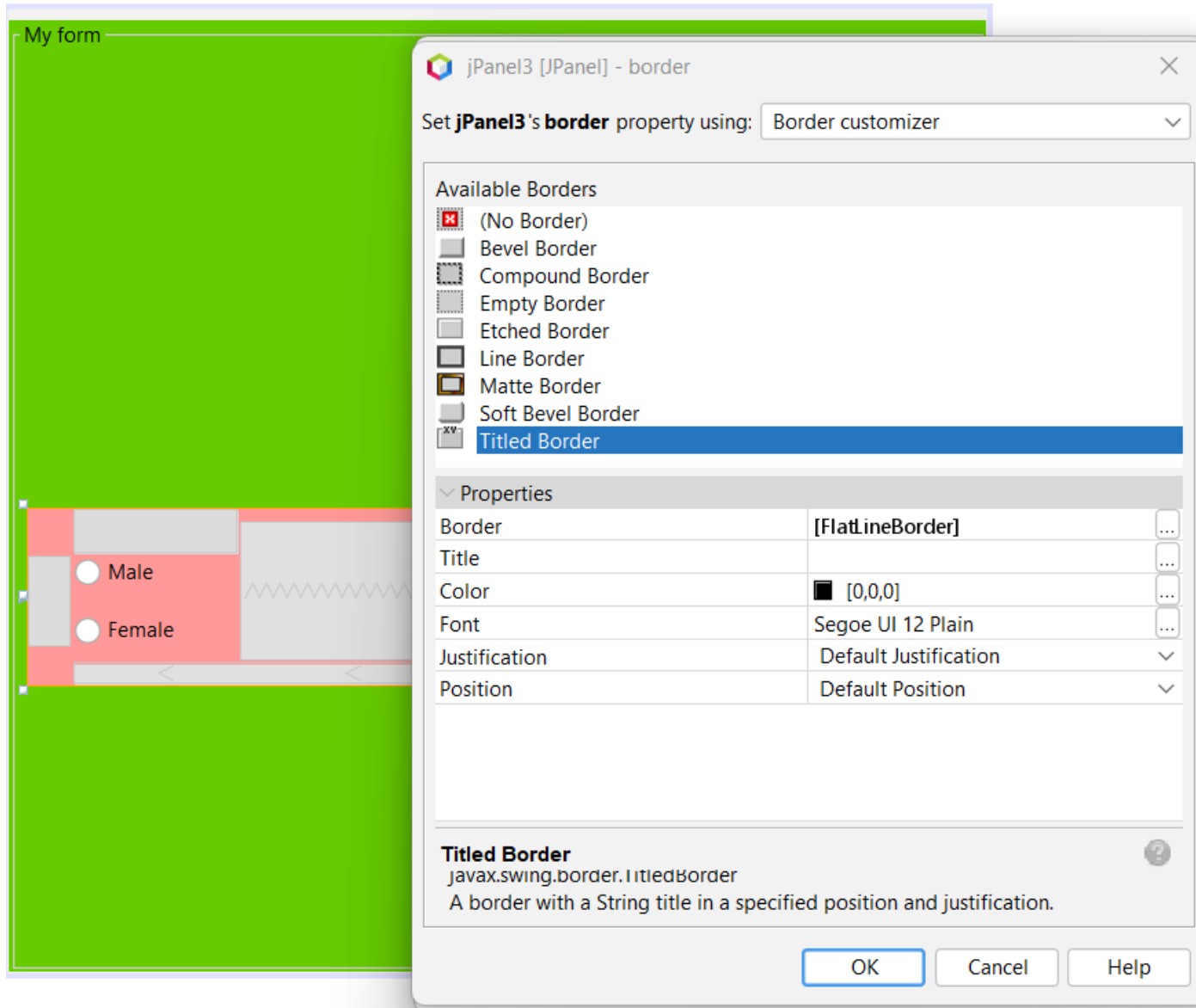
- 3. Change the properties of the JPanel component to add a border.
- In the **Properties** pane in the right and lower corner of the IDE, click the ellipsis button (...) next to the **border** property.



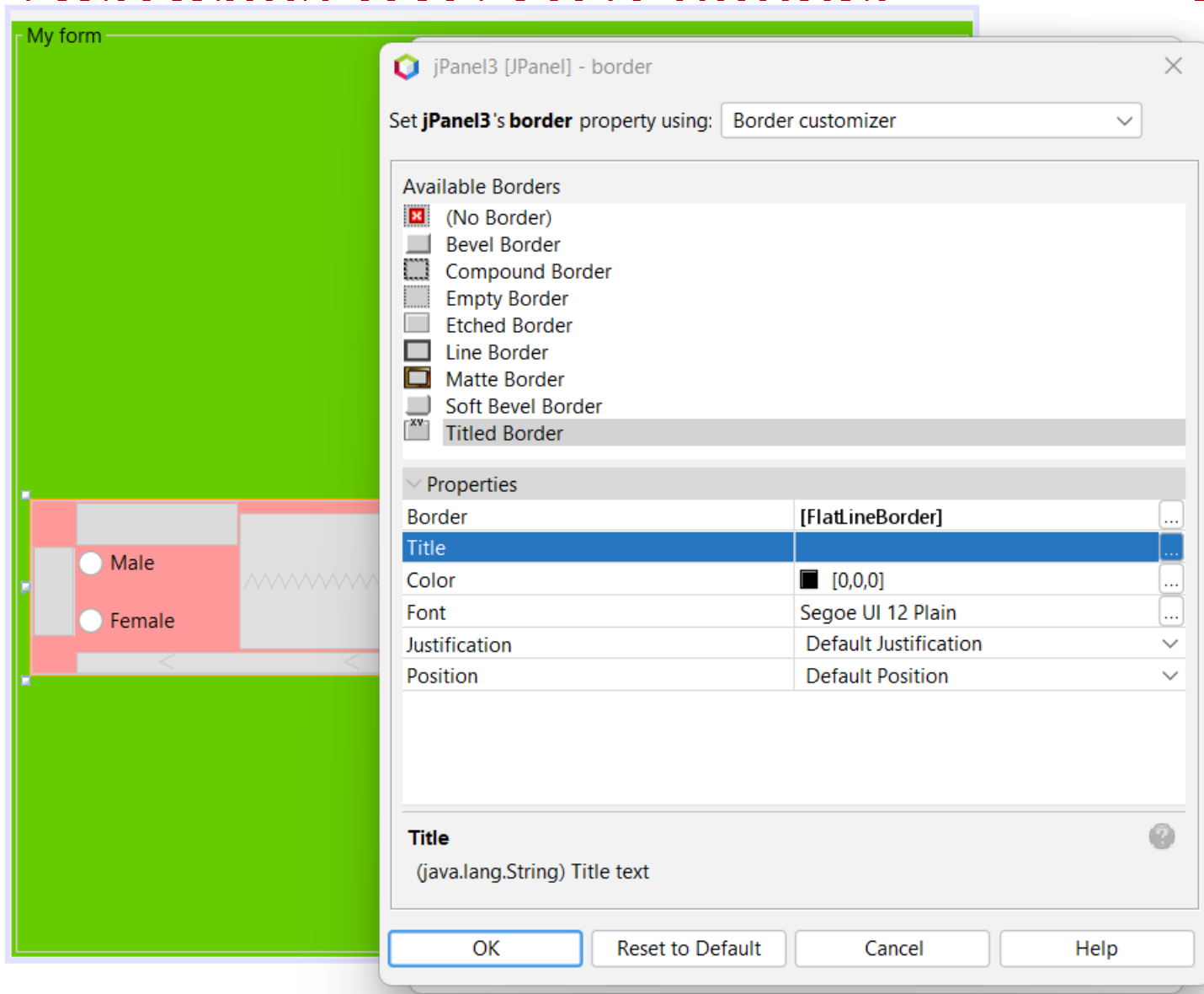
NetBeans IDE GUI builder



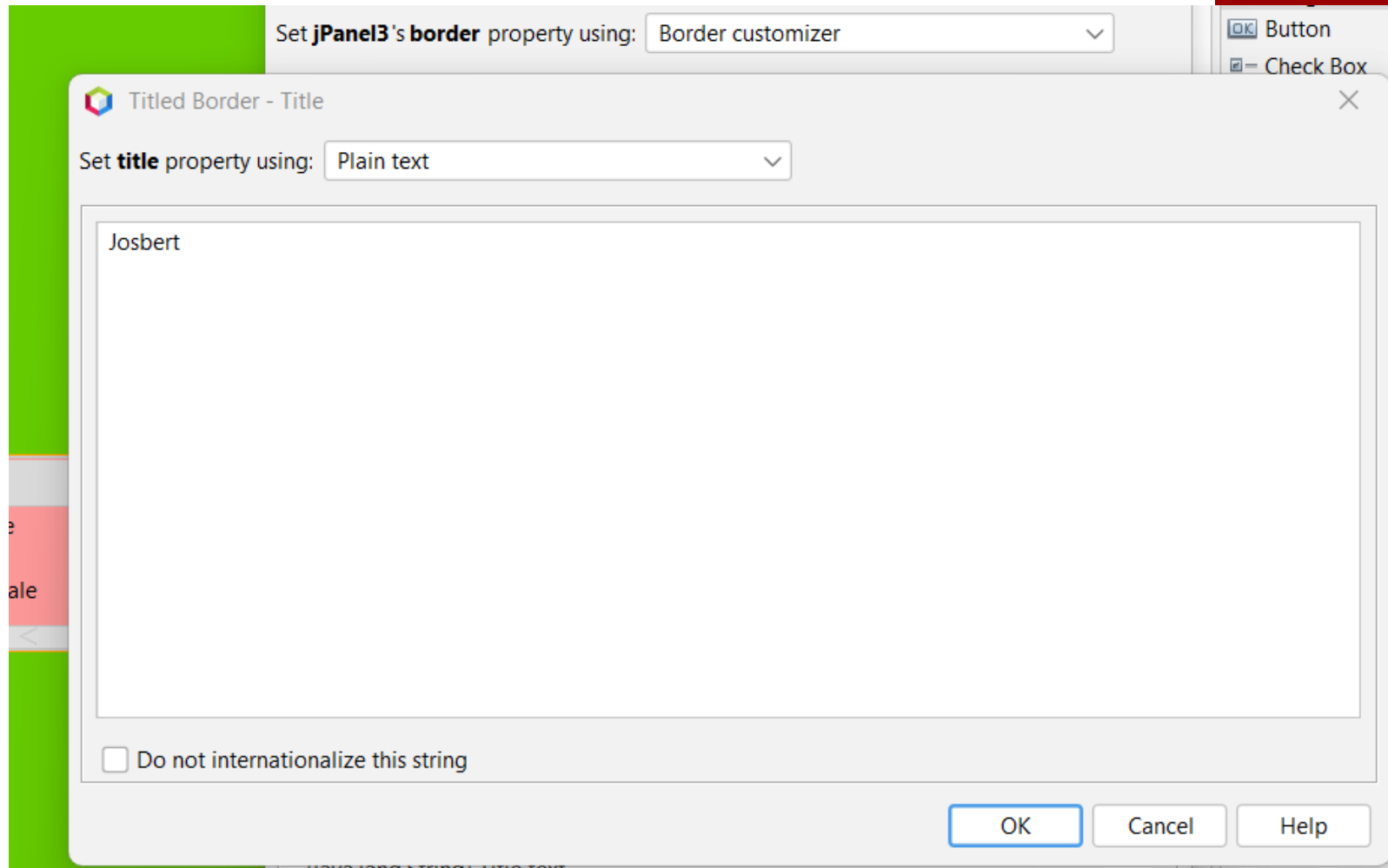
NetBeans IDE GUI builder



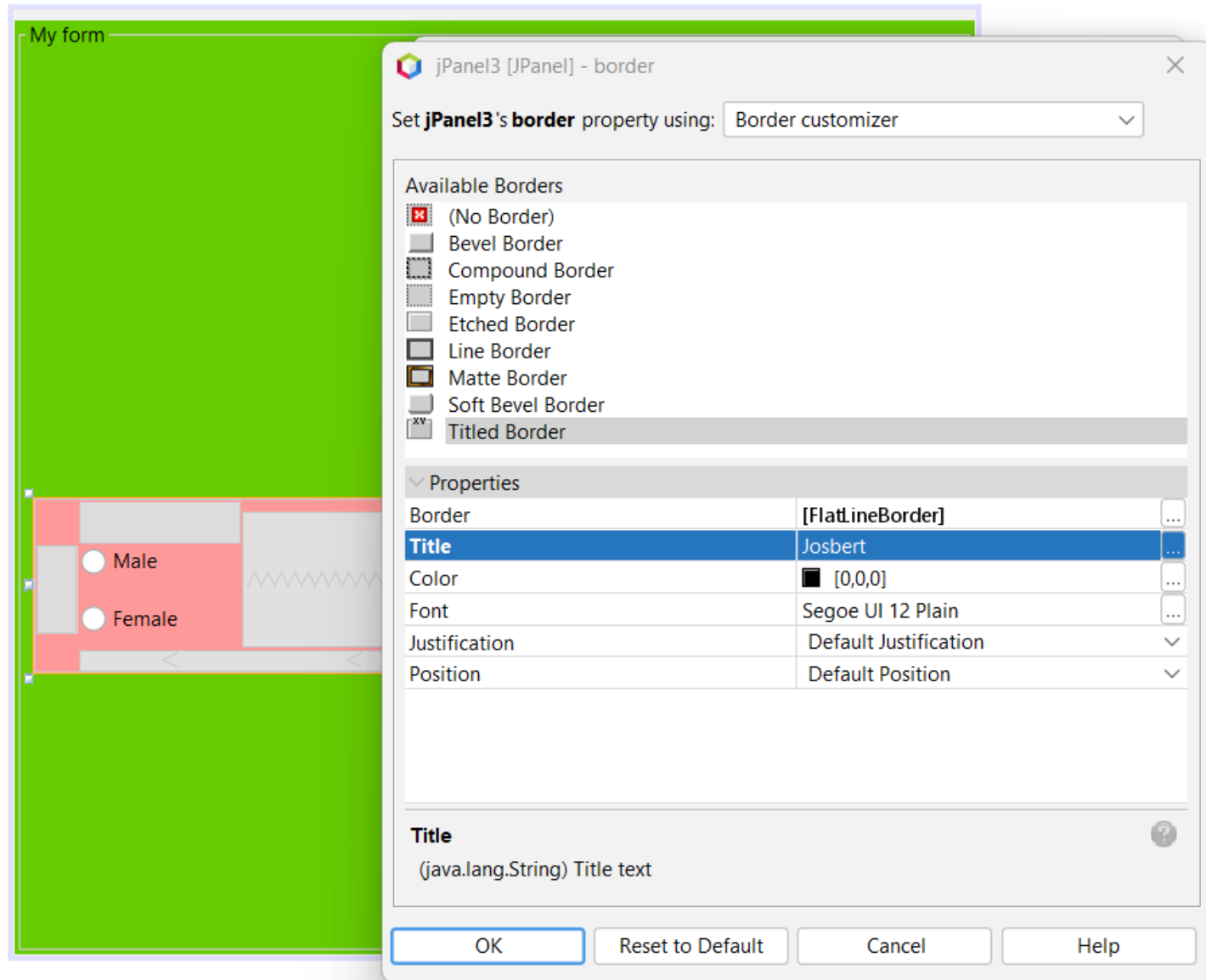
NetBeans IDE GUI builder



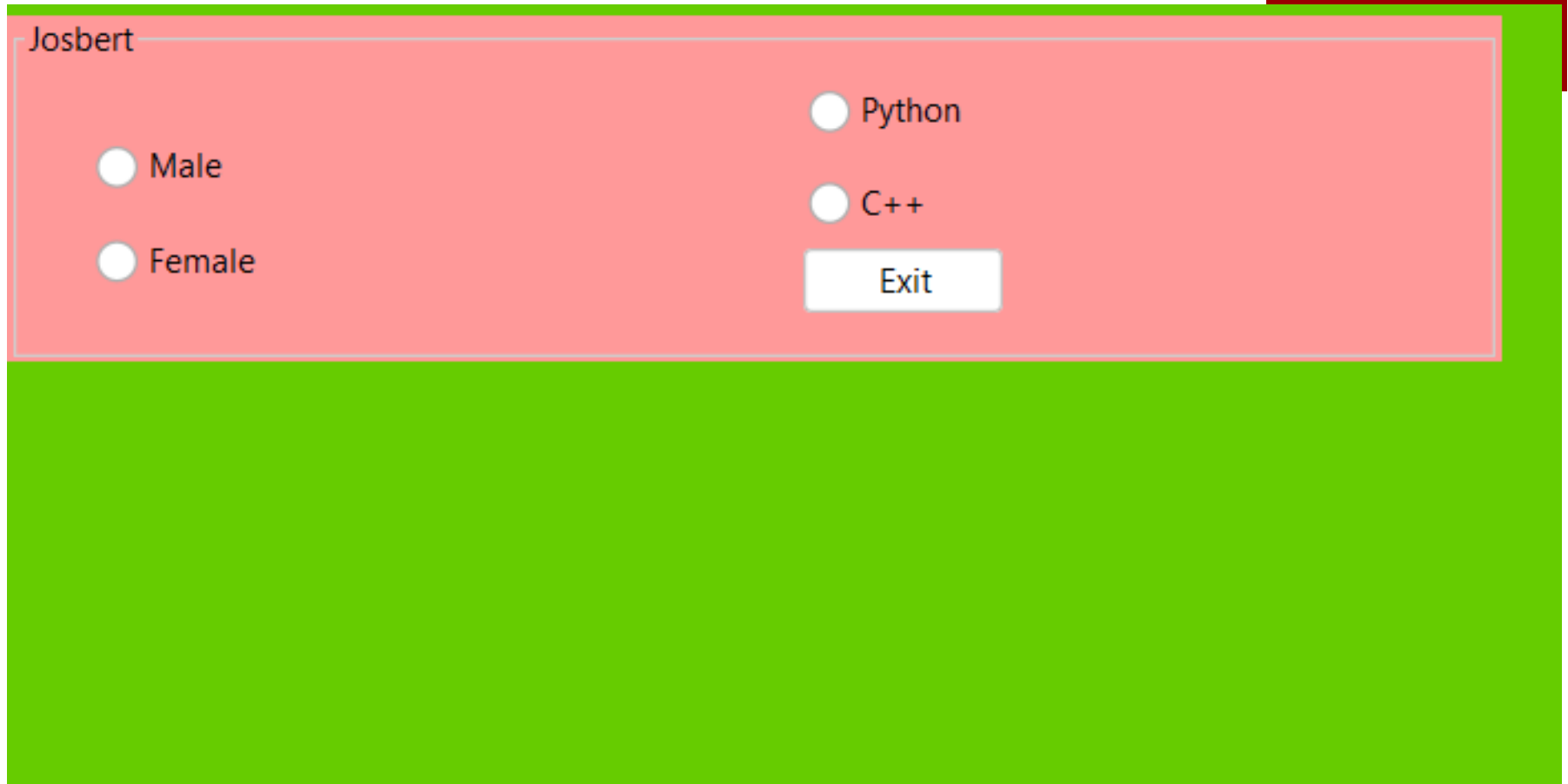
NetBeans IDE GUI builder



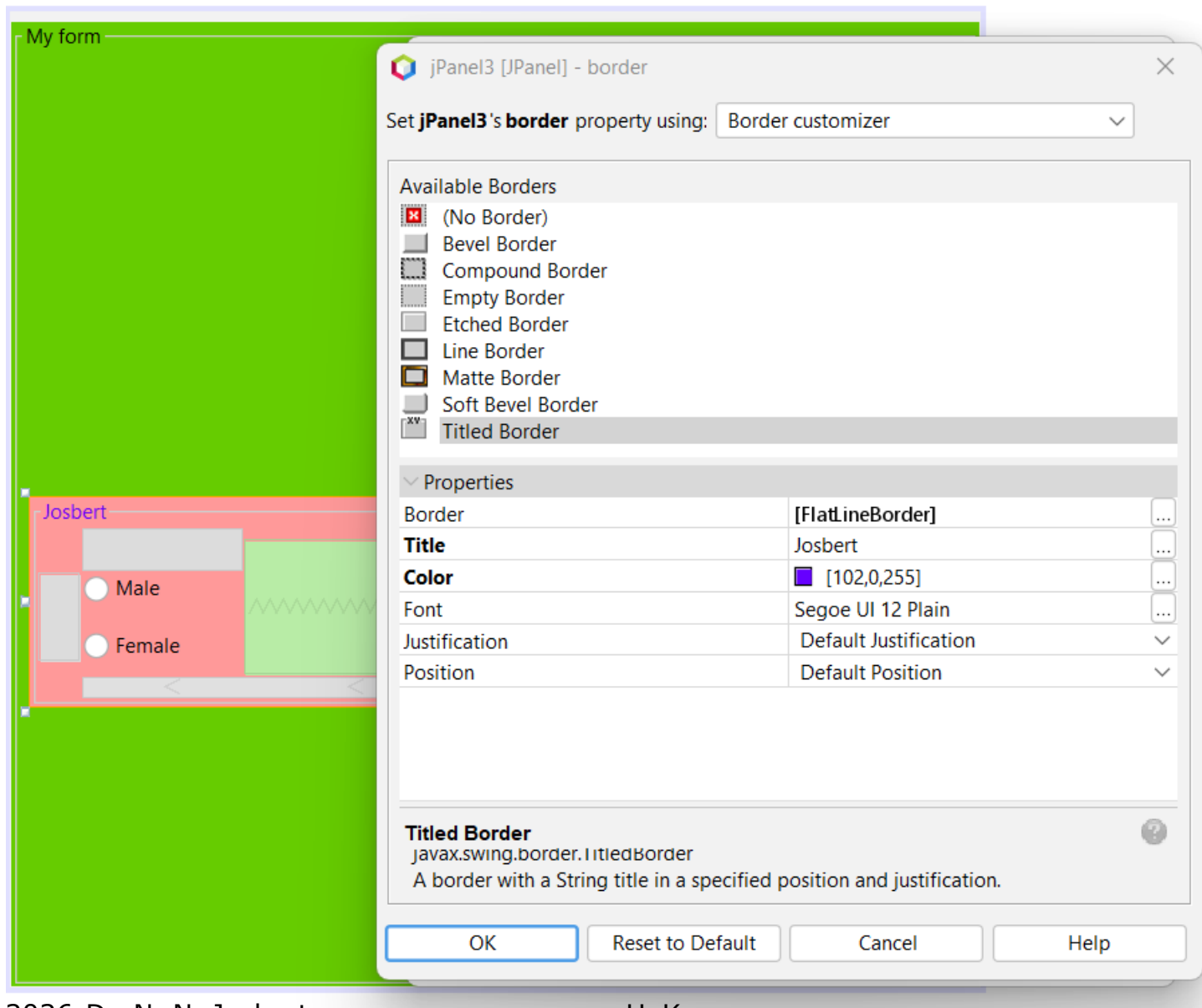
NetBeans IDE GUI builder



NetBeans IDE GUI builder

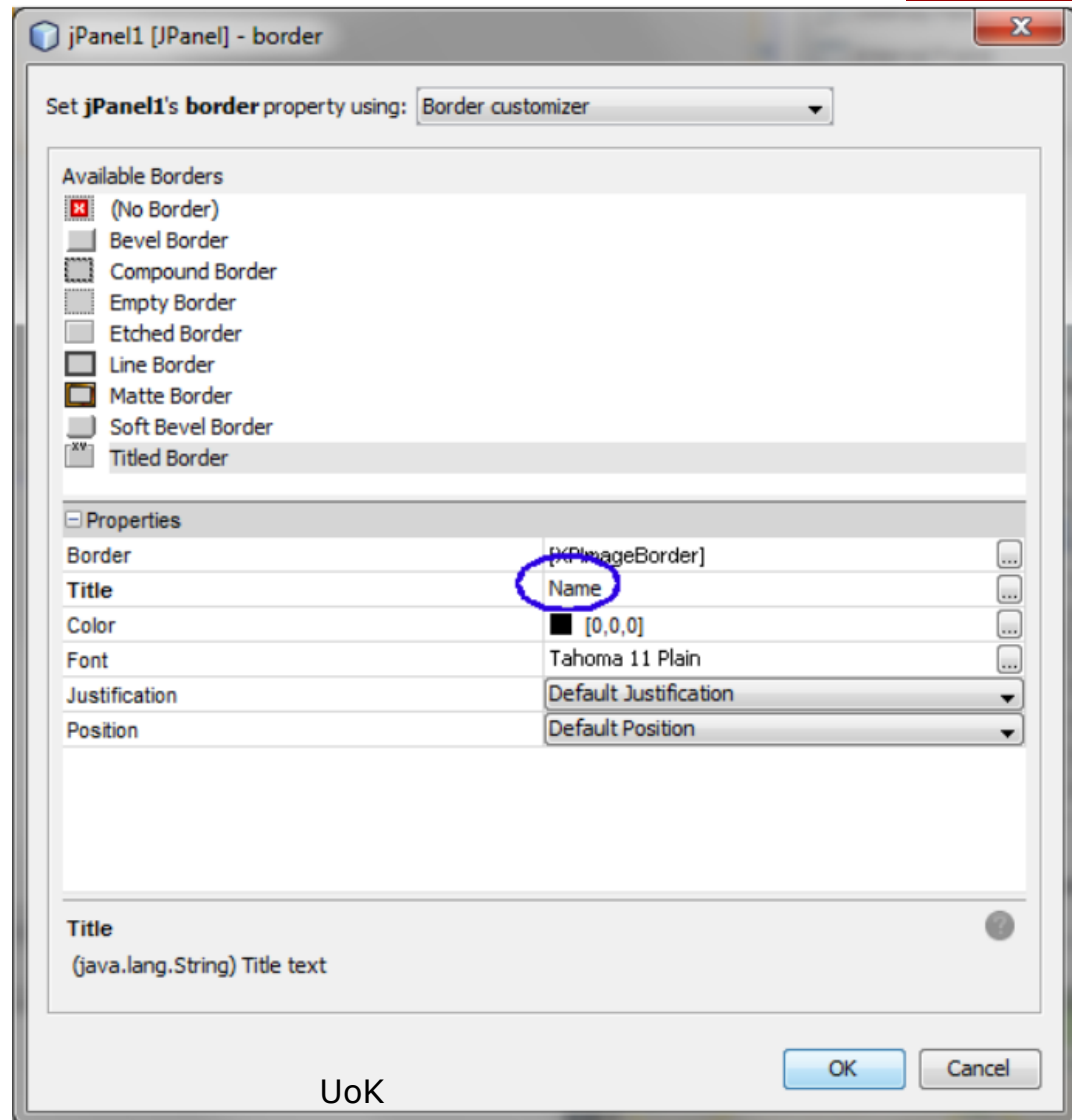


NetBeans IDE GUI builder



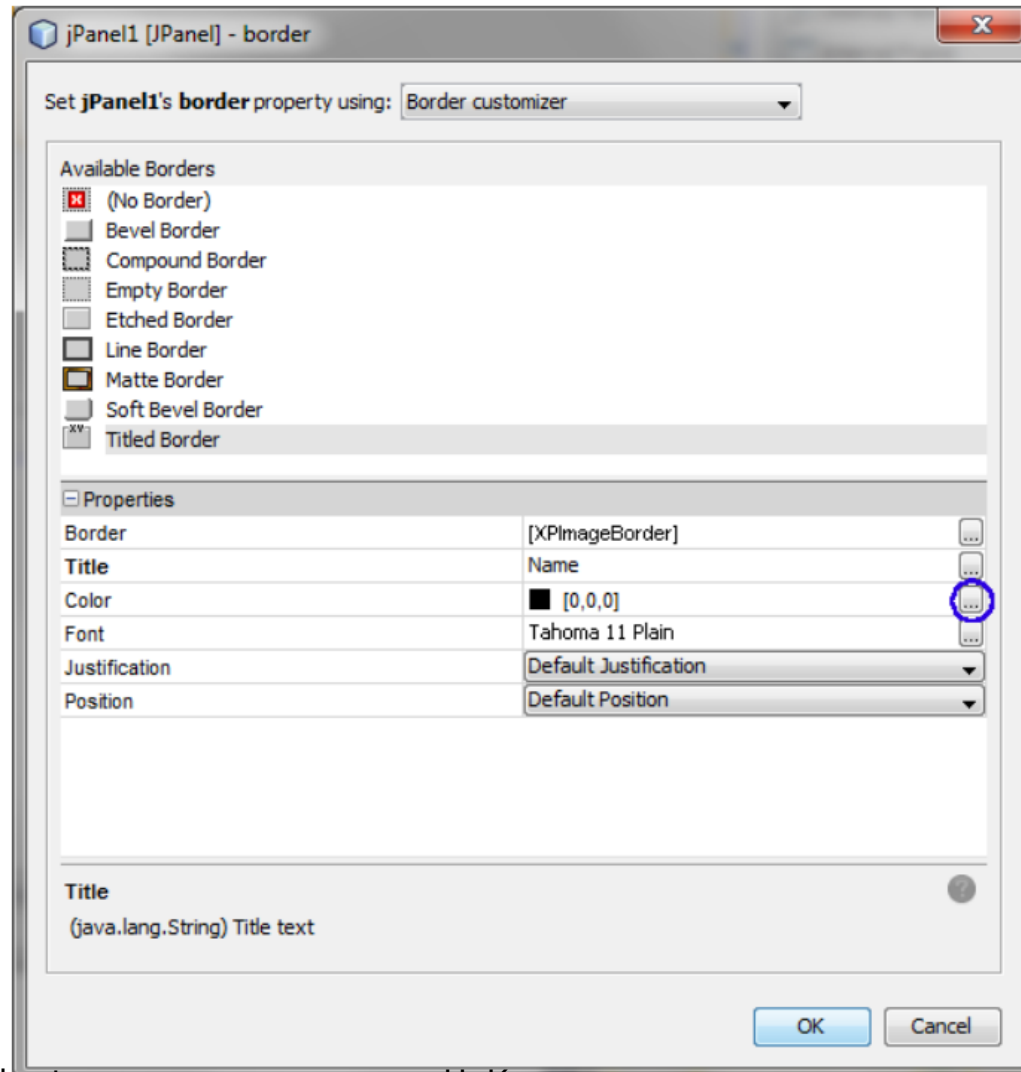
NetBeans IDE GUI builder

- Observe that the **jPanel1[JPanel] - border** property editor dialog box appears.
- **Select TitledBorder.**
- Set the value of the Title field to **Name**.

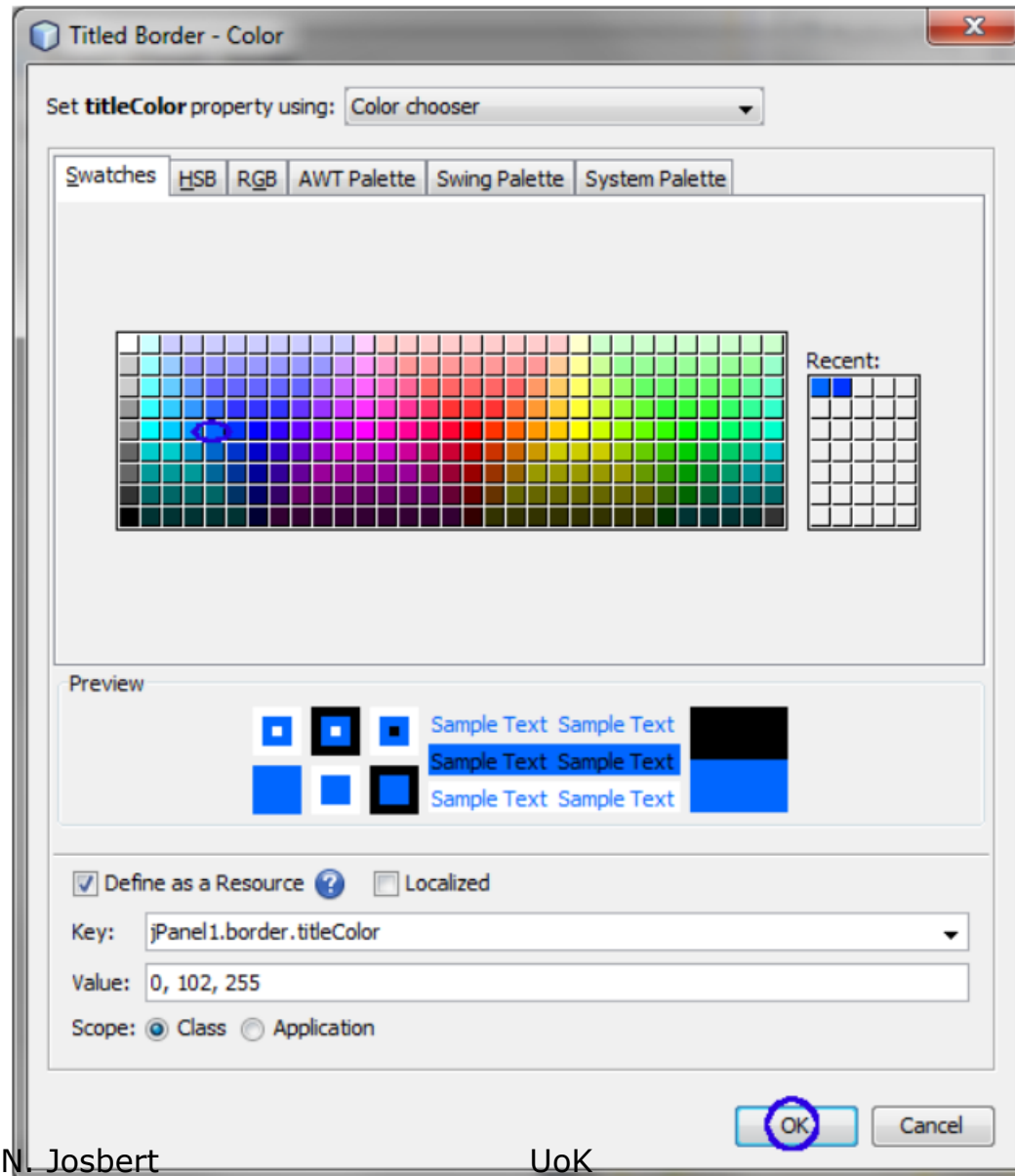


NetBeans IDE GUI builder

- Change the color to your favorite color.

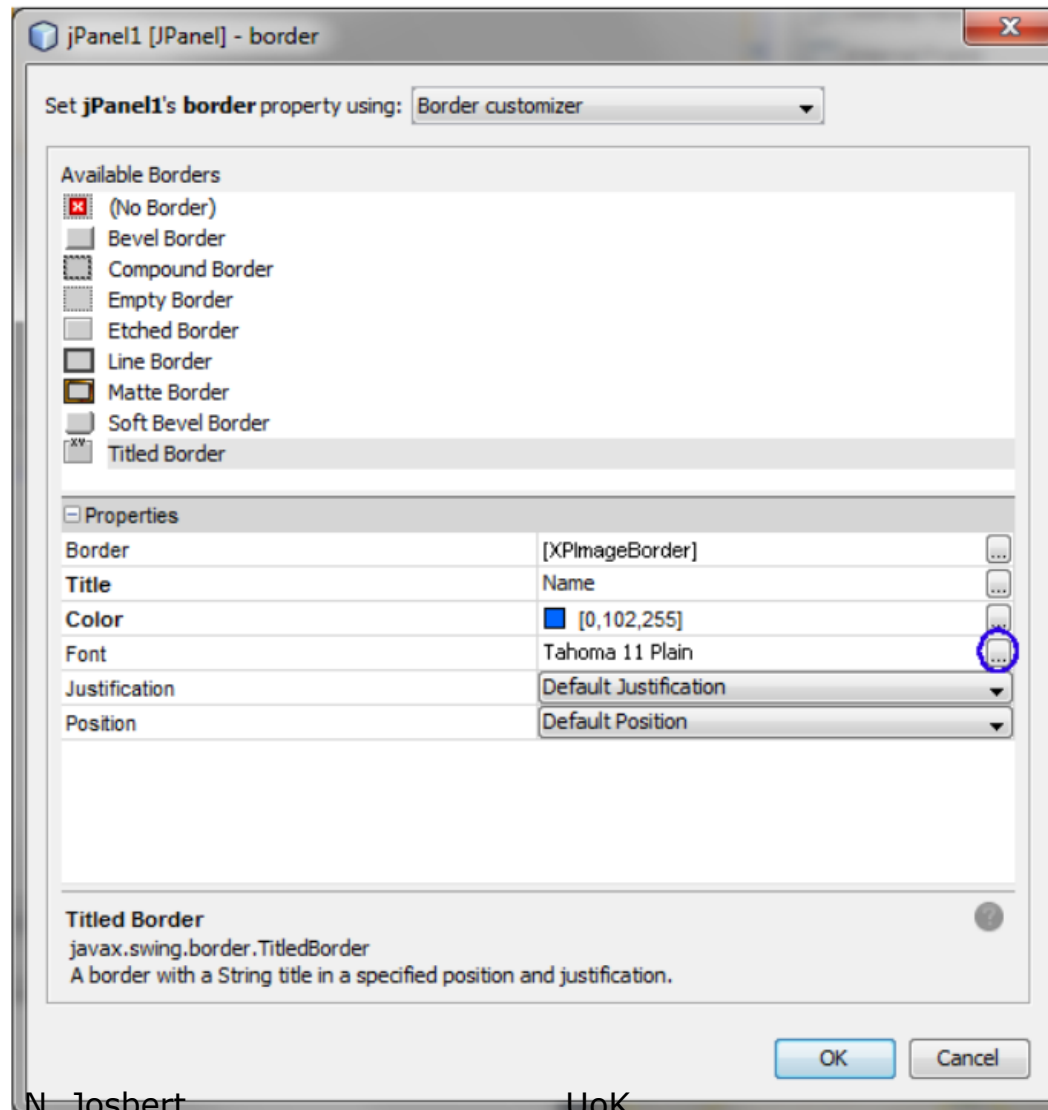


NetBeans IDE GUI builder

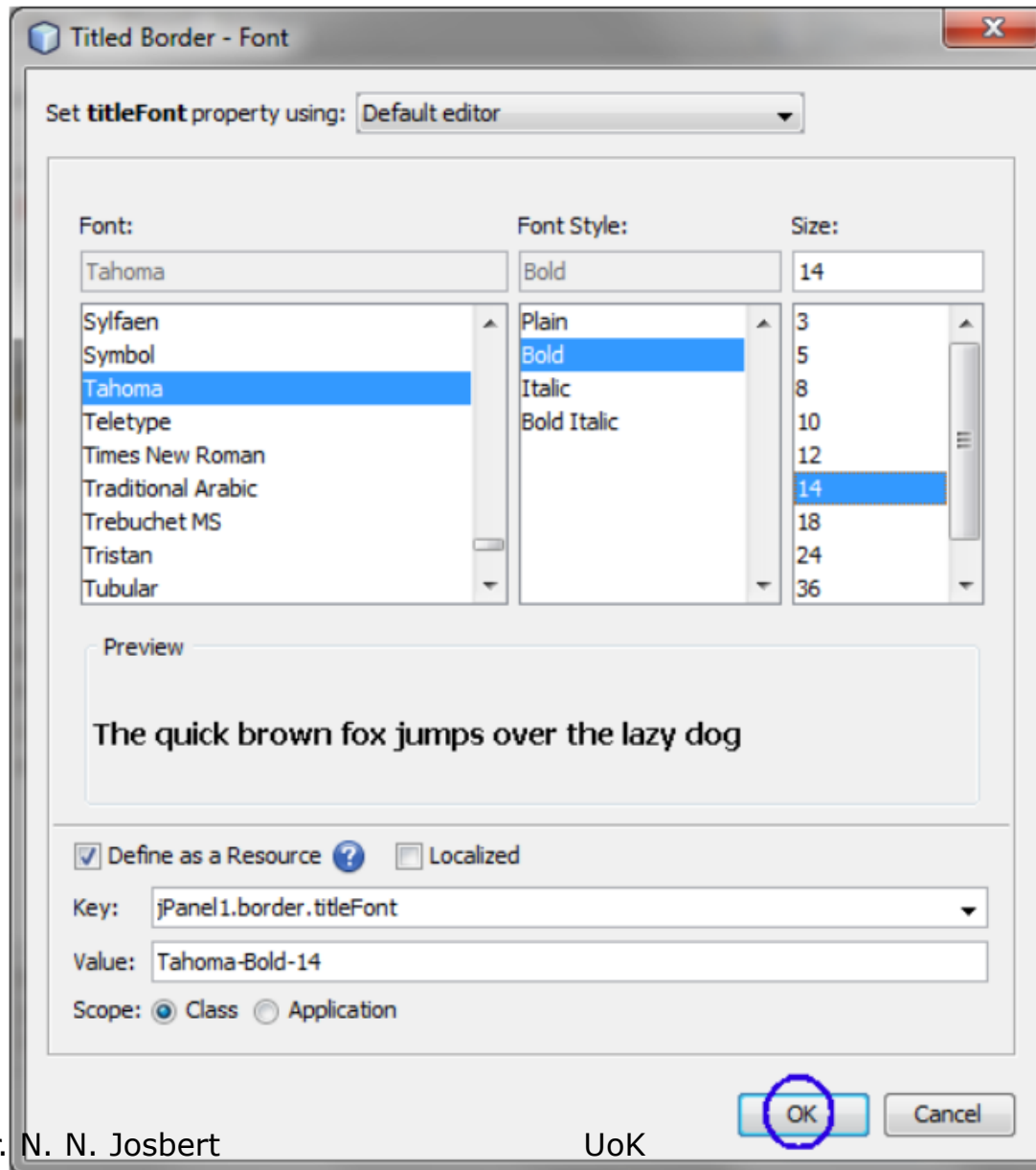


NetBeans IDE GUI builder

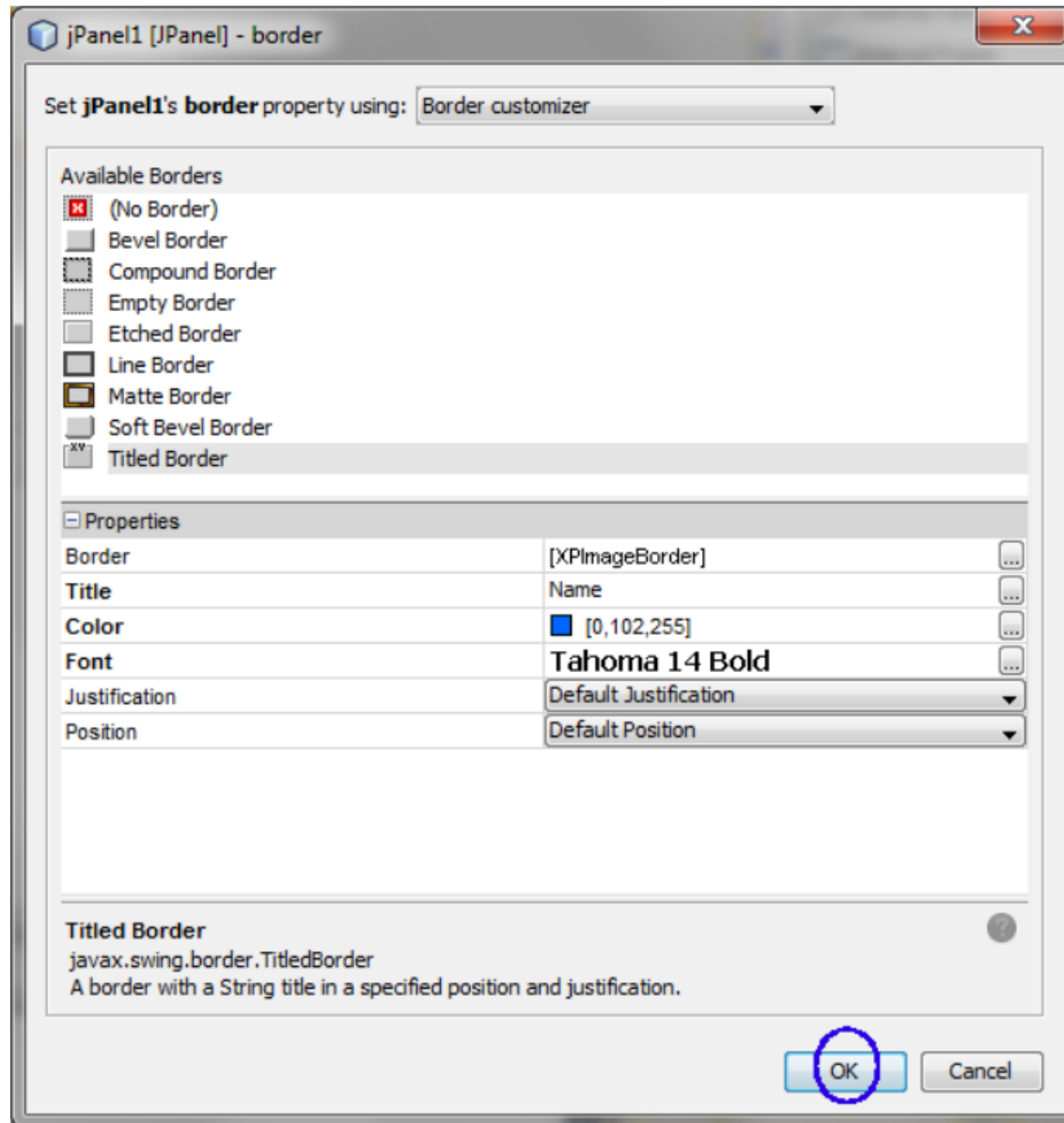
- Change the **Font** to Bold.



NetBeans IDE GUI builder

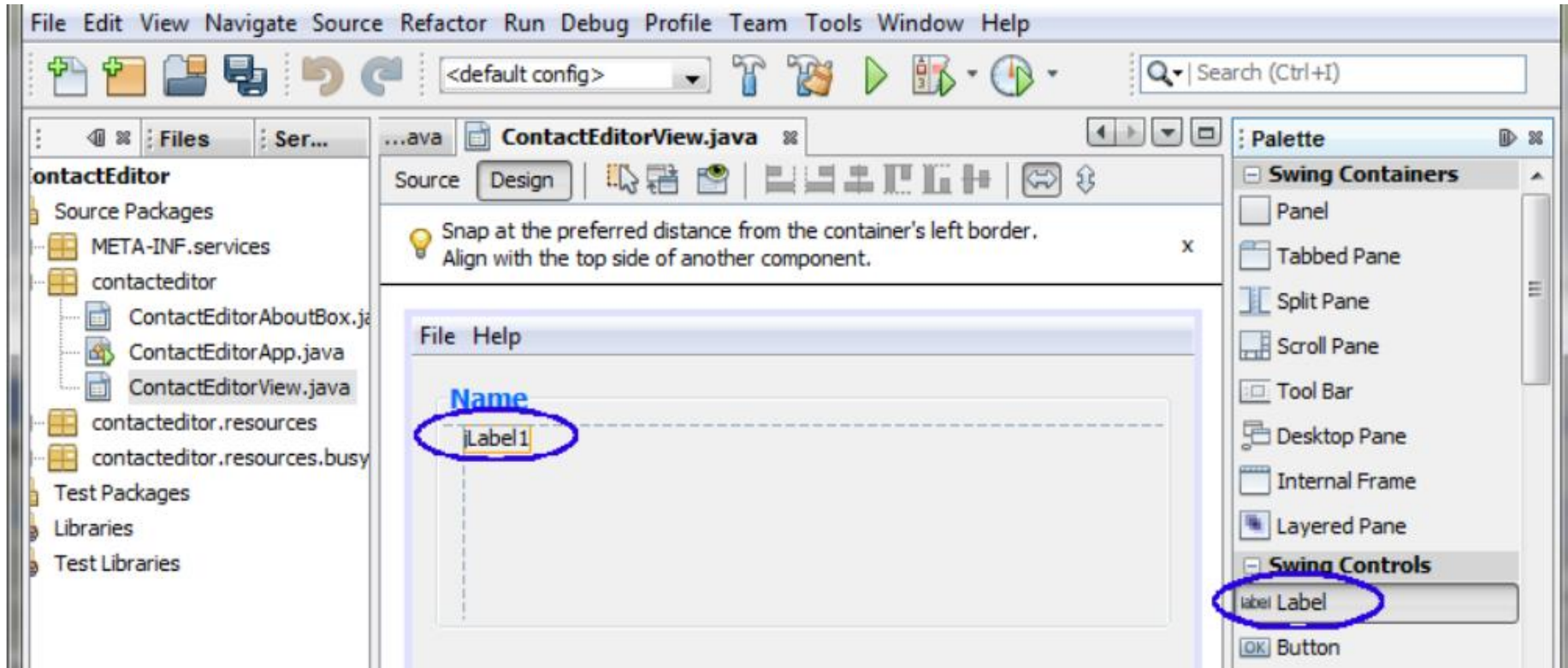


NetBeans IDE GUI builder



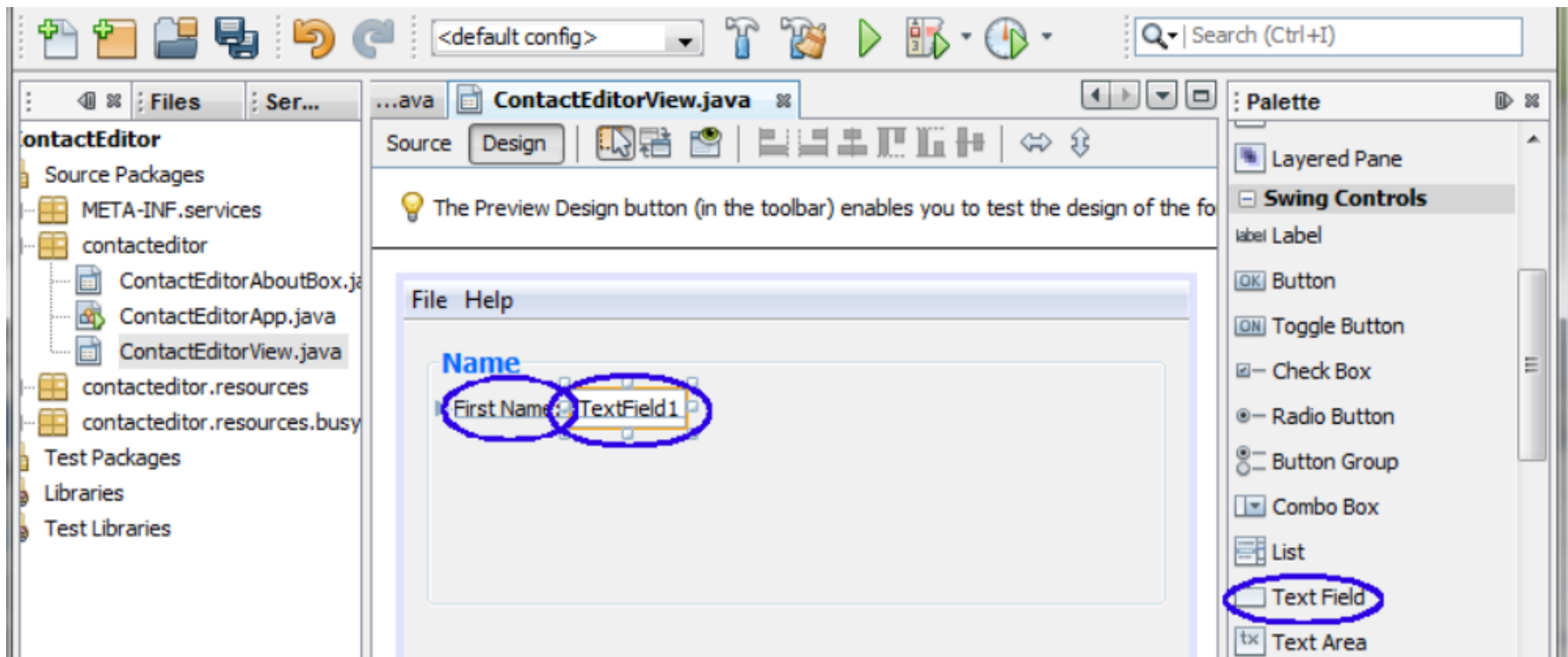
NetBeans IDE GUI builder

- (1.3) Add components to the Panel
- Add a **Label** and **Text Field** components to the **JPanel** container component. (In this document, the terms "Label" and "JLabel" are used interchangeably.)



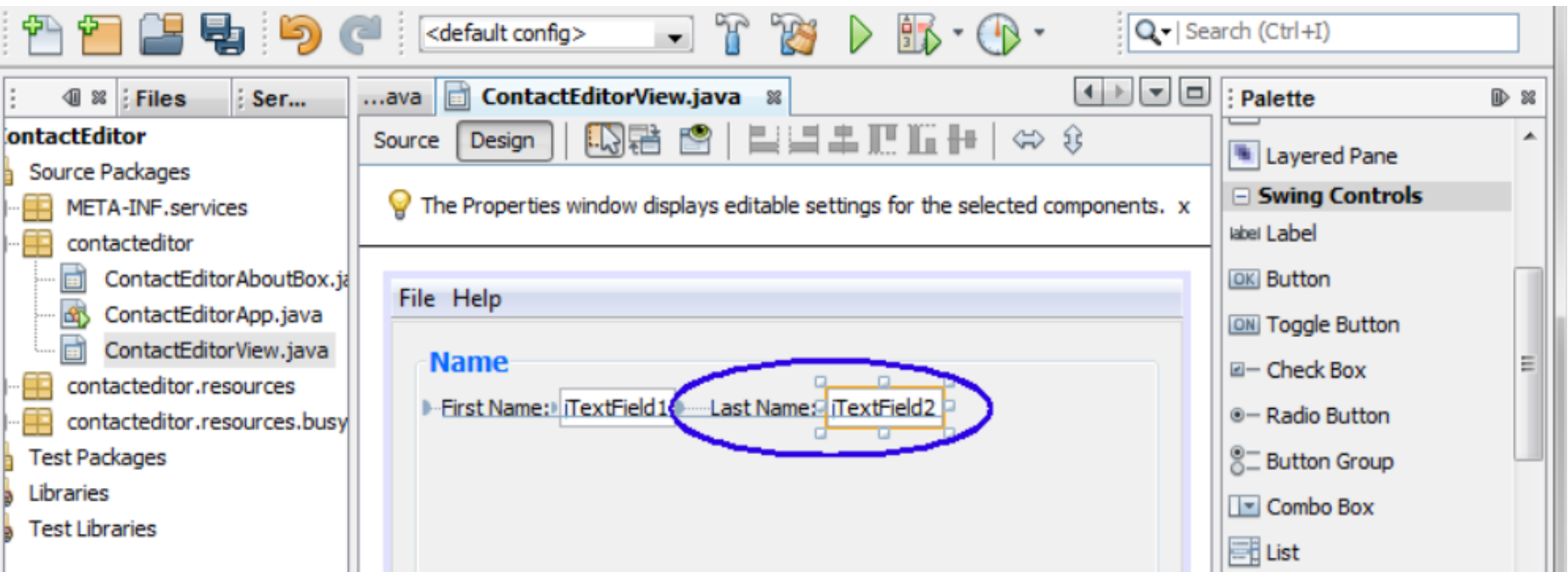
NetBeans IDE GUI builder

- Double click the newly added **Label** component and change the value to **First Name**:
- Drag the **Text Field** under **Swing Controls** section and drop it just next to the **First Name** Label component.

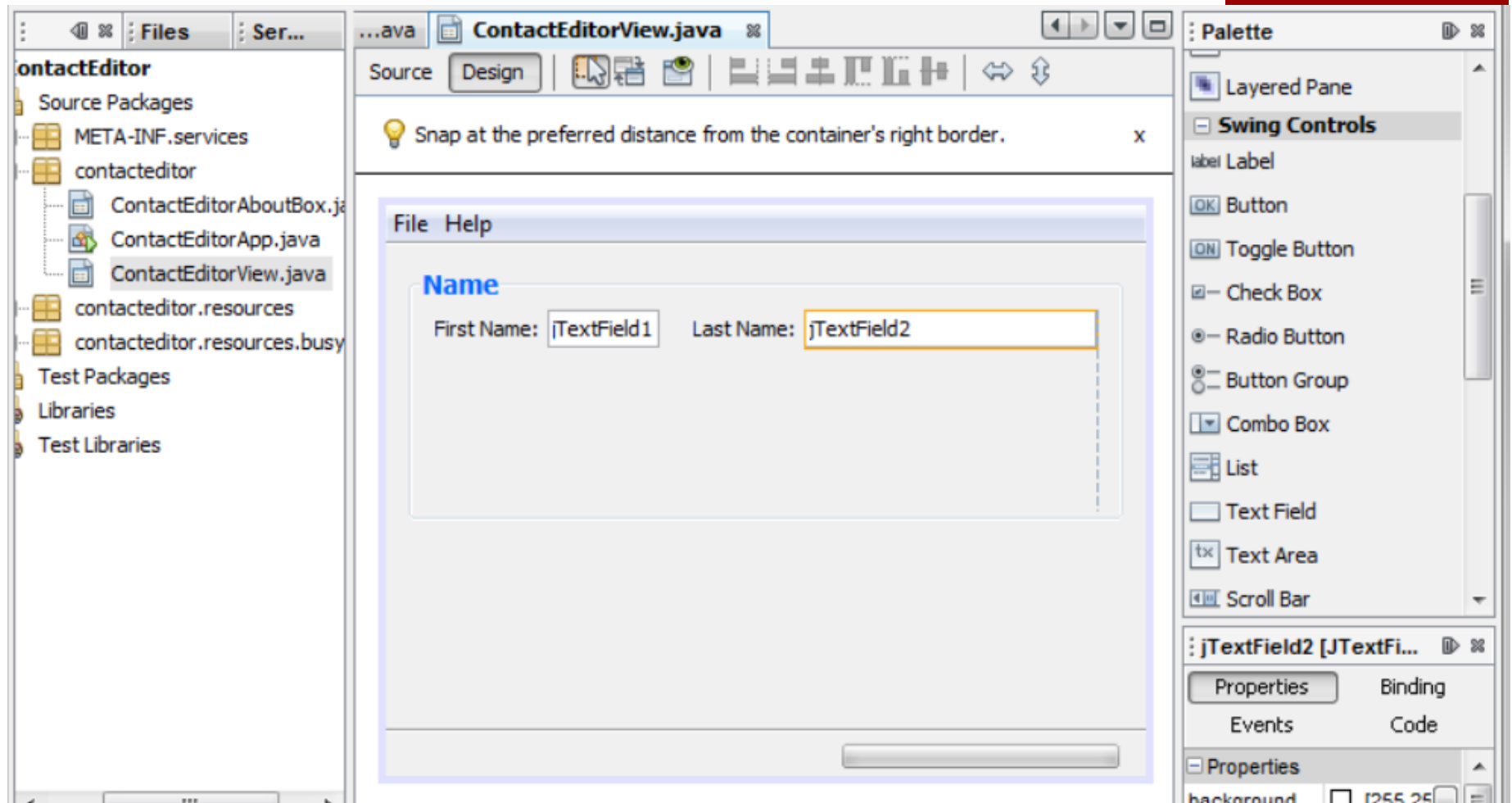


NetBeans IDE GUI builder

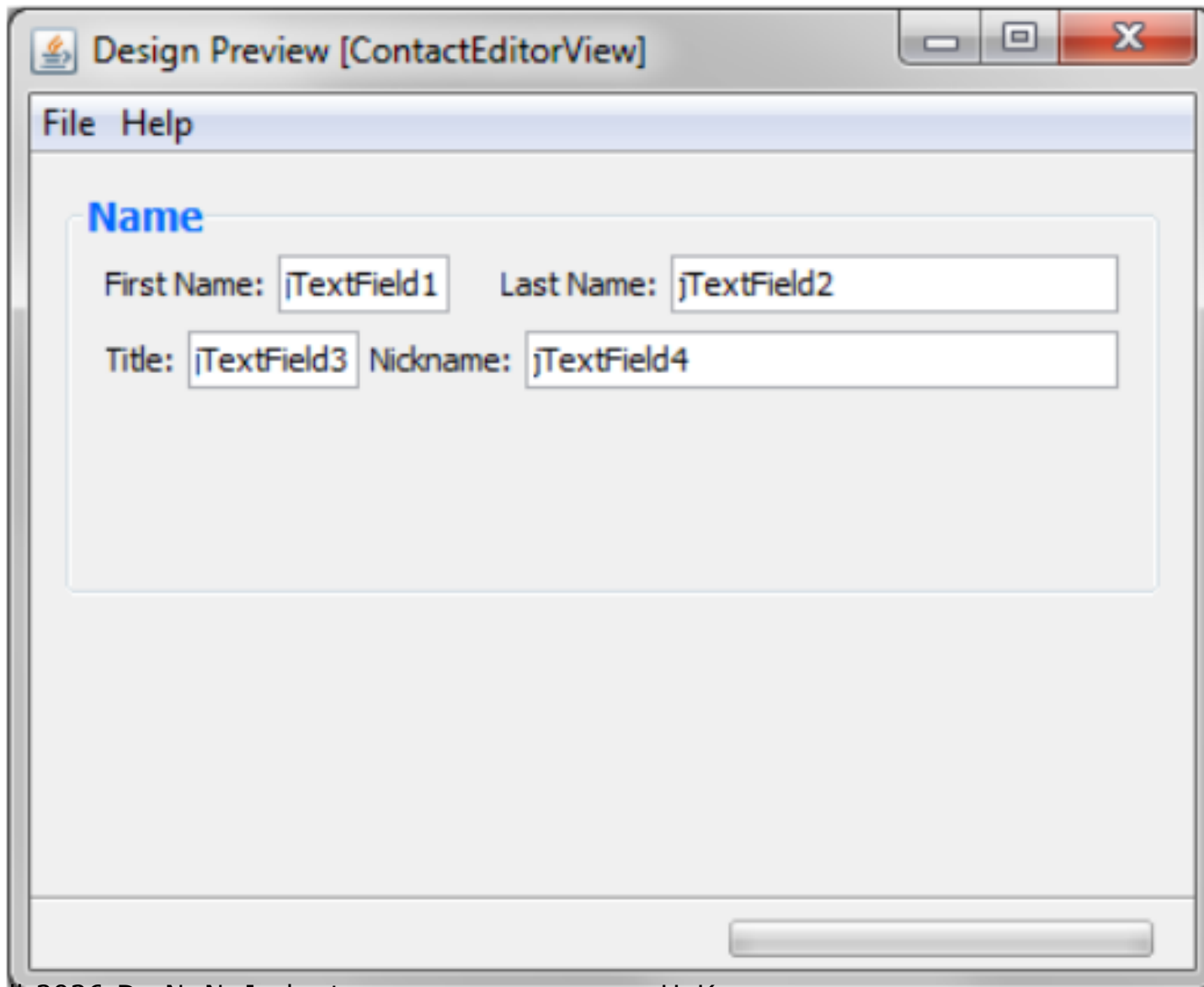
- Add another set of Label and Text Field components. (In this document, the terms "Text Field" and "JTextField" are used interchangeably.)
Drag and drop another set of Label and Text Field components just next to the first set.
- Change the name of the Label to **Last Name**:



NetBeans IDE GUI builder



NetBeans IDE GUI builder

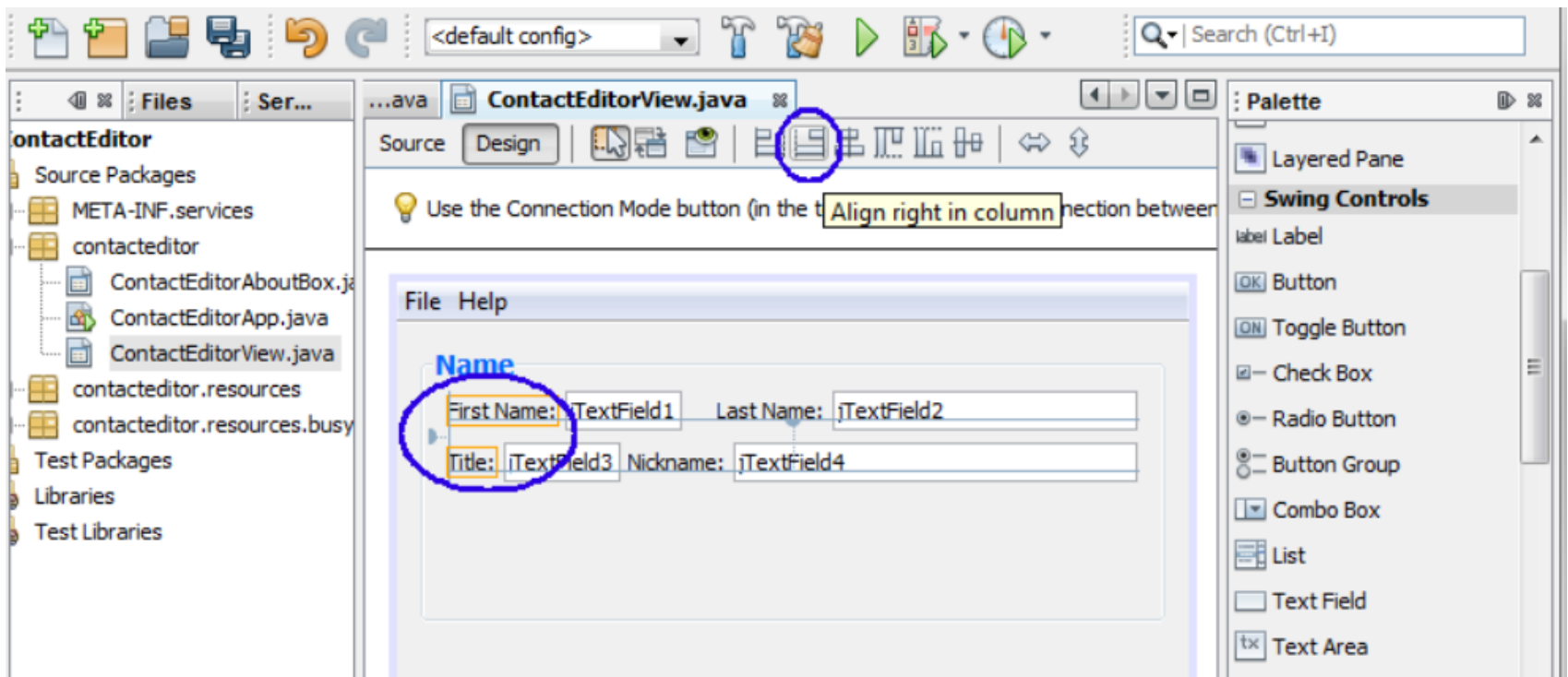


NetBeans IDE GUI builder

- Fix the abnormalities:

- Right align the Labels.

Select both **First Name:** and **Title:** JLabel components (using CTRL key) and select "Align right in column" icon.

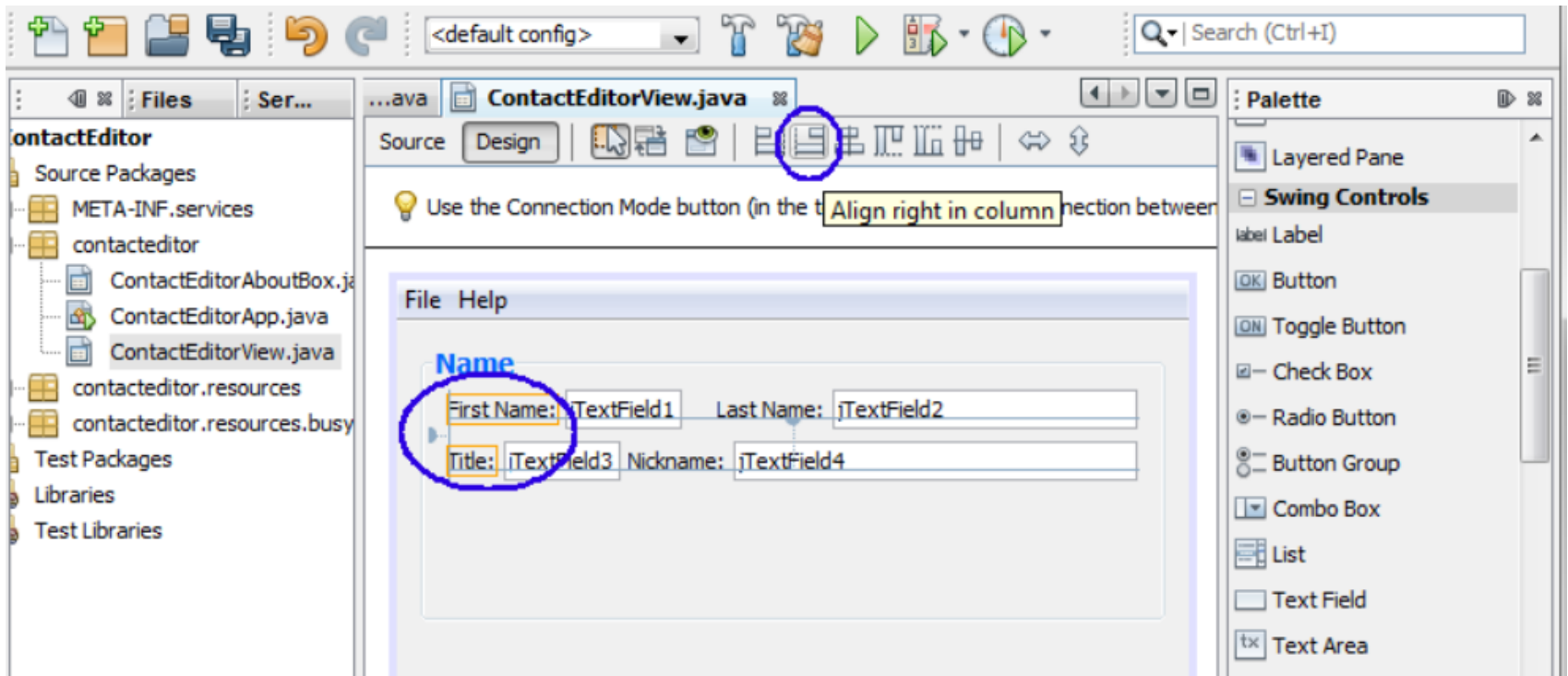


NetBeans IDE GUI builder

- Fix the abnormalities

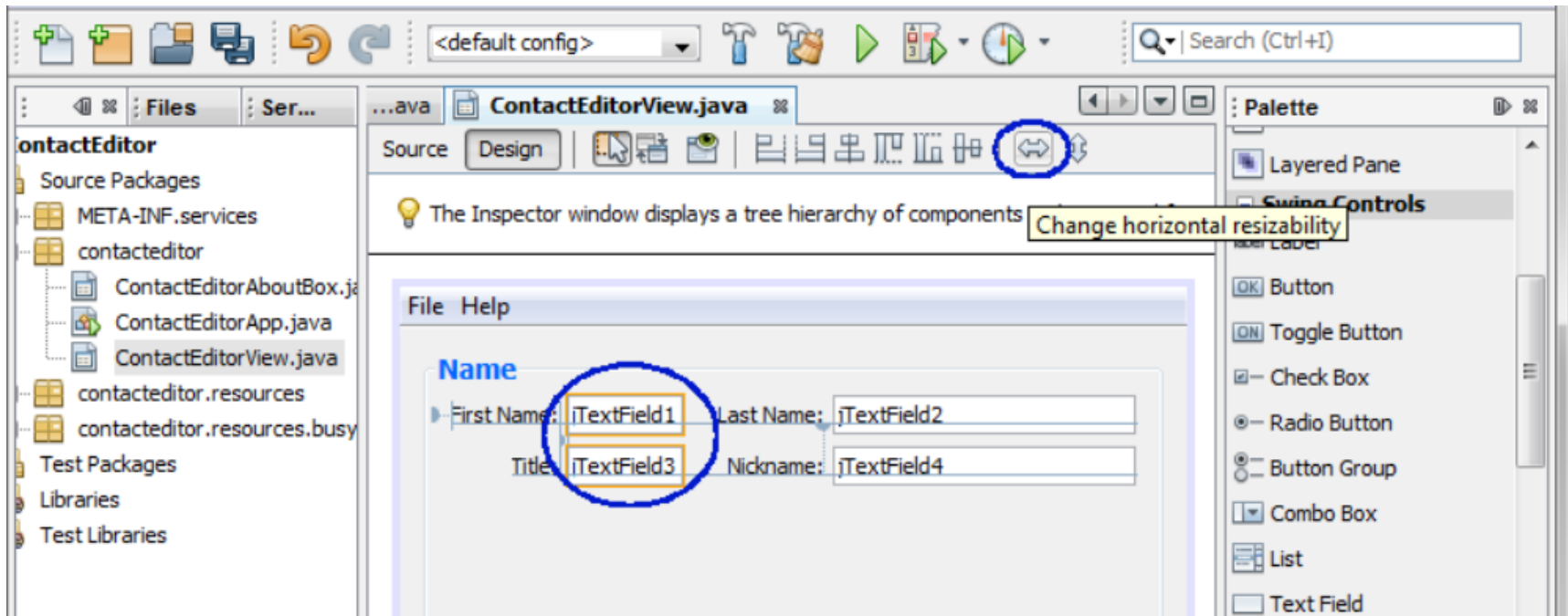
- Right align the Labels.

Select both **First Name:** and **Title:** JLabel components (using CTRL key) and select "Align right in column" icon.



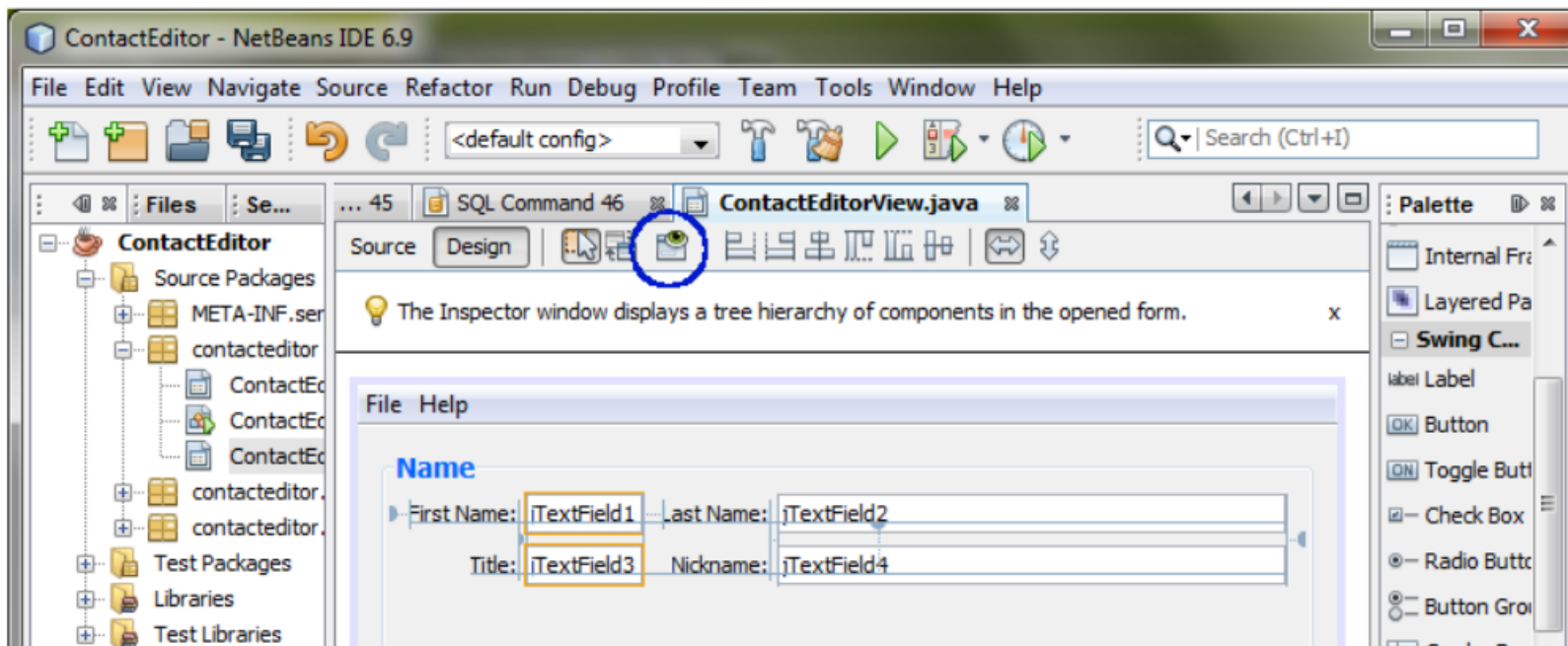
NetBeans IDE GUI builder

- Fix resizing problem
- Select **jTextField1** and **jTextField3** Text Field components (using CTRL key) and select **Change horizontal resizable** icon.



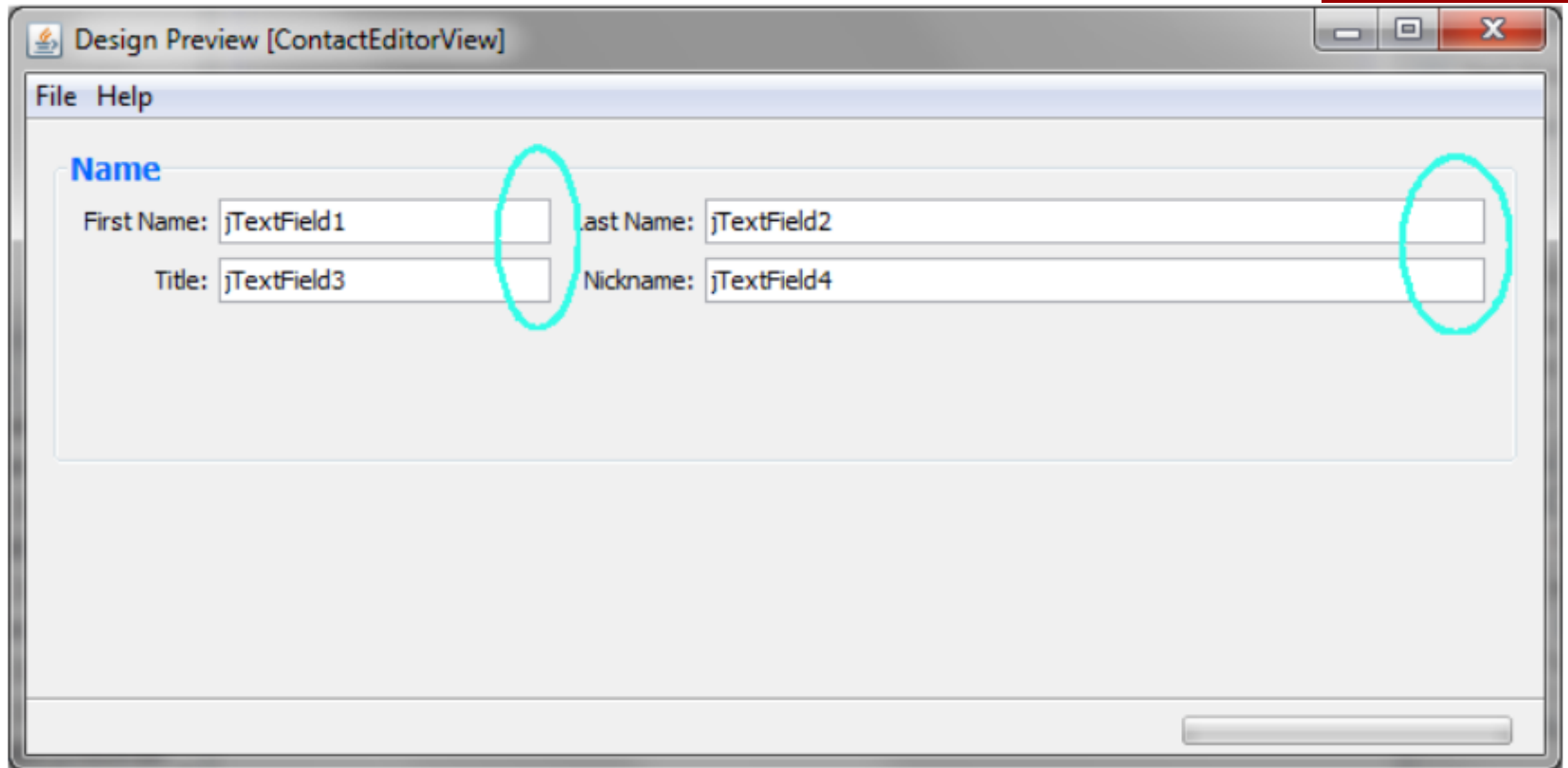
NetBeans IDE GUI builder

- Select **Preview Design** icon.



NetBeans IDE GUI builder

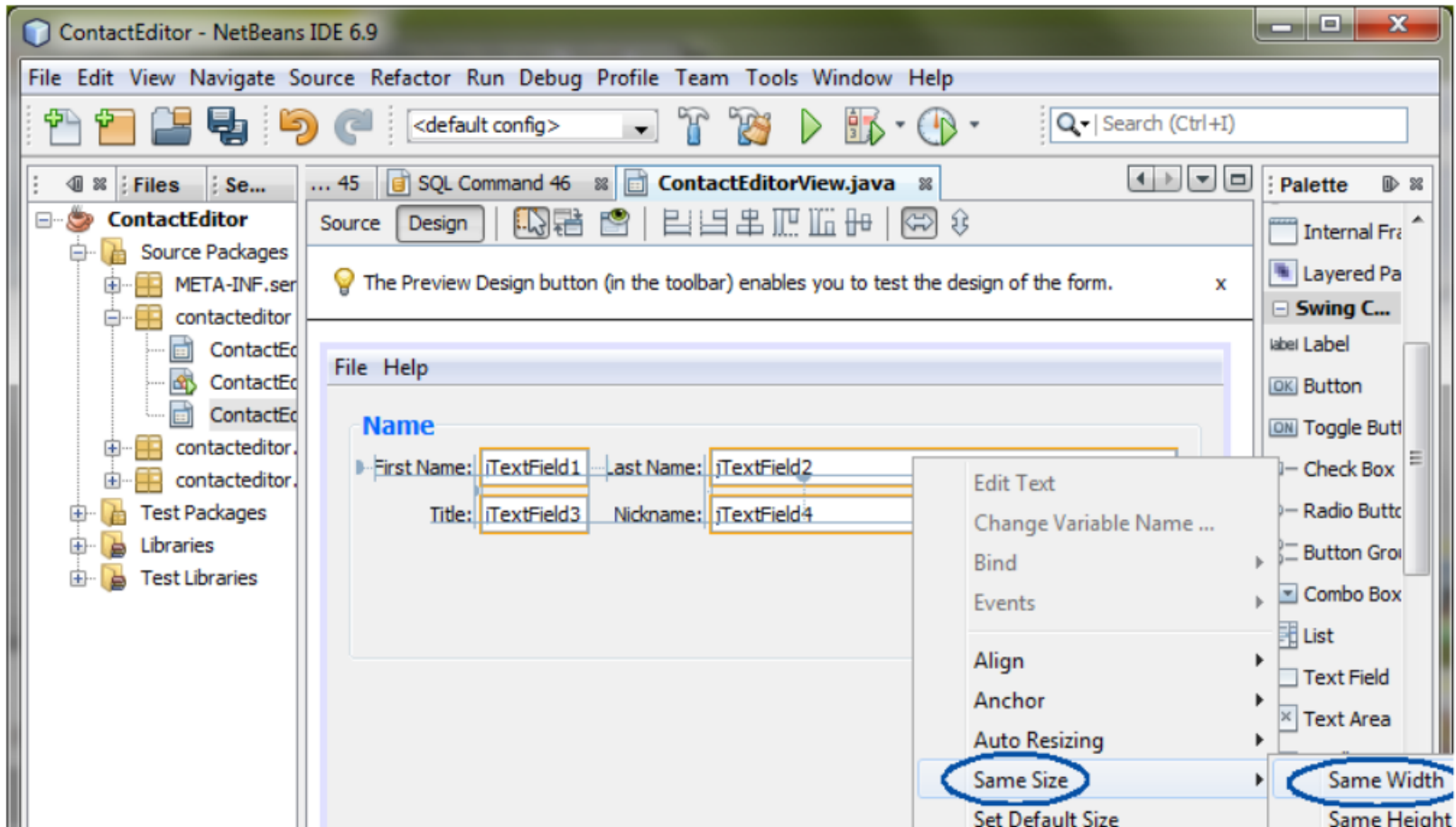
- Verify that now resizing works in all four Text Field components.



NetBeans IDE GUI builder

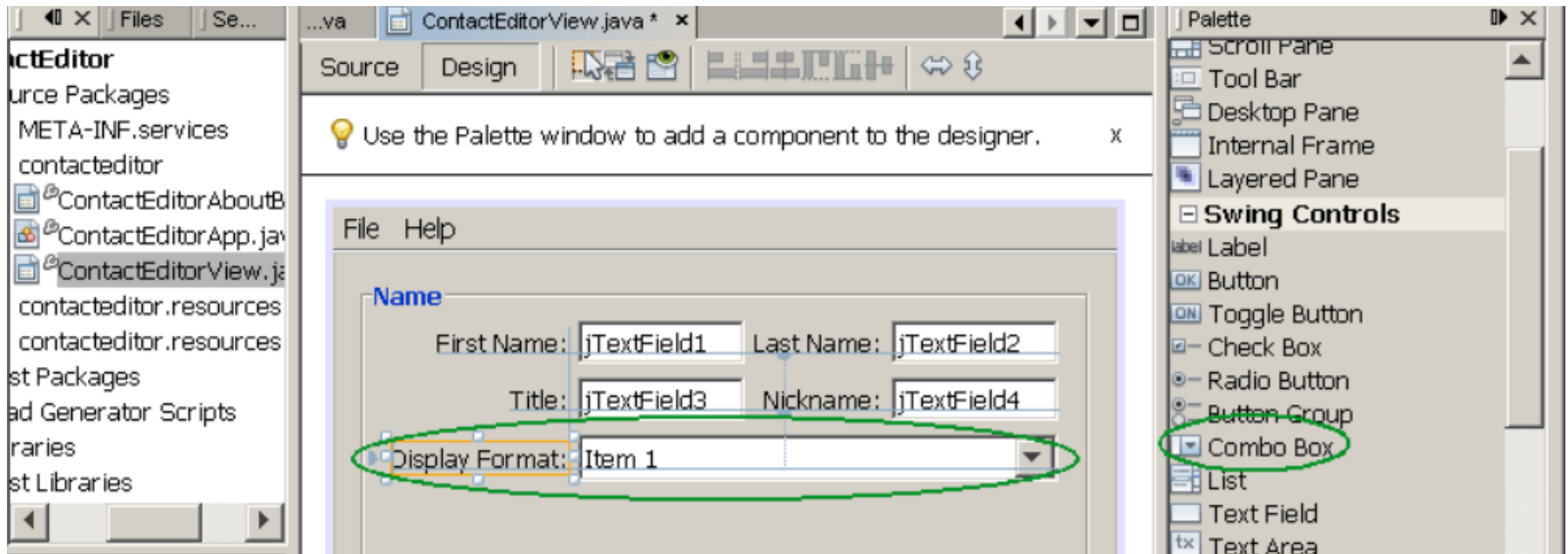
- **Make all Text Fields to be the same size:**

Select all four Text Field components and then select **Same Size->Same Width**.



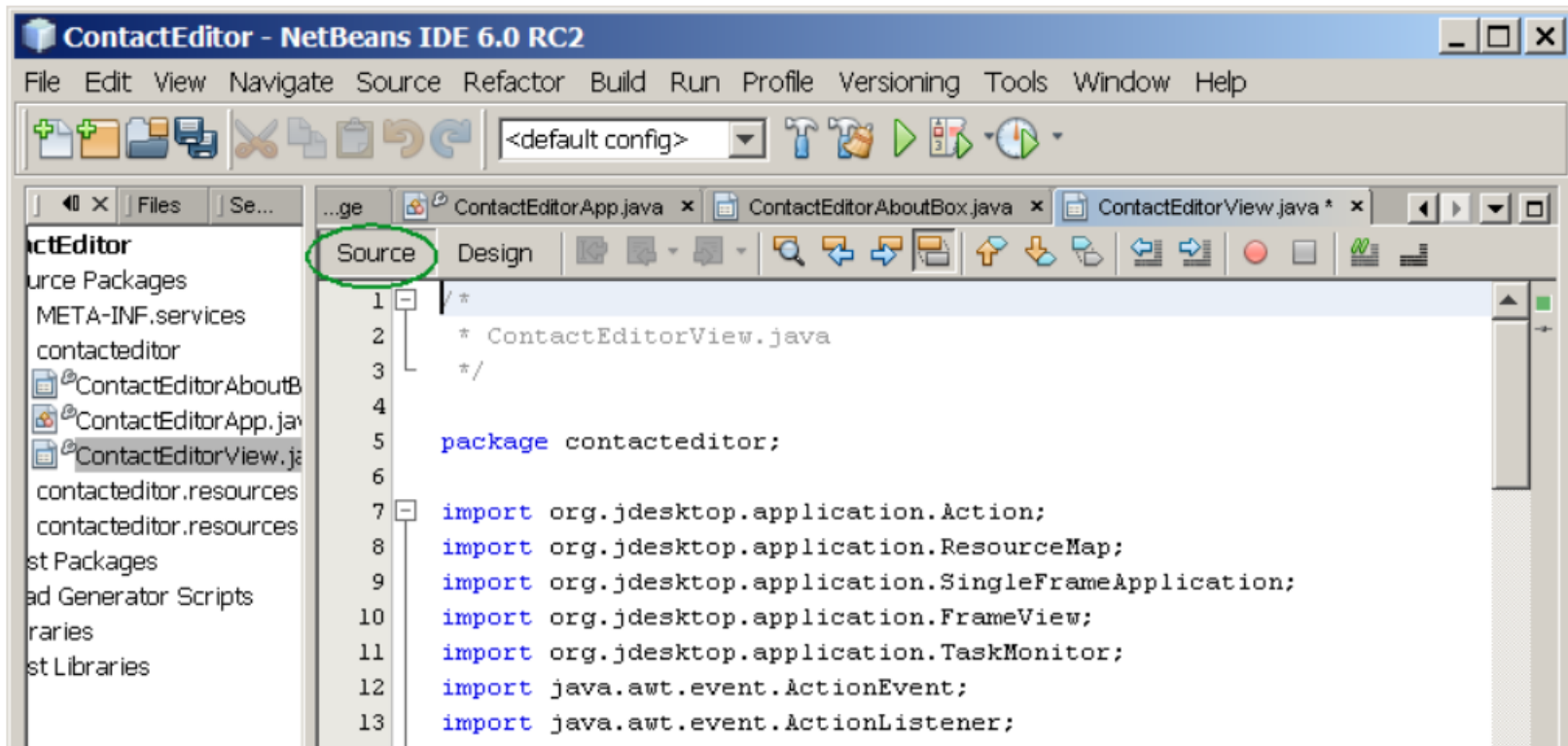
NetBeans IDE GUI builder

- Add a Combo Box component
- Add a new Label component. (Figure-1.71 below)
Drag the **Label** component and drop it under the Title: label right aligned.
- Change the value of the label to **Display Format**.

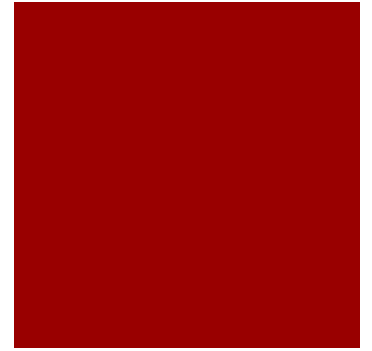


NetBeans IDE GUI builder

- **Observe the source code generated**
- Click Source tab window and observe the source code that was generated by the GUI Builder

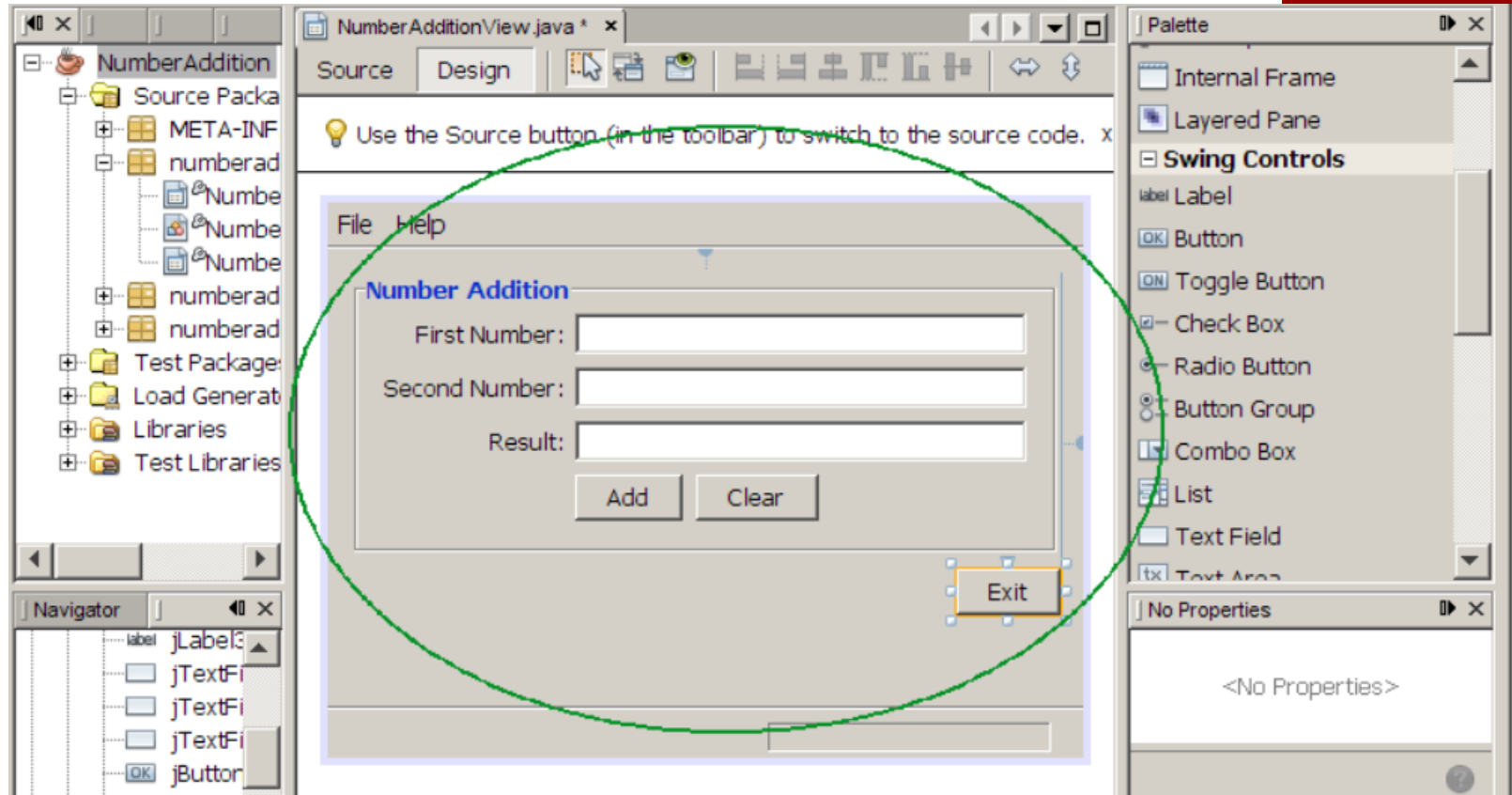


NetBeans IDE GUI builder



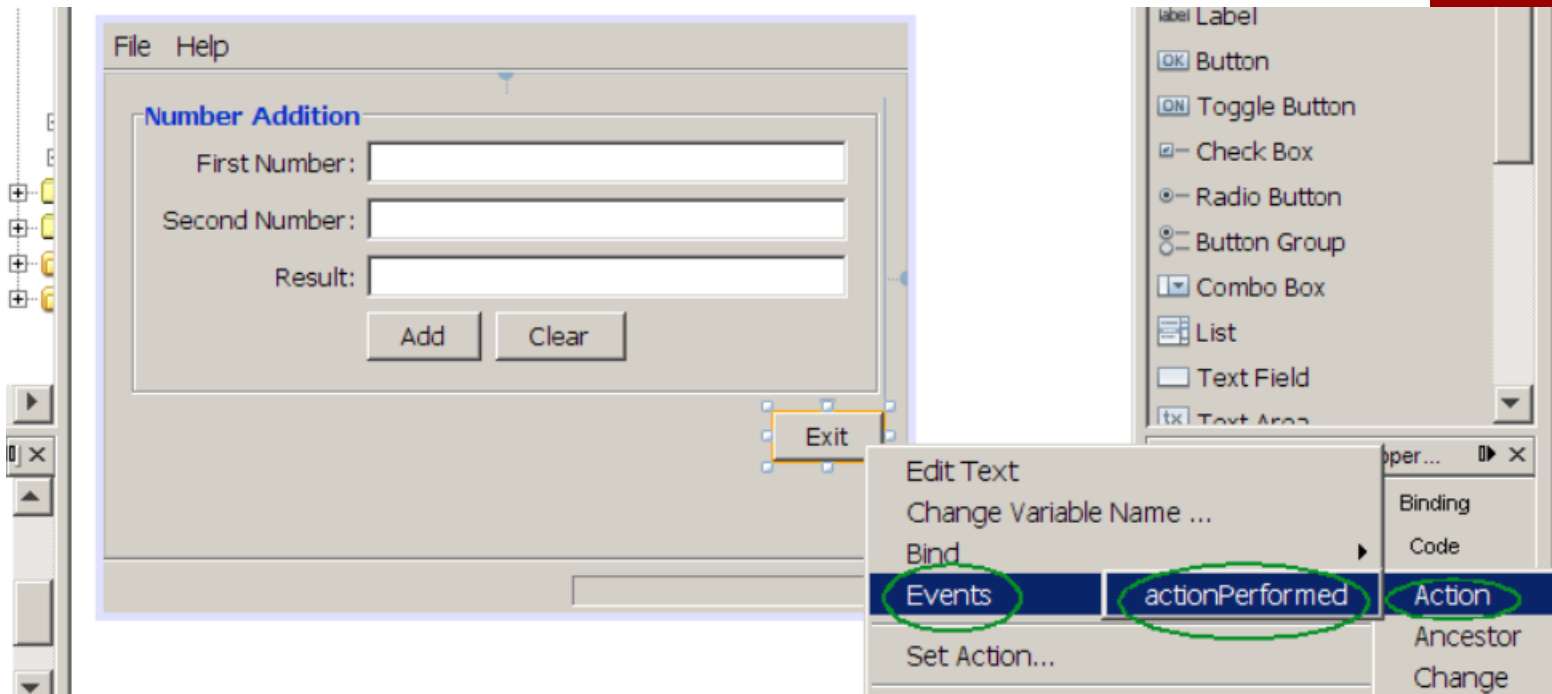
A CAS STUDY 3

NetBeans IDE GUI builder



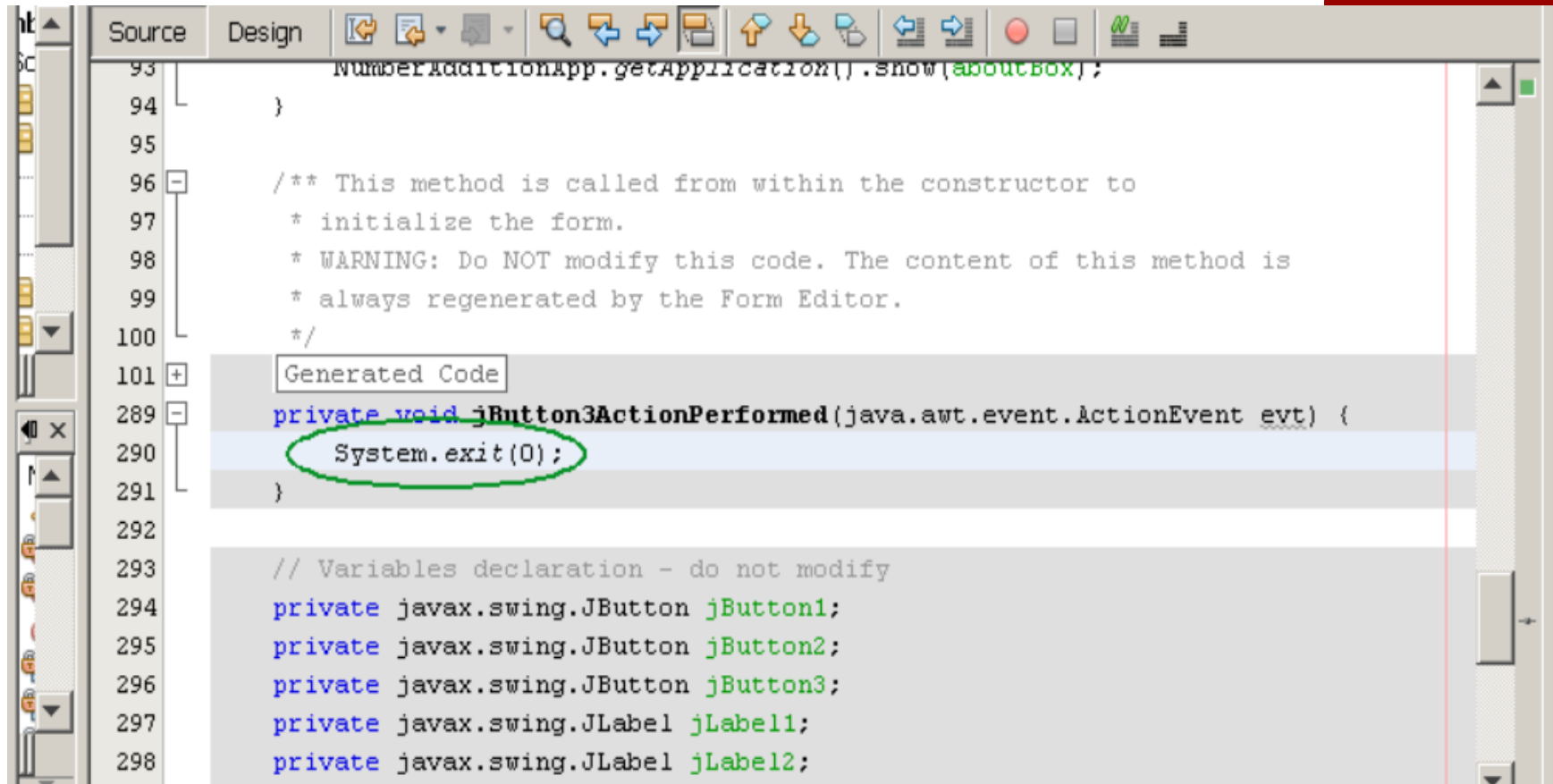
NetBeans IDE GUI builder

- Right click **Exit** button and select **Events->Action->actionPerformed**.



```
private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {  
    System.exit(0);  
}
```

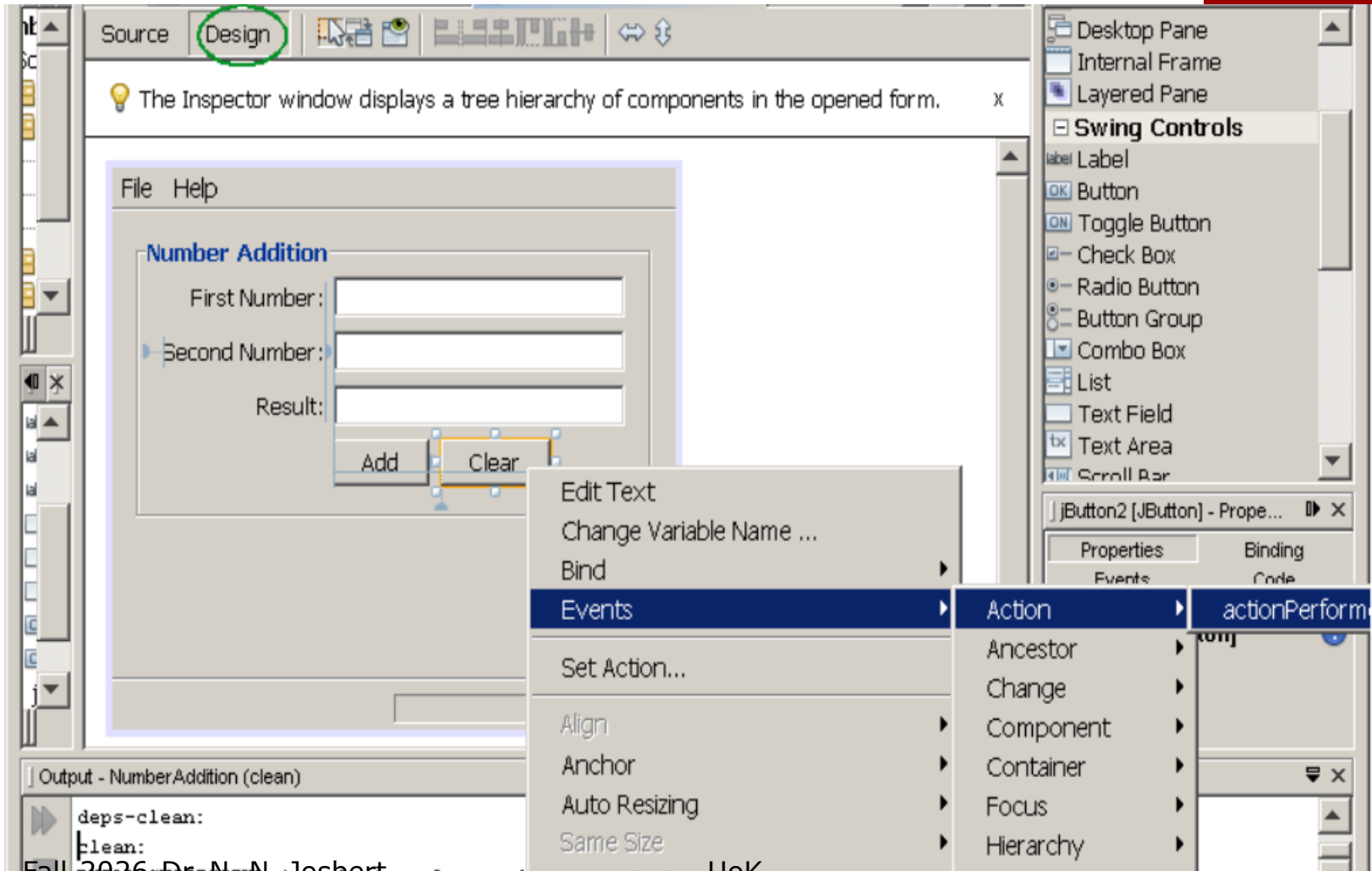
NetBeans IDE GUI builder



```
93      NumberAdditionApp.getApplication().show(aboutBox);
94  }
95
96  /** This method is called from within the constructor to
97   * initialize the form.
98   * WARNING: Do NOT modify this code. The content of this method is
99   * always regenerated by the Form Editor.
100  */
101  Generated Code
289  private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {
290      System.exit(0);
291  }
292
293  // Variables declaration - do not modify
294  private javax.swing.JButton jButton1;
295  private javax.swing.JButton jButton2;
296  private javax.swing.JButton jButton3;
297  private javax.swing.JLabel jLabel1;
298  private javax.swing.JLabel jLabel2;
```

NetBeans IDE GUI builder

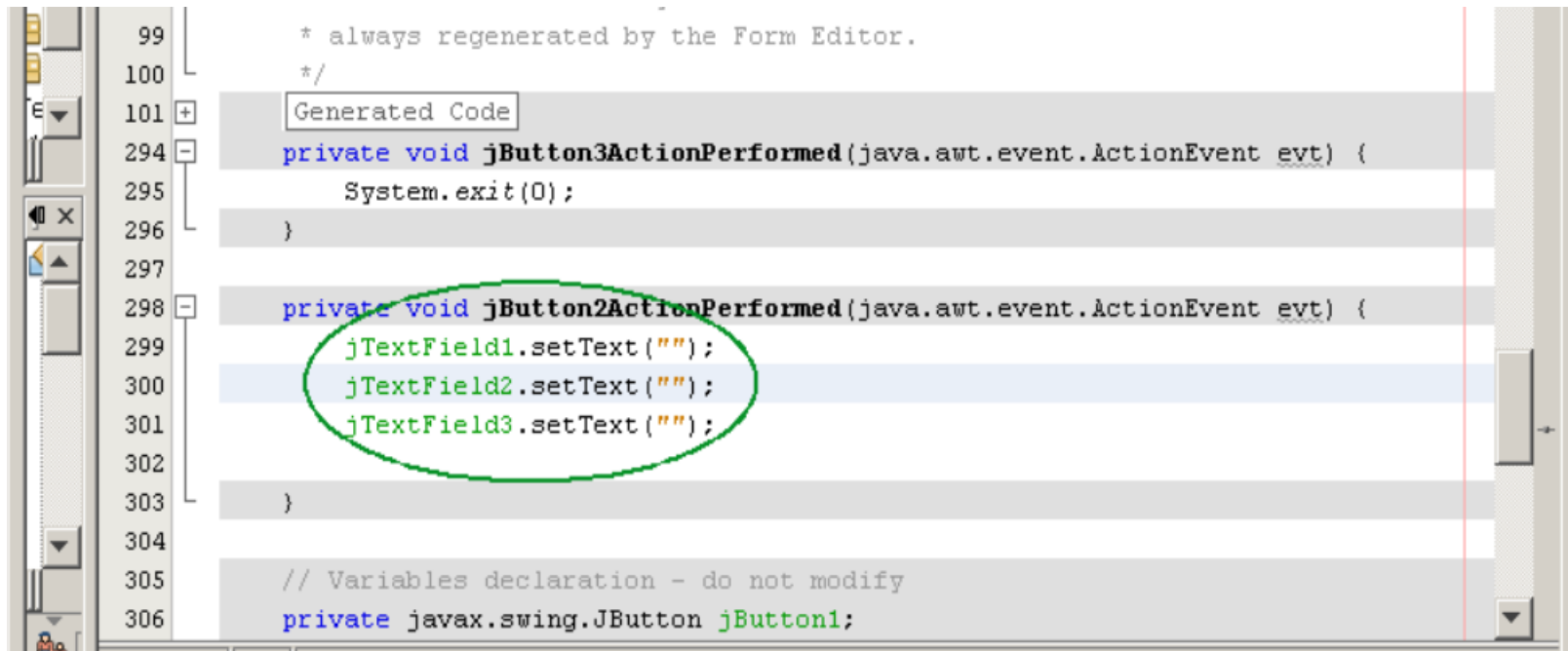
- Right click **Clear** button and select **Events->Action->actionPerformed**.



NetBeans IDE GUI builder

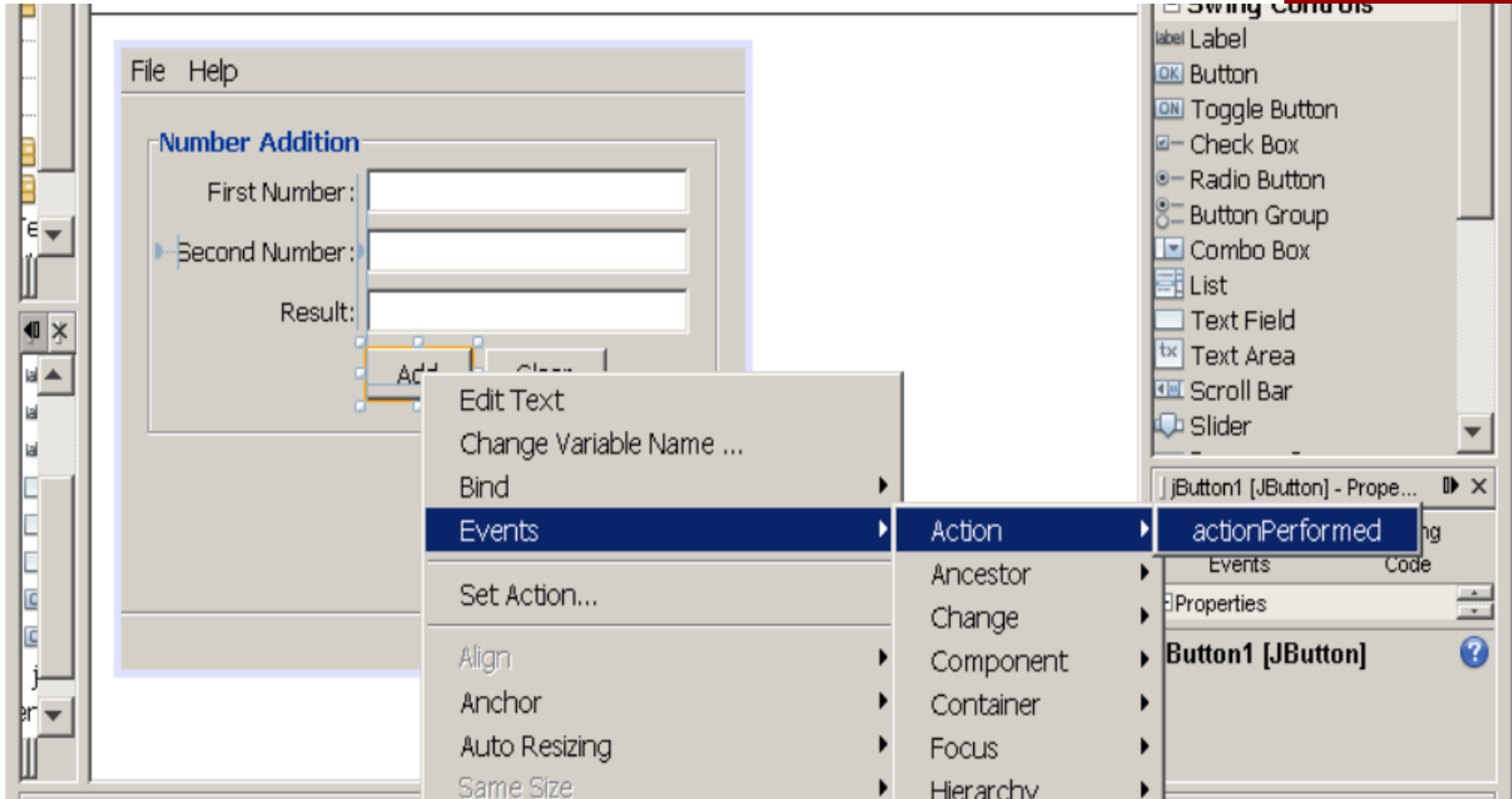
- Modify the event handler as shown in Code and Figure below. The code fragment that needs be added is high-lighted in bold-font.

```
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt){  
    jTextField1.setText("");  
    jTextField2.setText("");  
    jTextField3.setText("");  
}
```



NetBeans IDE GUI builder

- Right click **Add** button and select **Events->Action->actionPerformed**.



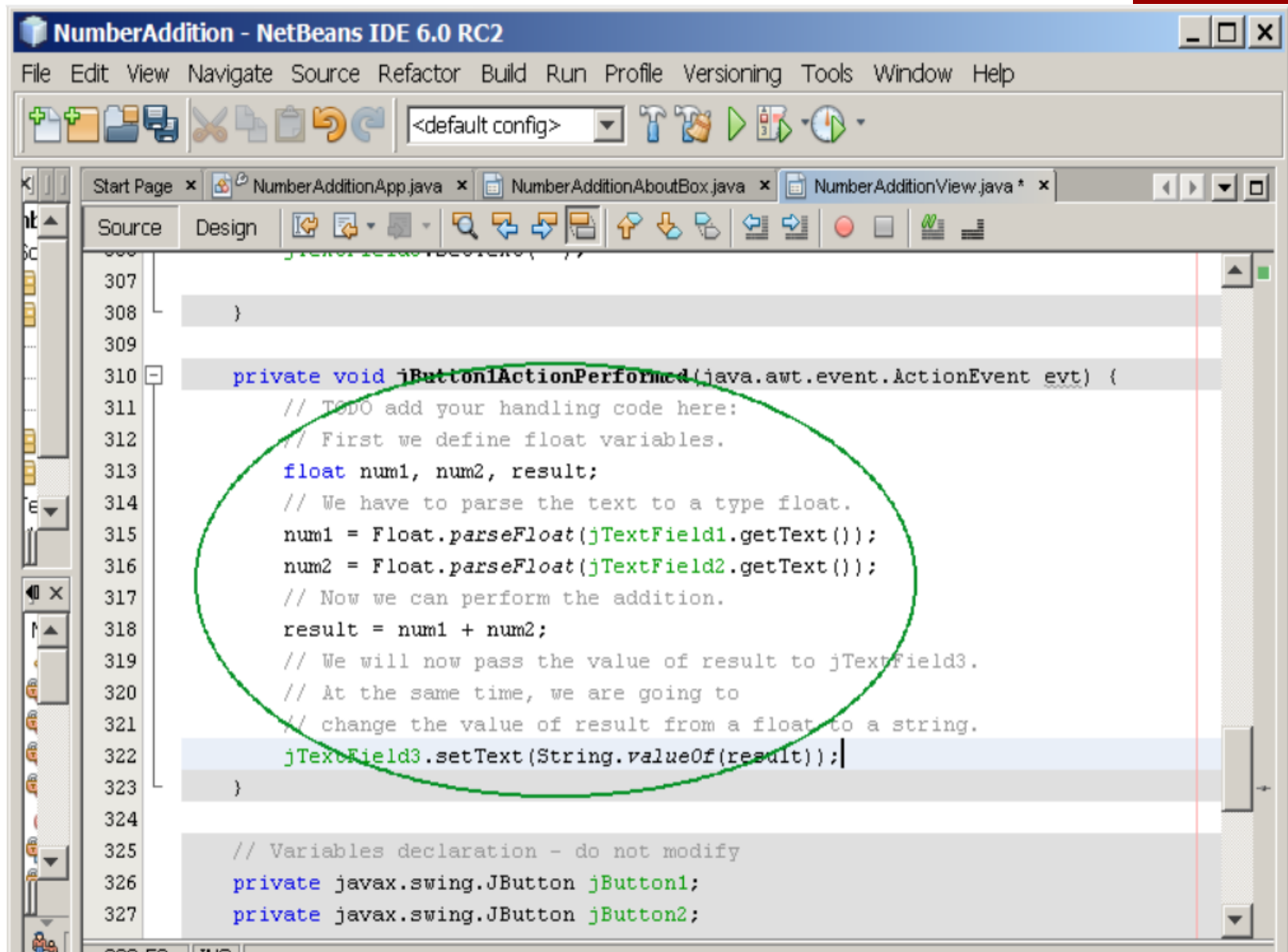
NetBeans IDE GUI builder

- Modify the event handler as shown in Code and Figure . The code fragment that needs be added is high-lighted in bold-font.

```
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt){  
    // First we define float variables.  
    float num1, num2, result;  
    // We have to parse the text to a type float.  
    num1 = Float.parseFloat(jTextField1.getText());  
    num2 = Float.parseFloat(jTextField2.getText());  
    // Now we can perform the addition.  
    result = num1+num2;  
    // We will now pass the value of result to jTextField3.  
    // At the same time, we are going to  
    // change the value of result from a float to a string.  
    jTextField3.setText(String.valueOf(result));  
}
```

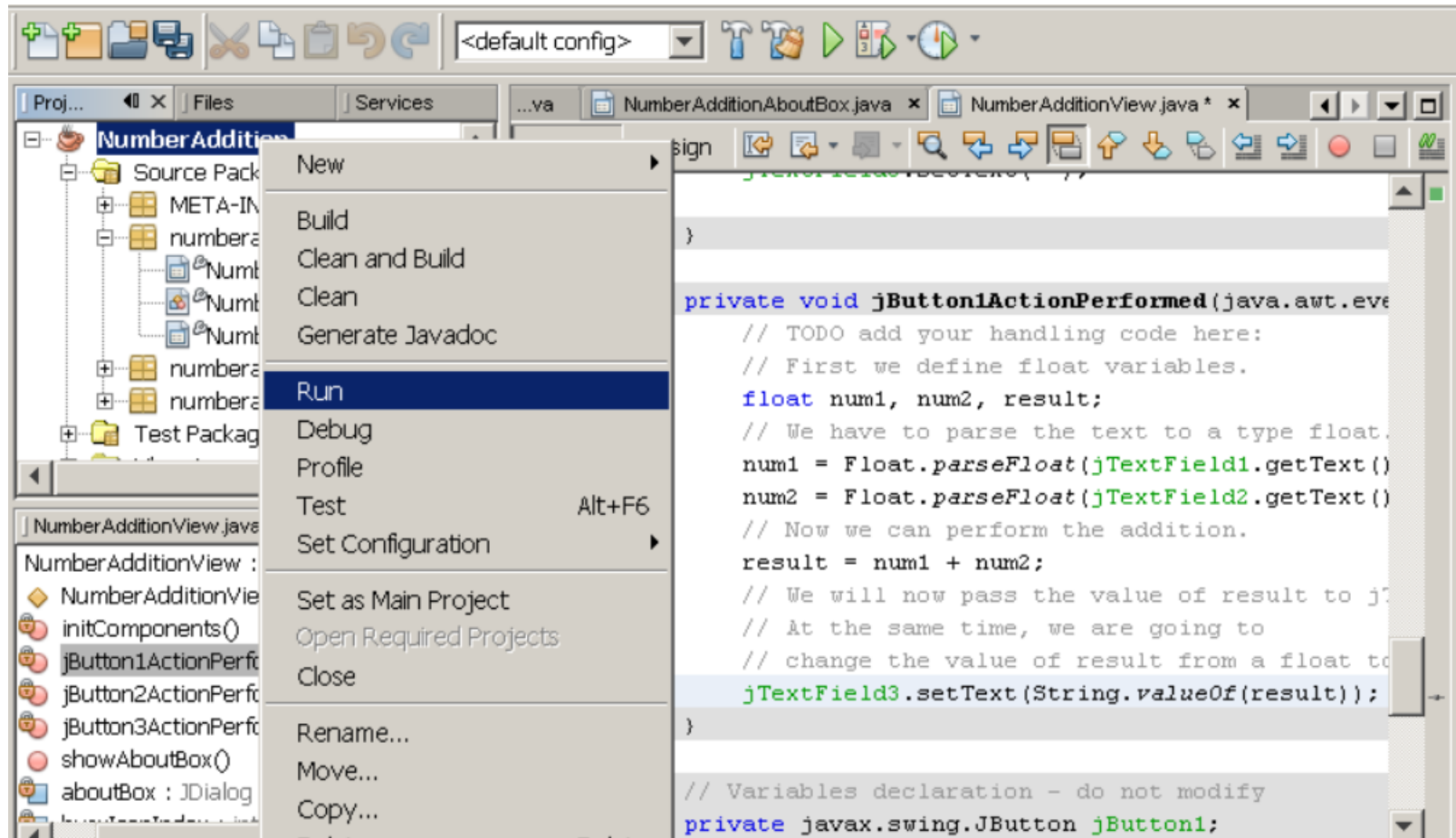
```
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt){  
    // First we define float variables.  
    float num1, num2, result;  
    // We have to parse the text to a type float.  
    num1 = Float.parseFloat(jTextField1.getText());  
    num2 = Float.parseFloat(jTextField2.getText());  
    // Now we can perform the addition.  
    result = num1+num2;  
    // We will now pass the value of result to jTextField3.  
    // At the same time, we are going to  
    // change the value of result from a float to a string.  
    jTextField3.setText(String.valueOf(result));  
}
```

NetBeans IDE GUI builder



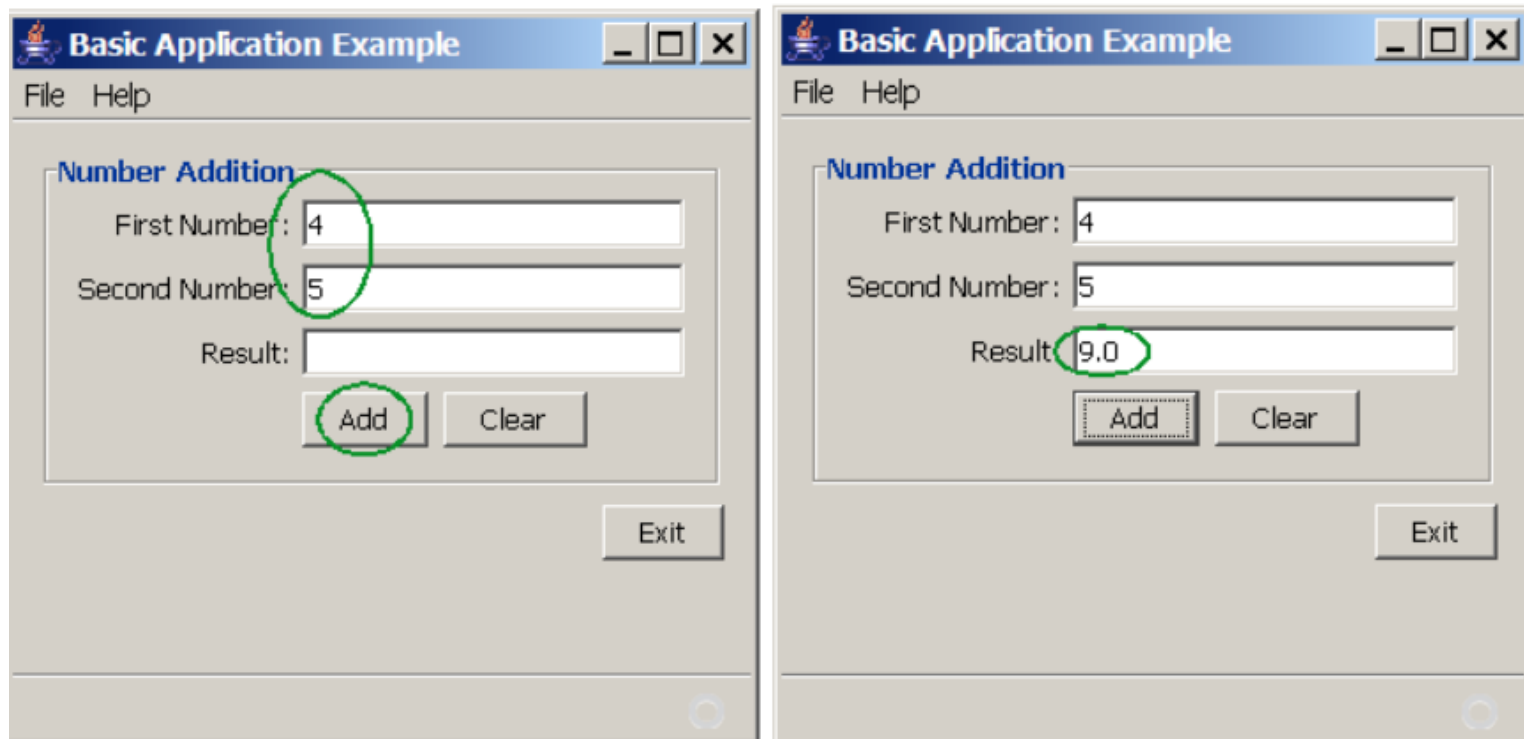
NetBeans IDE GUI builder

- Build and run the application:
- Right click the **NumberAddition** project and select **Run**.



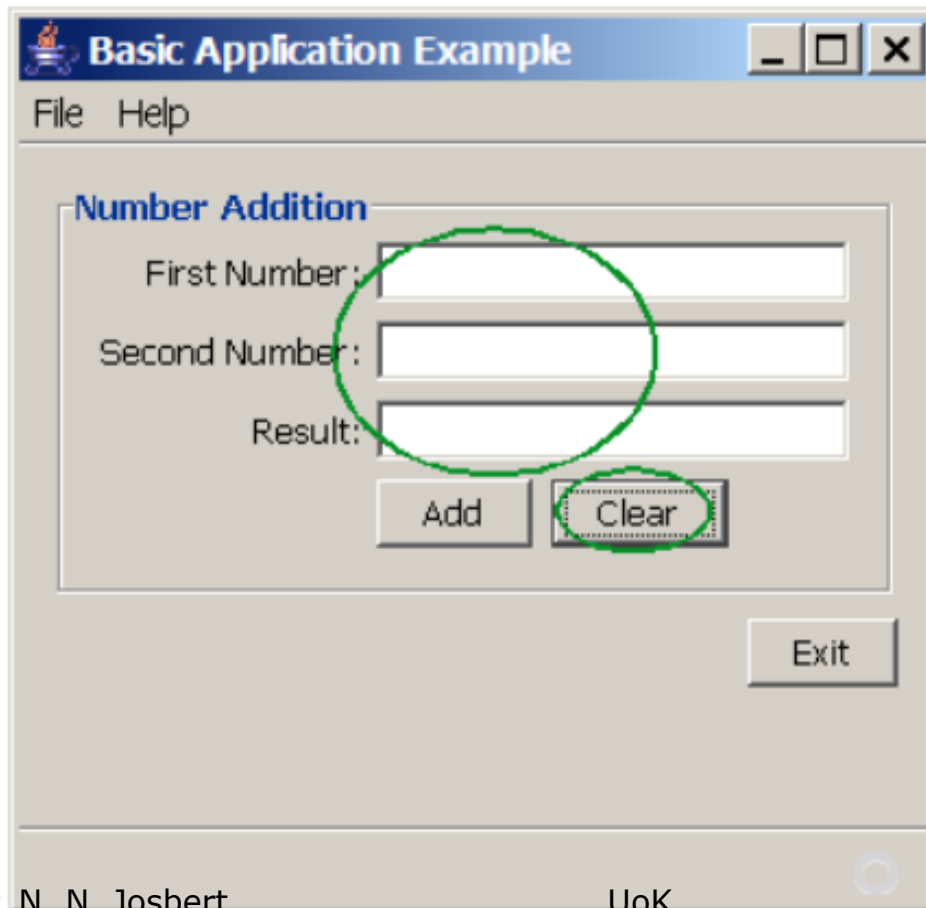
NetBeans IDE GUI builder

- **Perform Add operation:**
Enter numbers into the **First Number** and **Second Number** fields.
- Click **Add** button.



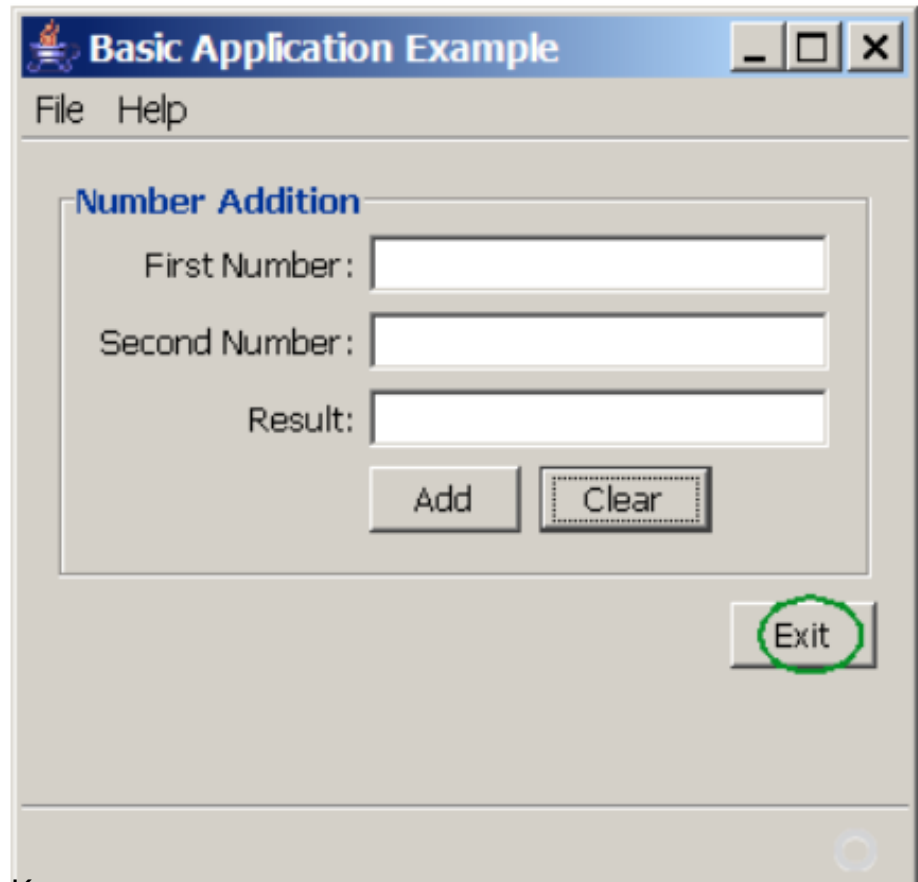
NetBeans IDE GUI builder

- **Perform Clear operation:**
Click Clear button.
- Observe that the fields are cleared.



NetBeans IDE GUI builder

- **Perform Exit operation:**
- Click Exit button.
- Observe the application closes.



NetBeans IDE GUI builder

■ Homework

- The homework is to modify the **NumberAddition** project to add **Subtract** button. The **Subtract** button subtracts the value of the Second Number field from the value of the First Number field and displays the result in the Result field. (You might want to create a new project by copying the **MyNumberComputer** project. You can name the homework project in any way you want but here I am going to call it **MyNumberComputer**.)

Example

106

NUMBER1		NUMBER3	
NUMBER2			
Sum		Average	
	Clear	End	

Codes



```
private void btnSumActionPerformed(java.awt.event.ActionEvent evt) {  
    int a,b,c,s;  
    a=Integer.parseInt(s: n1.getText());  
    b=Integer.parseInt(s: n2.getText());  
    c=Integer.parseInt(s: n3.getText());  
    s=a+b+c;  
    sum.setText(t: String.valueOf(i: s));  
}
```

```
private void btnAverageActionPerformed(java.awt.event.ActionEvent evt)  
    int a,b,c,s;  
    a=Integer.parseInt(s: n1.getText());  
    b=Integer.parseInt(s: n2.getText());  
    c=Integer.parseInt(s: n3.getText());  
    s=a+b+c;  
    double avg=Math.round((double)s/3);  
    sum.setText(t: String.valueOf(i: s));  
    average.setText(t: String.format(format: "%.2f", args: avg));  
}
```

```
private void btnClearActionPerformed(java.awt.event.ActionEvent evt)  
    n1.setText(t: "");  
    n2.setText(t: "");  
    sum.setText(t: "");  
    average.setText(t: "");  
}
```

```
private void btnEndActionPerformed(java.awt.event.ActionEvent evt) {  
    System.exit(status: 0);  
}
```

Example

Enter your marks

Your grade is :

Check **Clear** **End**

Marks	Grade
0 - 10	U
11 - 20	S
21 - 30	F
31 - 40	E
41 - 50	D
51 - 60	C
61 - 74	B
75 - 100	A

Codes

```
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {  
    double marks;  
    marks=Double.parseDouble(s: txtMarks.getText());  
    String grade;  
    if(marks>=75&& marks<=100)grade="A";  
        else if (marks>=61&& marks<75) grade="B";  
            else if (marks>=61&& marks<75) grade="B";  
                else if (marks>=51&& marks<60) grade="C";  
                    else if (marks>=41&& marks<500) grade="D";  
        else if (marks>=31&& marks<40) grade="E";  
        else if (marks>=21&& marks<30) grade="F";  
        else if (marks>=11&& marks<20) grade="S";  
        else if (marks>=0&& marks<10) grade="U";  
    else  
        grade="Invalid grade";  
    txtgrade.setText(t: grade);  
}  
  
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {  
    txtMarks.setText(t: "");  
    txtgrade.setText(t: "");  
}  
  
private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {  
    System.exit(status: 0);  
}
```

Form

Frame1

INPUT AND OUTPUT

Value of X

RESULT

ACTION

☒ NONE ☐ tan(x)

☐ sin(x) ☐ sqrt(x)

☐ cos(x) ☐ sqrt(x)

FIND Clear

```
private void btnFindActionPerformed(java.awt.event.ActionEvent evt) {  
    try {  
        double x = Double.parseDouble(txtValue.getText());  
        double result = 0.0;  
  
        if (rbtnSin.isSelected()) {  
            result = Math.sin(Math.toRadians(x));  
        } else if (rbtnCos.isSelected()) {  
            result = Math.cos(Math.toRadians(x));  
        } else if (rbtnTan.isSelected()) {  
            result = Math.tan(Math.toRadians(x));  
        } else if (rbtnSqr.isSelected()) {  
            result = x * x;  
        } else if (rbtnSqrt.isSelected()) {  
            result = Math.sqrt(x);  
        } else if (rbtnNone.isSelected()) {  
            txtResult.setText("No operation selected");  
            return;  
        }  
  
        txtResult.setText(String.format("%.4f", result));  
  
    } catch (NumberFormatException e) {  
        JOptionPane.showMessageDialog(this, "Please enter a valid number for X.");  
    }  
}
```

UoK STUDENTS FORM

Name:

Age:

Departement:

Gender:

☐

Male

☐

Female

Nationality:

Comments:

SUBMIT

DELETE

SAVE



UoK STUDENTS FORM

Name:

Age:

Departement:

Gender: ☐ Male ☐ Female

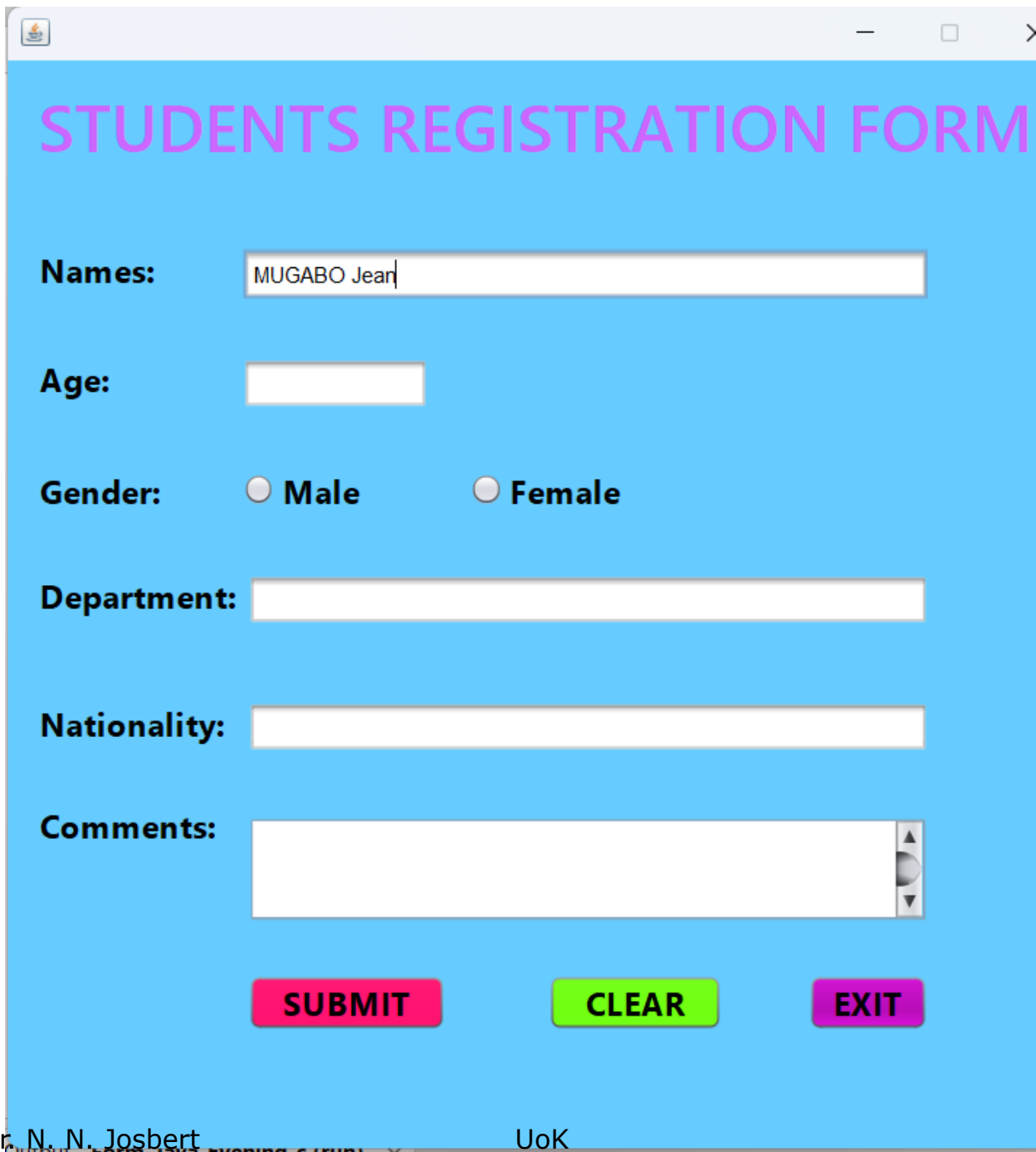
Nationality:

Comments:

SUBMIT

DELETE

SAVE



STUDENTS REGISTRATION FORM

Names:

Age:

Gender: ☐ **Male** ☐ **Female**

Department:

Nationality:

Comments:

SUBMIT **CLEAR** **EXIT**





UoK STUDENTS FORM

Name:

textField1

Age:

textField2

Nationality

textField3

Departement

Item 1

Gender:

☐ Male

☐ Female

Comments:

SUBMIT



GREEN VALLEY SECONDARY SCHOOL - FEES CALCULATION SYSTEM

Student Name:

Basic Tuition Fee (RWF):

Meals Charge (20%):

Transport Fee (10%):

Sports & Activities (Fixed 5000):

Total Fees Before Discount:

Scholarship Discount:

Other Deductions (RWF):

Amount to Pay:

Calculate

Clear

Exit



HOSPITAL PATIENT BILLING FORM

Patient Name:

Base Treatment Cost:

Ward Charge (15%):

Lab & Pharmacy (5%):

Gross Bill:

Insurance (5%):

VAT:

Other Deductions:

Net Payable:

Calculate

Clear

Exit